第 6 章 时序逻辑电路

Sequential Logic Circuits

§ 6.1 概述 Introduction

时序电路

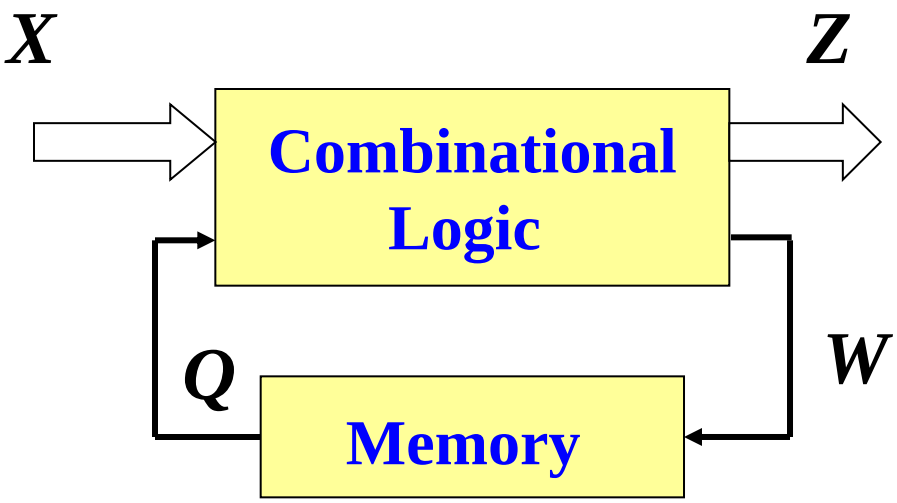
- 输出
  - 输入
  - 以前状态 记忆
- 基本单元: FF (逻辑门 + 反馈线)

逻辑电路

- 同步 所有的触发器在*CLK* 同一边沿触发
- 异步

时序电路结构:

组合电路 + 记忆元件



X：外输入

Z：外输出

W: 控制输入 -- *J, K, D, T*

Q: 触发器输出（状态）

外输入 X　　控制输入 W（J, K, D, T）

外输出 Z　　状态　Q

关系：

$$\begin{cases} \text{输出方程} & Z = F(X, Q) \\ \text{驱动方程} & W = H(X, Q) \\ \text{特征方程} & Q^{n+1} = G(W, Q^n) \end{cases}$$

按照电路中输出变量是否和输入变量直接相关

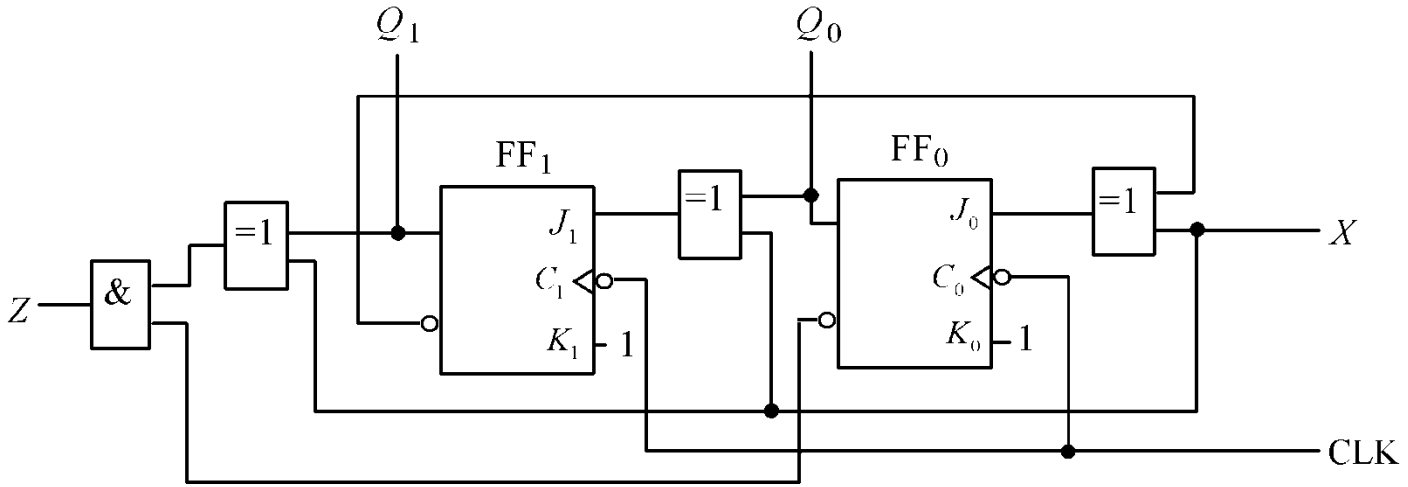
时序电路
- 米里型 (Mealy)　输出 Z $\begin{cases} Q^n \\ X \end{cases}$
- 莫尔型 (Moore)　输出 $Z \sim Q^n$

§6.2 同步时序电路分析

Sequential Logic Circuits Analysis

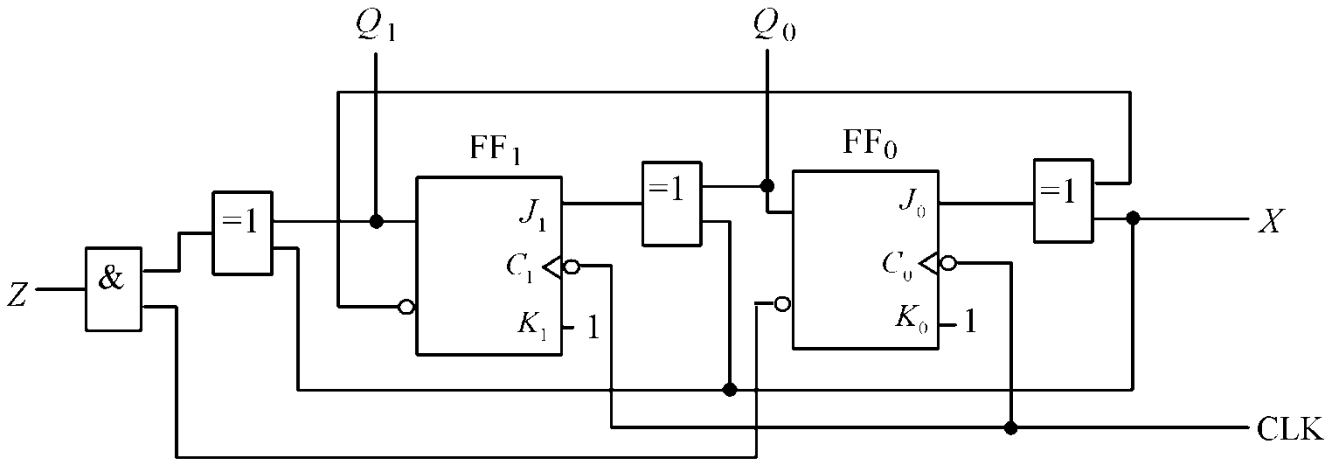
分析：已知电路，描述电路原理及功能.

例1: 分析下图时序逻辑电路



1) 输入 X 控制输入 J_0, K_0, J_1, K_1

输出 Z 状态 Q_1 (MSB), Q_0



2) 方程

输出方程 $$Z = (X \oplus Q_1^n) \cdot \overline{Q_0^n}$$

驱动方程 $$\begin{cases} J_0 = X \oplus \overline{Q_1^n} \\ K_0 = 1 \end{cases} \qquad \begin{cases} J_1 = X \oplus Q_0^n \\ K_1 = 1 \end{cases}$$

特征方程 $$\begin{cases} Q_0^{n+1} = J_0\overline{Q_0^n} + \overline{K_0}Q_0^n = (X \oplus \overline{Q_1^n}) \cdot \overline{Q_0^n} \\ Q_1^{n+1} = J_1\overline{Q_1^n} + \overline{K_1}Q_1^n = (X \oplus Q_0^n) \cdot \overline{Q_1^n} \end{cases}$$

3) 状态表和状态图

已知：输入 X，Q^n

求：输出 Z，Q^{n+1}

$$Q_1^{n+1} = (X \oplus Q_0^n) \cdot \overline{Q_1^n}$$

$$Q_0^{n+1} = (X \oplus \overline{Q_1^n})\overline{Q_0^n}$$

$$Z = (X \oplus Q_1^n) \cdot \overline{Q_0^n}$$

状态表

	X	Q_1^n	Q_0^n	Q_1^{n+1}	Q_0^{n+1}	Z
X=0	0	0	0	0	1	0
	0	0	1	1	0	0
	0	1	0	0	0	1
	0	1	1	0	0	0
X=1	1	0	0	1	0	1
	1	0	1	0	0	0
	1	1	0	0	1	0
	1	1	1	0	0	0

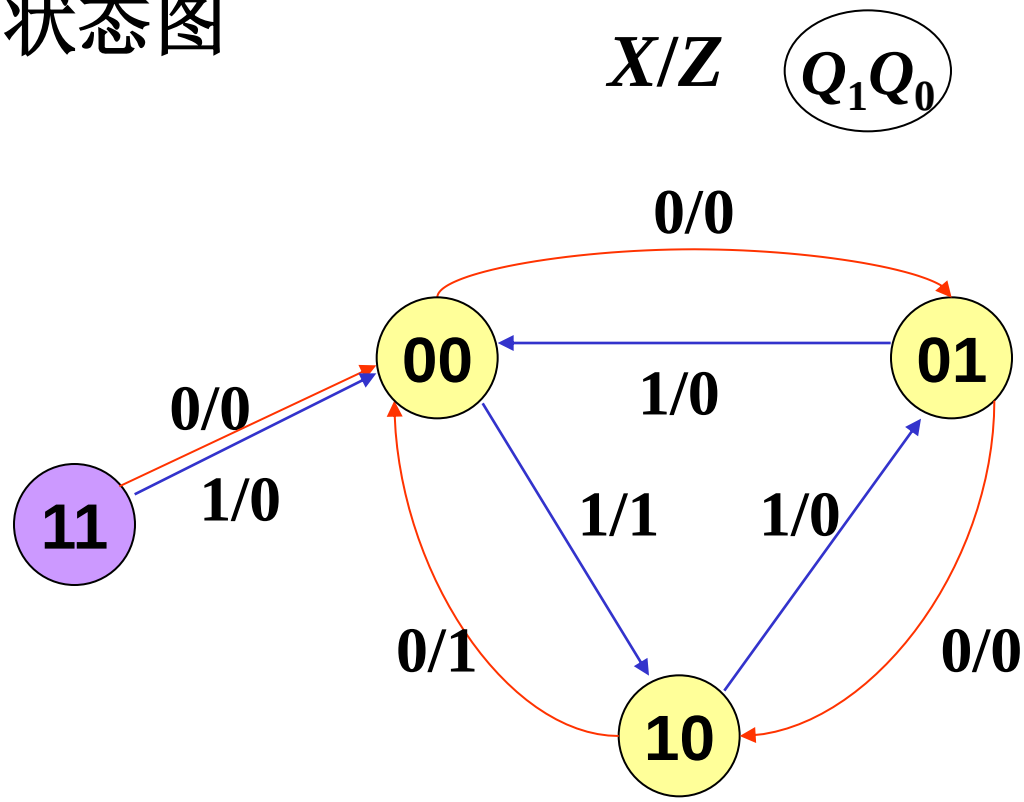
X=0

$$\begin{cases} Q_1^{n+1} = Q_0^n \cdot \overline{Q}_1^n \\ Q_0^{n+1} = \overline{Q_1^n} \cdot \overline{Q_0^n} = \overline{Q_1^n + Q_0^n} \\ Z = Q_1^n \cdot \overline{Q_0^n} \end{cases}$$

X=1

$$\begin{cases} Q_1^{n+1} = \overline{Q_0^n} \cdot \overline{Q_1^n} \\ Q_0^{n+1} = Q_1^n \cdot \overline{Q_0^n} \\ Z = \overline{Q_1^n} \cdot \overline{Q_0^n} \end{cases}$$

状态图



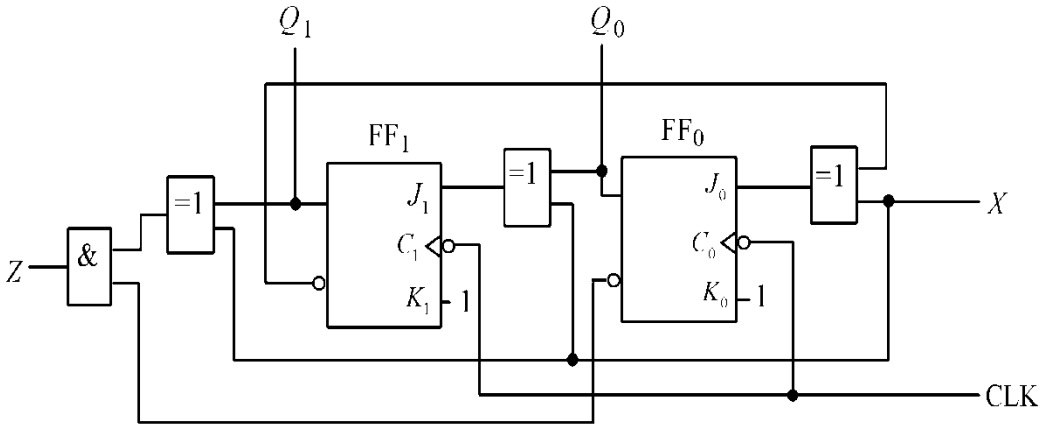
状态表

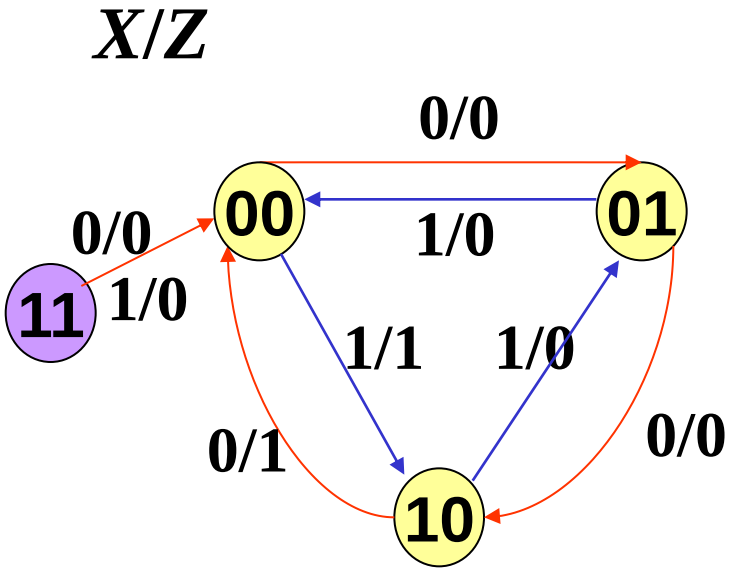
X	Q_1^n	Q_0^n	Q_1^{n+1}	Q_0^{n+1}	Z
0	0	0	0	1	0
0	0	1	1	0	0
0	1	0	0	0	1
0	1	1	0	0	0
1	0	0	1	0	1
1	0	1	0	0	0
1	1	0	0	1	0
1	1	1	0	0	0

⟶ 对应一个*CLK*　每一条线对应真值表的一行

输出*Z*是原状态下的输出。

4) 电路功能



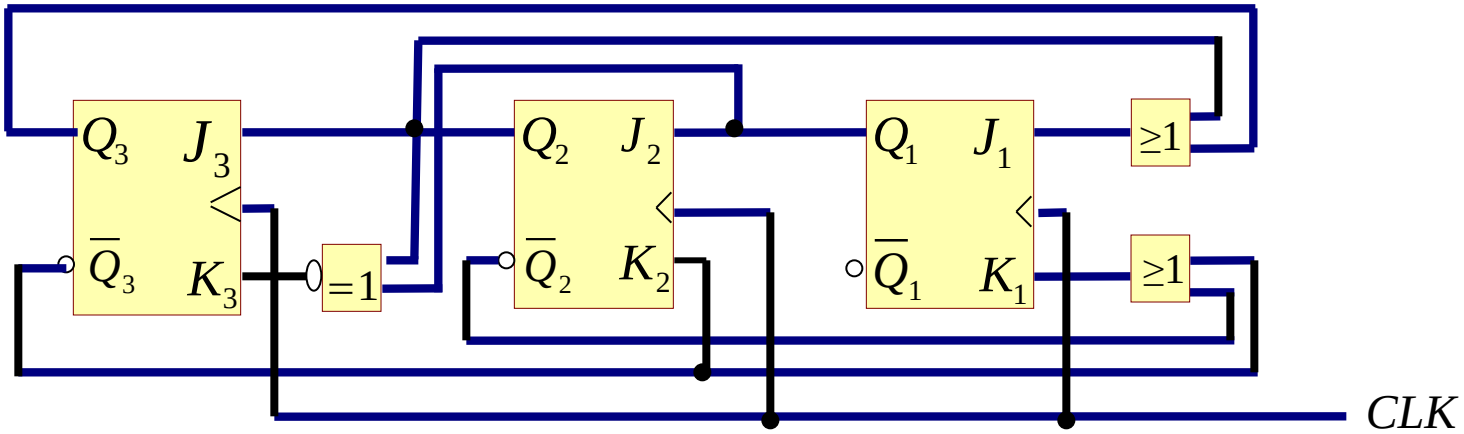


状态图主循环：模 3 加减双向计数器

X=0, M-3 Up counter：Z＝1，carry output；

X=1, M-3 Down counter：Z＝1，borrow output。

例 2. 分析下图时序电路



无外输入，无外输出

$$\begin{cases} J_3 = Q_2^{\,n} \\ K_3 = \overline{Q_2^{\,n} \oplus Q_1^{\,n}} \end{cases} \qquad \begin{cases} J_2 = Q_1^{\,n} \\ K_2 = \overline{Q}_3^{\,n} \end{cases} \qquad \begin{cases} J_1 = Q_2^{\,n} + Q_3^{\,n} \\ K_1 = \overline{Q}_2^{\,n} + \overline{Q}_3^{\,n} = \overline{Q_2^{\,n} Q_3^{\,n}} \end{cases}$$

$$\begin{cases} J_3 = Q_2^{\,n} \\ K_3 = \overline{Q_2^{\,n} \oplus Q_1^{\,n}} \end{cases} \qquad \begin{cases} J_2 = Q_1^{\,n} \\ K_2 = \overline{Q_3^{\,n}} \end{cases} \qquad \begin{cases} J_1 = Q_2^{\,n} + Q_3^{\,n} \\ K_1 = \overline{Q}_2^{\,n} + \overline{Q}_3^{\,n} = \overline{Q_2^{\,n} Q_3^{\,n}} \end{cases}$$

$$Q_3^{n+1} = J_3 \overline{Q_3^n} + \overline{K}_3 Q_3^n = Q_2^n \overline{Q_3^n} + (Q_2^n \oplus Q_1^n) Q_3^n$$

$$Q_2^{n+1} = J_2 \overline{Q_2^n} + \overline{K}_2 Q_2^n = Q_1^n \overline{Q_2^n} + Q_3^n Q_2^n$$

$$Q_1^{n+1} = J_1 \overline{Q_1^n} + \overline{K}_1 Q_1^n = (Q_2^n + Q_3^n) \overline{Q_1^n} + Q_2^n Q_3^n Q_1^n$$

状态表

Q_3^n	Q_2^n	Q_1^n	Q_3^{n+1}	Q_2^{n+1}	Q_1^{n+1}
0	0	0	0	0	0
0	0	1	0	1	0
0	1	0	1	0	1
0	1	1	1	0	0
1	0	0	0	0	1
1	0	1	1	1	0
1	1	0	1	1	1
1	1	1	0	1	1

$$Q_3^{n+1} = Q_2^n \overline{Q_3^n} + (Q_2^n \oplus Q_1^n) Q_3^n$$

$$Q_2^{n+1} = Q_1^n \overline{Q_2^n} + Q_3^n Q_2^n$$

$$Q_1^{n+1} = (Q_2^n + Q_3^n) \overline{Q_1^n} + Q_2^n Q_3^n Q_1^n$$

$$Q_3^{n+1} \begin{cases} Q_2^n & Q_3^n = 0, \\ Q_2^n \oplus Q_1^n & Q_3^n = 1, \end{cases}$$

$$Q_2^{n+1} \begin{cases} Q_1^n & Q_2^n = 0, \\ Q_3^n & Q_2^n = 1, \end{cases}$$

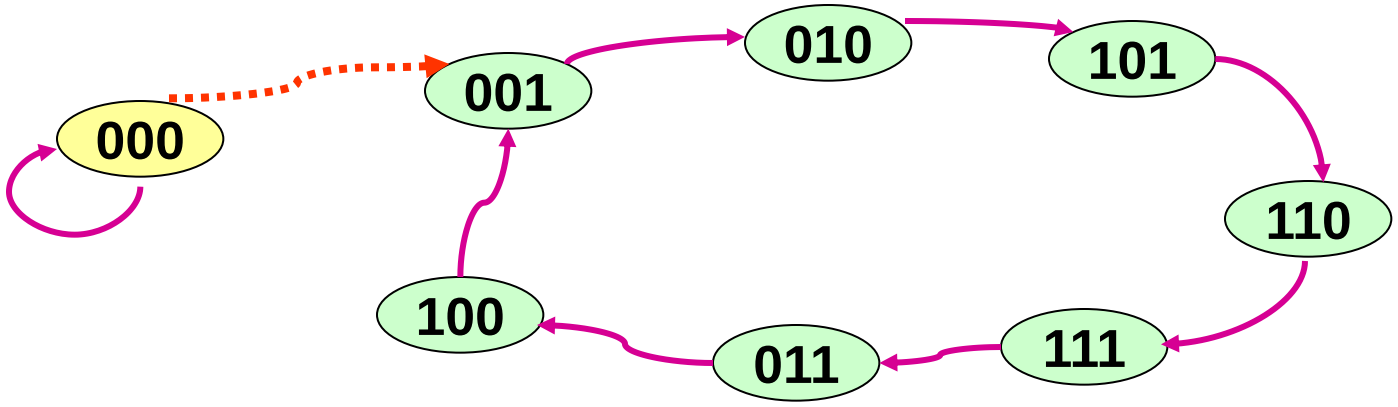
$$Q_1^{n+1} \begin{cases} Q_2^n + Q_3^n & Q_1^n = 0, \\ Q_2^n Q_3^n & Q_1^n = 1, \end{cases}$$

Q_3^n	Q_2^n	Q_1^n	Q_3^{n+1}	Q_2^{n+1}	Q_1^{n+1}
0	0	0	0	0	0
0	0	1	0	1	0
0	1	0	1	0	1
0	1	1	1	0	0
1	0	0	0	0	1
1	0	1	1	1	0
1	1	0	1	1	1
1	1	1	0	1	1

状态图

000 孤立状态

自启动



§6.3 同步时序电路设计

Synchronous Sequential Circuit Design

已知 → 功能或状态图

求 → 电路

设计步骤:

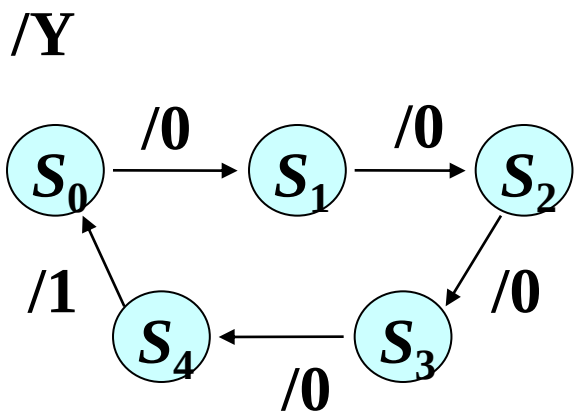
1) 确定状态和状态图
2) 状态化简
3) 状态分配 (编码)
4) 选择触发器类型
5) 状态方程Q^{n+1}及控制输入– ***J, K, D, T***
6) 画出电路
7) 自启动

例 1. 设计同步5进制加法计数器 (例6.4)

1) 确定状态及状态图

M-5 计数器，5 个状态：S_0, S_1, S_2, S_3, S_4

在计数脉冲*CLK*作用下，5 个状态周期性变换, 在 S_4 状态下进位输出 $Y = 1$ 。



2) 状态化简

M-5, 5 个状态， 不须再化简

3) 状态分配、编码

$2^{n-1} \le$状态数 $\le 2^n$　　n:二进制位数　　5个状态需3位

S_0 ⟹ 000

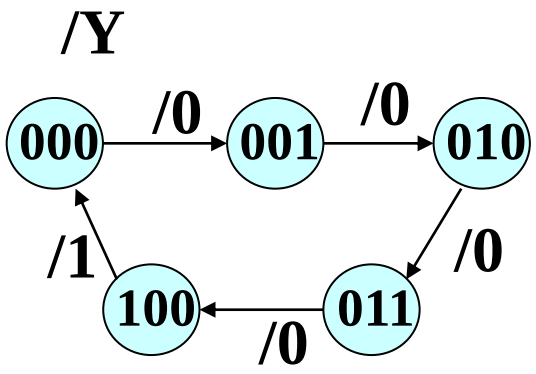
S_1 ⟹ 001

S_2 ⟹ 010

S_3 ⟹ 011

S_4 ⟹ 100

状态图



状态表

Q_3^n Q_2^n Q_1^n	Q_3^{n+1} Q_2^{n+1} Q_1^{n+1}	Y
0 0 0	0 0 1	0
0 0 1	0 1 0	0
0 1 0	0 1 1	0
0 1 1	1 0 0	0
1 0 0	0 0 0	1

4) 选择 FF，确定状态方程Q^{n+1}及输入

方法 1：先不确定用哪种触发器

由状态表填卡诺图

状态表

Q_3^n	Q_2^n	Q_1^n	Q_3^{n+1}	Q_2^{n+1}	Q_1^{n+1}	Y
0	0	0	0	0	1	0
0	0	1	0	1	0	0
0	1	0	0	1	1	0
0	1	1	1	0	0	0
1	0	0	0	0	0	1

Q_3^{n+1}

Q_1^n \ $Q_3^nQ_2^n$	00	01	11	10
0	0	0	Φ	0
1	0	1	Φ	Φ

Q_2^{n+1}

Q_1^n \ $Q_3^nQ_2^n$	00	01	11	10
0	0	1	Φ	0
1	1	0	Φ	Φ

Q_1^{n+1}

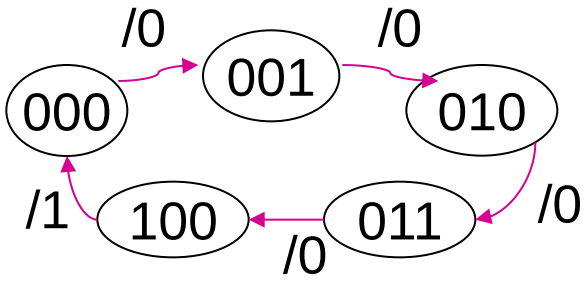
Q_1^n \ $Q_3^nQ_2^n$	00	01	11	10
0	1	1	Φ	0
1	0	0	Φ	Φ

Y

Q_1^n \ $Q_3^nQ_2^n$	00	01	11	10
0	0	0	Φ	1
1	0	0	Φ	Φ

也可以直接填卡诺图

直接填卡诺图



5个 有效状态　$5 < 2^3$　3 个FF

$Q_3^nQ_2^nQ_1^n$ /Y $Q_3^{n+1}Q_2^{n+1}Q_1^{n+1}$

Y

Q_1^n \ $Q_3^nQ_2^n$	00	01	11	10
0	0	0	Φ	1
1	0	0	Φ	Φ

Q_3^{n+1}

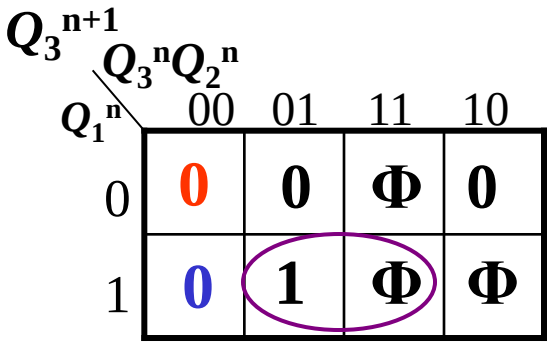
Q_1^n \ $Q_3^nQ_2^n$	00	01	11	10
0	0	0	Φ	0
1	0	1	Φ	Φ

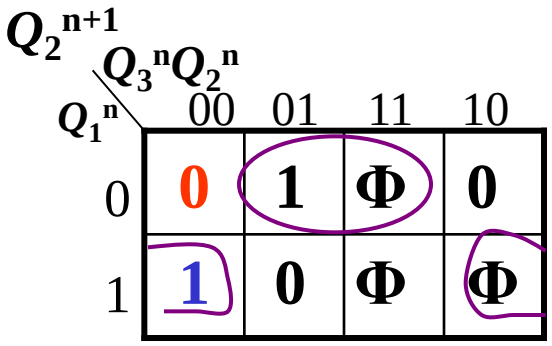
Q_2^{n+1}

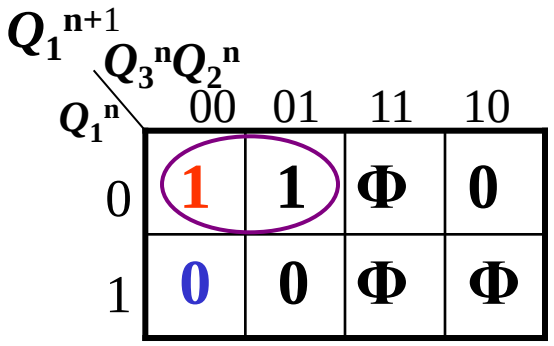
Q_1^n \ $Q_3^nQ_2^n$	00	01	11	10
0	0	1	Φ	0
1	1	0	Φ	Φ

Q_1^{n+1}

Q_1^n \ $Q_3^nQ_2^n$	00	01	11	10
0	1	1	Φ	0
1	0	0	Φ	Φ







$$Q_3^{n+1} = Q_2^nQ_1^n$$

$$= D_3$$

$$D_3 = Q_2^nQ_1^n$$

$$Q_2^{n+1} = Q_1^n\bar{Q}_2^n + \bar{Q}_1^nQ_2^n$$

$$= Q_1^n \oplus Q_2^n$$

$$= T_2 \oplus Q_2^n$$

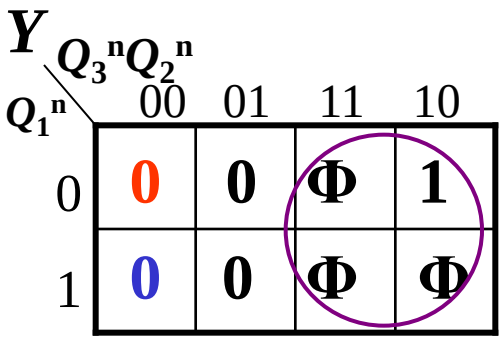
$$T_2 = Q_1^n$$

$$Q_1^{n+1} = \bar{Q}_3^n\bar{Q}_1^n$$

$$= D_1$$

或

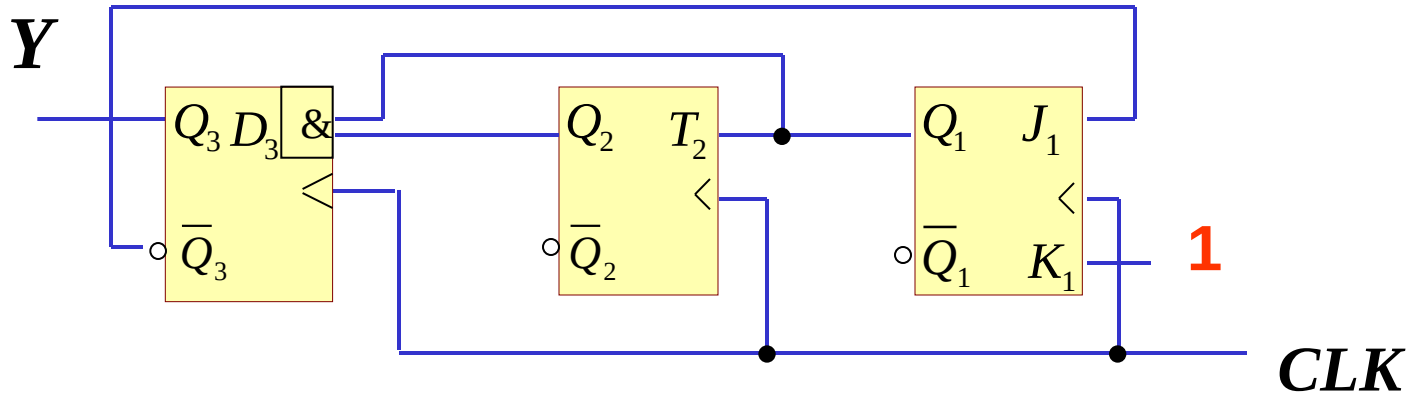
$$\begin{cases} J_1 = \overline{Q_3^n} \\ K_1 = 1 \end{cases}$$



$$Y = Q_3^n$$

$D_3 = Q_2^n Q_1^n$　　$T_2 = Q_1^n$　　$\begin{cases} J_1 = \overline{Q}_3^n \\ K_1 = 1 \end{cases}$　　$Y = Q_3^n$

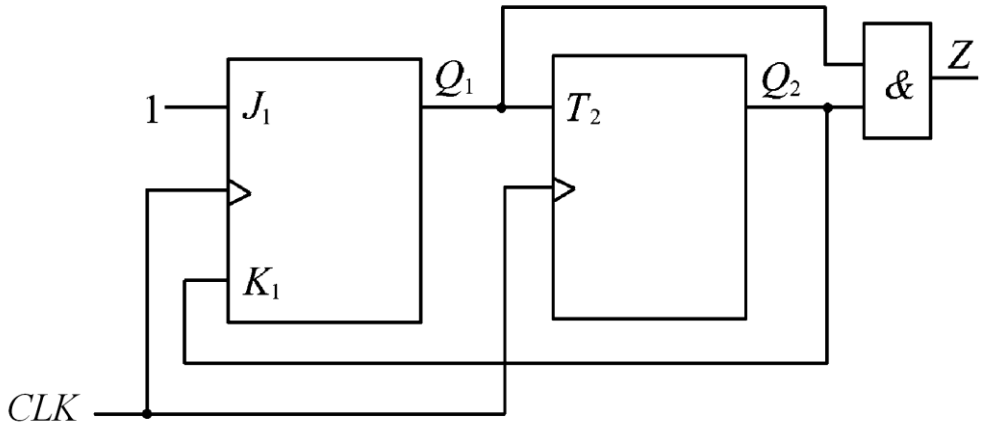
5）电路



与门可以省略　　同步5进制加法计数器

课堂练习

分析题图所示的同步时序电路，画出状态图。



解： $J_1 = 1, \quad K_1 = Q_2^n,$

$$T_2 = Q_1^n,$$

$$Z = Q_2^n Q_1^n$$

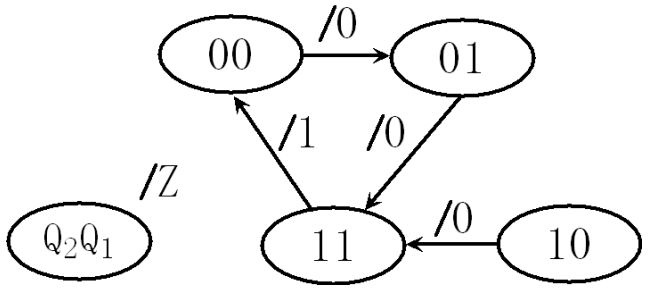
$$Q_1^{n+1} = J_1\overline{Q_1^n} + \overline{K}_1 Q_1^n$$

$$= \overline{Q_1^n} + \overline{Q_2^n} Q_1^n$$

$$= \overline{Q_1^n} + \overline{Q_2^n}$$

$$Q_2^{n+1} = T_2 \oplus Q_2^n = Q_1^n \oplus Q_2^n$$

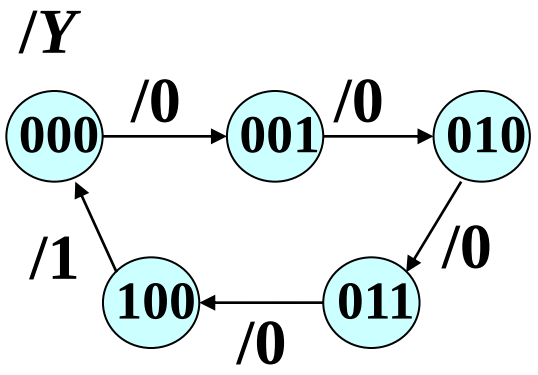
Q_2^n	Q_1^n	Q_2^{n+1}	Q_1^{n+1}	Z
0	0	0	1	0
0	1	1	1	0
1	0	1	1	0
1	1	0	0	1



设计同步5进制加法计数器

方法 2：确定用哪种触发器

状态图



4) 选择 FF

选 JK-FFs

5) 状态方程Q^{n+1}及控制输入– *J*, *K*

状态表

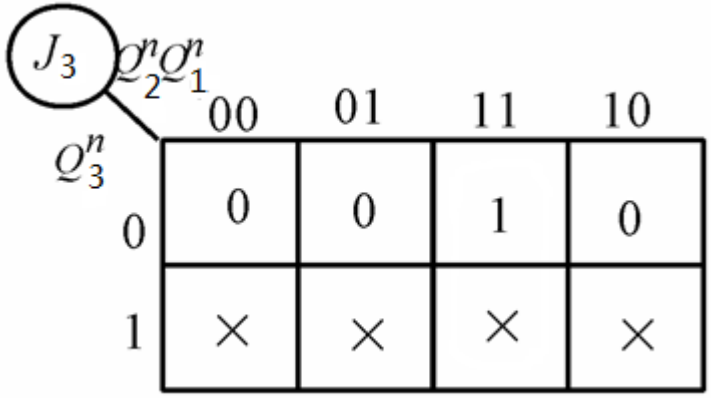
$Q_3^nQ_2^nQ_1^n$	$Q_3^{n+1}Q_2^{n+1}Q_1^{n+1}$	Y
0 0 0	0 0 1	0
0 0 1	0 1 0	0
0 1 0	0 1 1	0
0 1 1	1 0 0	0
1 0 0	0 0 0	1

JK-FF 驱动表

$Q^n \to Q^{n+1}$	J	K
0	0	0 ×
0	1	1 ×
1	0	× 1
1	1	× 0

$Q_3^n \Rightarrow Q_3^{n+1}$		J_3
0	0	0
0	0	0
0	0	0
0	1	1
1	0	X
X	X	X
X	X	X
X	X	X

得到 $3^{\#}$-FF 控制输入 J_3 驱动卡诺图



状态表

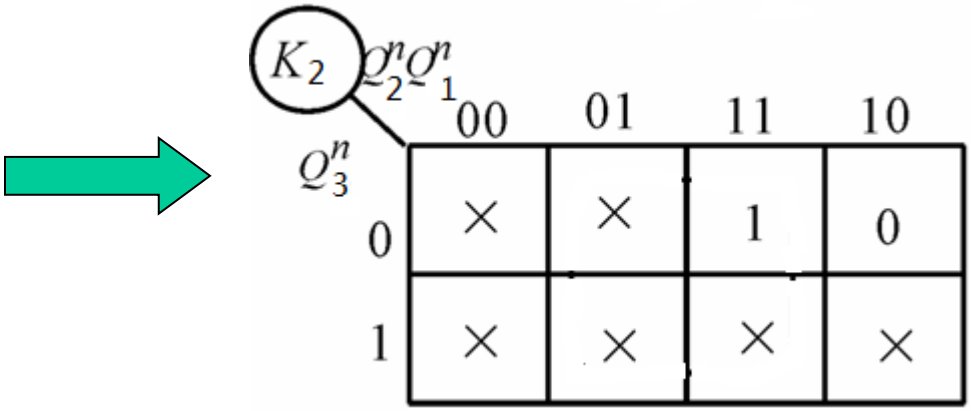
$Q_3^nQ_2^nQ_1^n$	$Q_3^{n+1}Q_2^{n+1}Q_1^{n+1}$	Y
0 0 0	0 0 1	0
0 0 1	0 1 0	0
0 1 0	0 1 1	0
0 1 1	1 0 0	0
1 0 0	0 0 0	1

JK-FF 驱动表

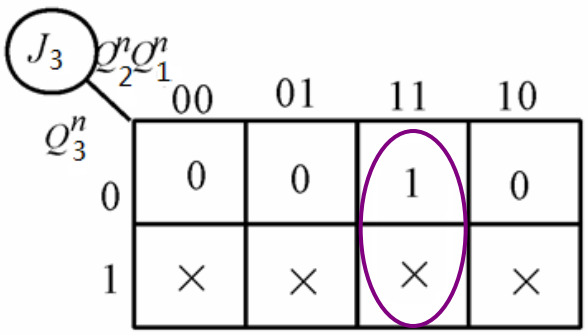
$Q^n \to Q^{n+1}$	J	K
0	0	0 ×
0	1	1 ×
1	0	× 1
1	1	× 0

$Q_2^n \Longrightarrow Q_2^{n+1}$		K_2
0	0	X
0	1	X
1	1	0
1	0	1
0	0	X
X	X	X
X	X	X
X	X	X

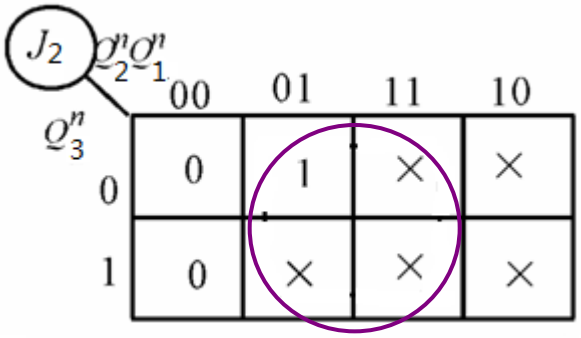
得到 $2^{\#}$-FF 控制输入 K_2 驱动卡诺图



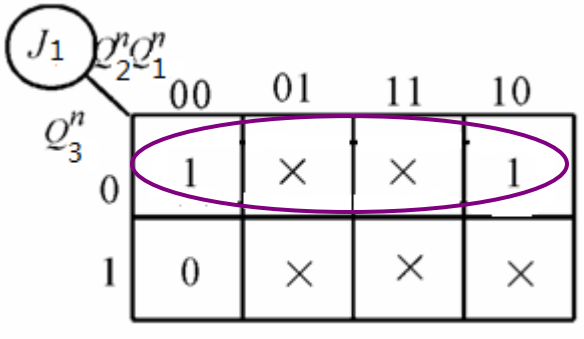
得到各个触发器控制输入驱动卡诺图及控制输入



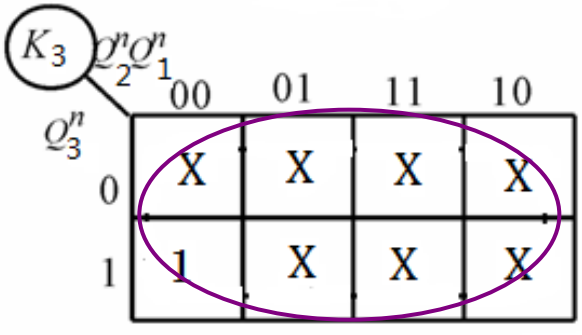
$J_3 = Q_2^n Q_1^n$



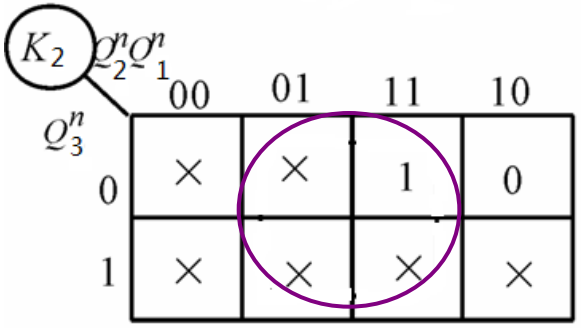
$J_2 = Q_1^n$



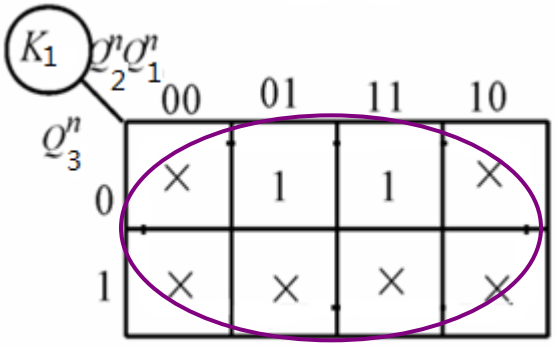
$J_1 = \overline{Q_3^n}$



$K_3 = 1$

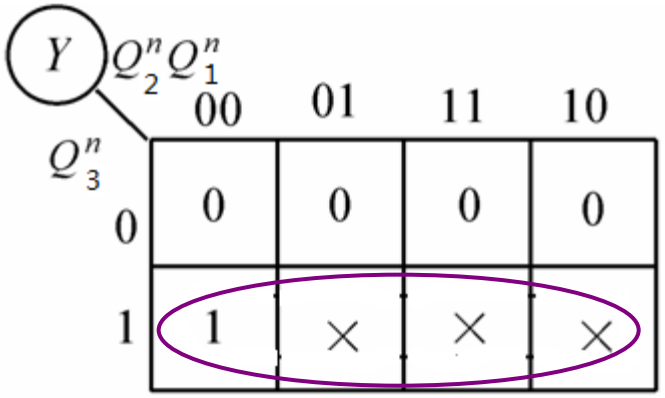


$K_2 = Q_1^n$



$K_1 = 1$

输出卡诺图



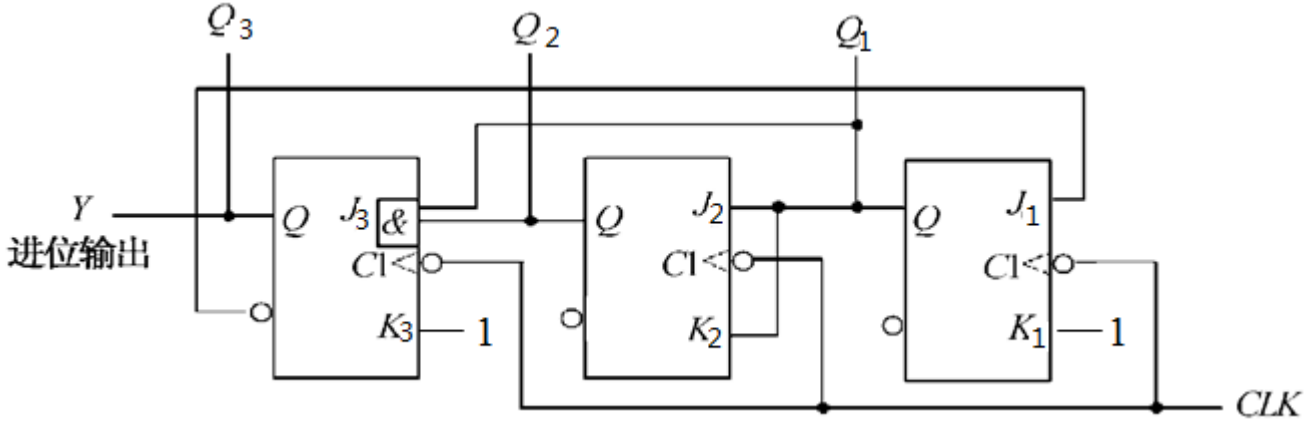
状态表

$Q_3^nQ_2^nQ_1^n$	$Q_3^{n+1}Q_2^{n+1}Q_1^{n+1}$	Y
0 0 0	0 0 1	0
0 0 1	0 1 0	0
0 1 0	0 1 1	0
0 1 1	1 0 0	0
1 0 0	0 0 0	1

$$Y = Q_3^n$$

$$\begin{cases} J_3 = Q_2^nQ_1^n \\ K_3 = 1 \end{cases} \qquad \begin{cases} J_2 = Q_1^n \\ K_2 = Q_1^n \end{cases} \qquad \begin{cases} J_1 = \overline{Q_3^n} \\ K_1 = 1 \end{cases}$$

6) 电路



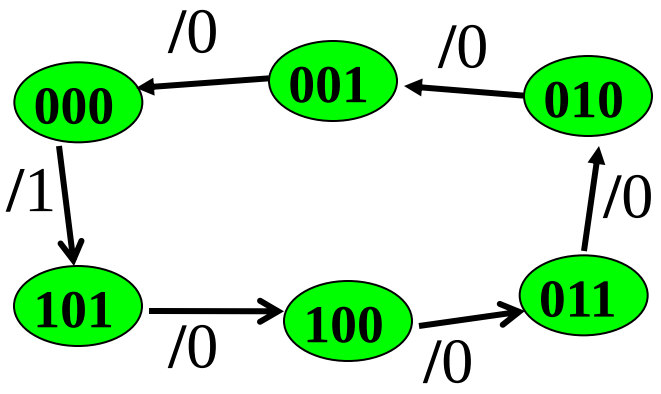
用JK-FF实现同步5进制加法计数器　（与方法1结果相同）

例 2.　设计 M-6 减法计数器

6 个状态　　直接用3位数编码　　$Q_3Q_2Q_1$

借位输出 Z

/ Z



Z

Q_1^n \ $Q_3^nQ_2^n$	00	01	11	10
0	1	0	Φ	0
1	0	0	Φ	0

Q_3^{n+1}

Q_1^n \ $Q_3^nQ_2^n$	00	01	11	10
0	1	0	Φ	0
1	0	0	Φ	1

Q_2^{n+1}

Q_1^n \ $Q_3^nQ_2^n$	00	01	11	10
0	0	0	Φ	1
1	0	1	Φ	0

Q_1^{n+1}

Q_1^n \ $Q_3^nQ_2^n$	00	01	11	10
0	1	1	Φ	1
1	0	0	Φ	0

Q_3^{n+1}

Q_1^n \ $Q_3^nQ_2^n$	00	01	11	10
0	1	0	Φ	0
1	0	0	Φ	1

$$Q_3^{n+1} = \overline{Q_3}\ \overline{Q_2}\ \overline{Q_1} + Q_3Q_1$$

$$D_3 = \overline{Q_3}\ \overline{Q_2}\ \overline{Q_1} + Q_3Q_1$$

Q_2^{n+1}

Q_1^n \ $Q_3^nQ_2^n$	00	01	11	10
0	0	0	Φ	1
1	0	1	Φ	0

$$Q_2^{n+1} = Q_2Q_1 + Q_3\overline{Q_1}$$

$$D_2 = Q_2Q_1 + Q_3\overline{Q_1}$$

Q_1^{n+1}

Q_1^n \ $Q_3^nQ_2^n$	00	01	11	10
0	1	1	Φ	1
1	0	0	Φ	0

$$Q_1^{n+1} = \overline{Q_1}$$

$$D_1 = \overline{Q_1}$$

Z

Q_1^n \ $Q_3^nQ_2^n$	00	01	11	10
0	1	0	Φ	0
1	0	0	Φ	0

$$Z = \overline{Q_3}\ \overline{Q_2}\ \overline{Q_1}$$

电路图略

例 3. (例6.5)

设计一个串行数据检测器。该检测器有一个输入端X。电路的功能是对输入信号进行检测。当连续输入三个1（以及三个以上1）时，该电路输出$Y = 1$，否则输出$Y = 0$。

1）根据设计要求，设定状态，画出状态转换图。

X=1，收到一个“1”

S_0—初始状态或没有收到1时的状态；

S_1—收到一个1后的状态；

S_2—连续收到两个1后的状态；

S_3—连续收到三个1（以及三个以上1）后的状态。

S_0—初始状态或没有收到1时的状态；

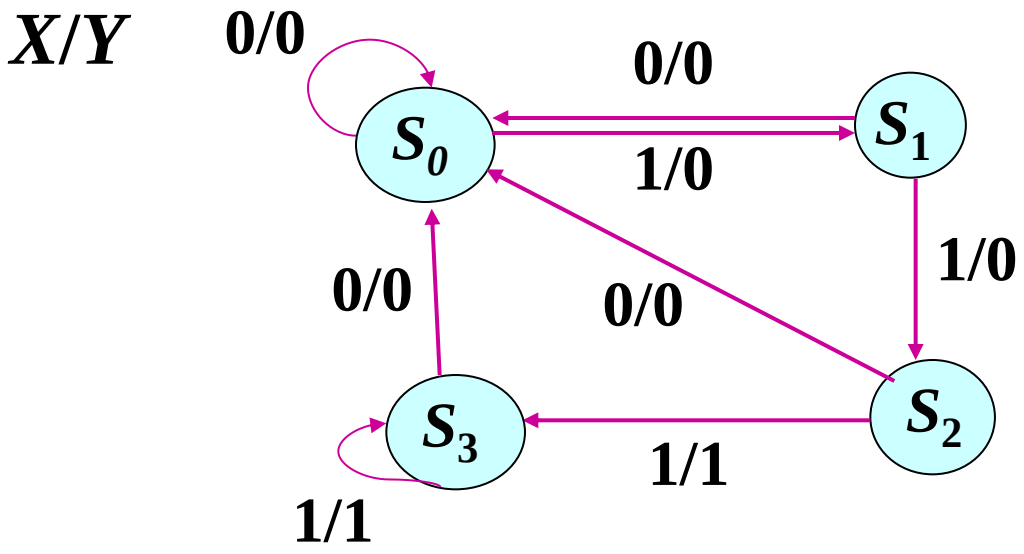
S_1—收到一个1后的状态；

S_2—连续收到两个1后的状态；

S_3—连续收到三个1（以及三个以上1）后的状态。

X=1, 收到一个 “1”

输入三个1（以及三个以上1）时，输出Y = 1

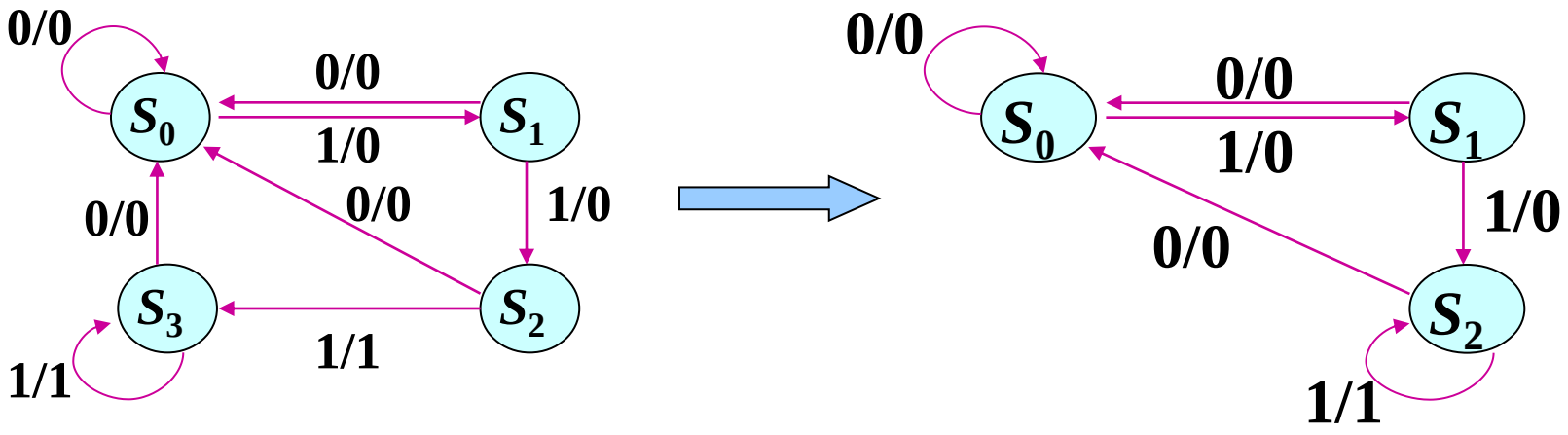


2）状态化简

状态化简： 合并等效状态

等效状态：在相同的输入条件下，
输出相同、次态也相同的状态。

S_2 和 S_3 是等效状态，将S_2 和 S_3 合并为S_2



3）状态分配、编码

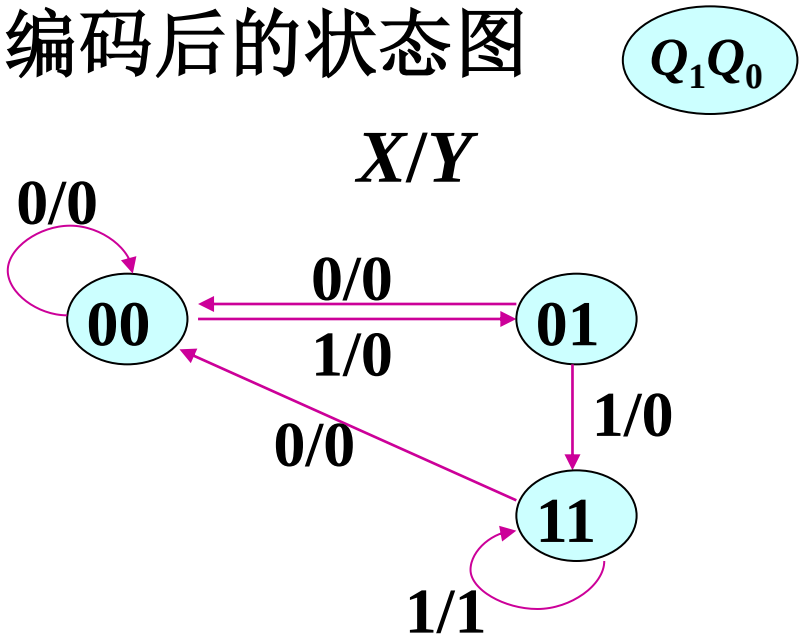
设 $S_0 = 00$

$S_1 = 01$

$S_2 = 11$

编码可以不连续

编码后的状态图



状态表

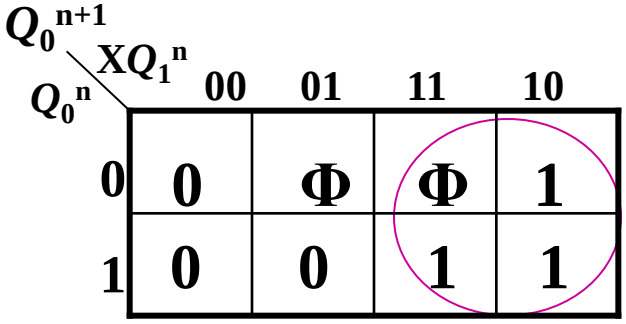
X	Q_1^n	Q_0^n	Q_1^{n+1}	Q_0^{n+1}	Y
0	0	0	0	0	0
0	0	1	0	0	0
0	1	0	Φ	Φ	Φ
0	1	1	0	0	0
1	0	0	0	1	0
1	0	1	1	1	0
1	1	0	Φ	Φ	Φ
1	1	1	1	1	1

4) 选触发器及控制输入

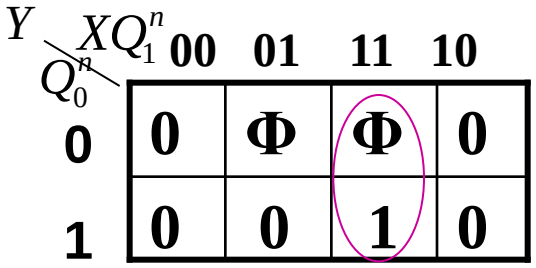
X	Q_1^n	Q_0^n	Q_1^{n+1}	Q_0^{n+1}	Y
0	0	0	0	0	0
0	0	1	0	0	0
0	1	0	Φ	Φ	Φ
0	1	1	0	0	0
1	0	0	0	1	0
1	0	1	1	1	0
1	1	0	Φ	Φ	Φ
1	1	1	1	1	1



$$Q_1^{n+1} = XQ_0^n = D_1 \qquad D_1 = XQ_0^n$$



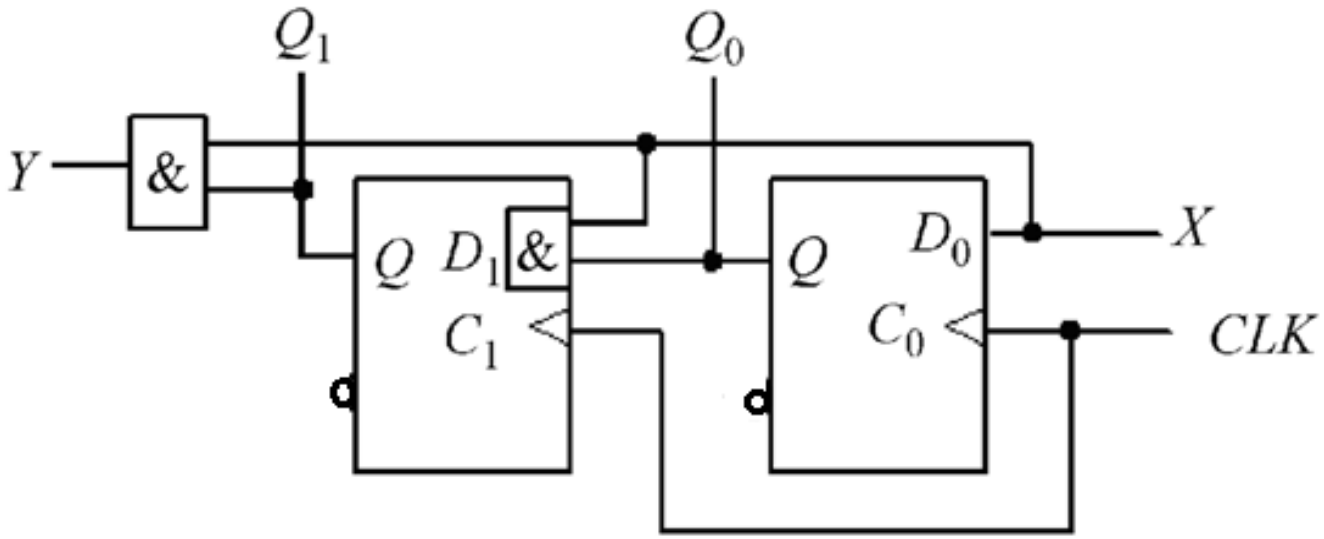
$$Q_0^{n+1} = X = D_0 \qquad D_0 = X$$



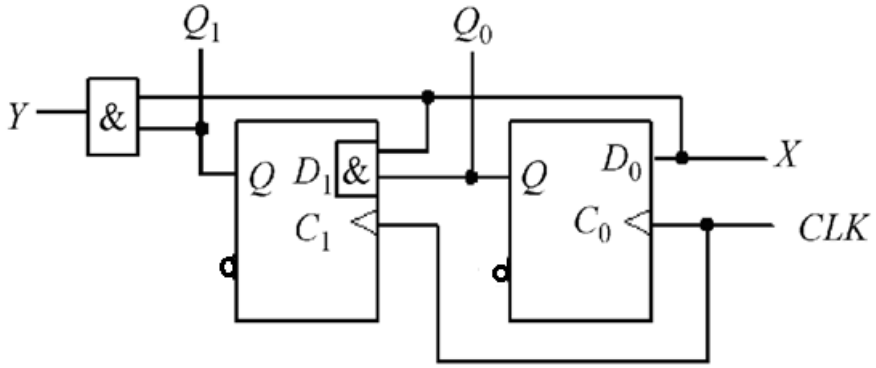
$$Y = XQ_1^n$$

5) 电路

2 D-FFs $D_1 = XQ_0^n$ $D_0 = X$ $Y = XQ_1^n$



6）自启动

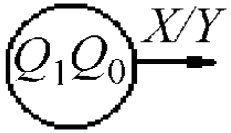


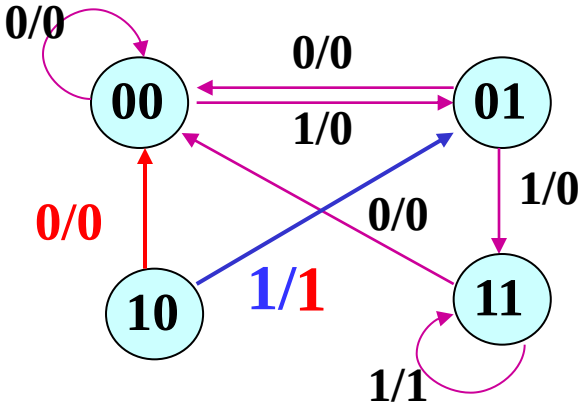
$$Q_1^{n+1} = XQ_0^n$$

$$Q_0^{n+1} = X$$

$$Y = XQ_1^n$$

从电路的状态图分析





可以自启动

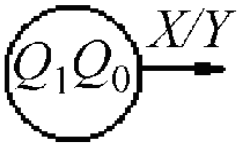
但其功能错误，

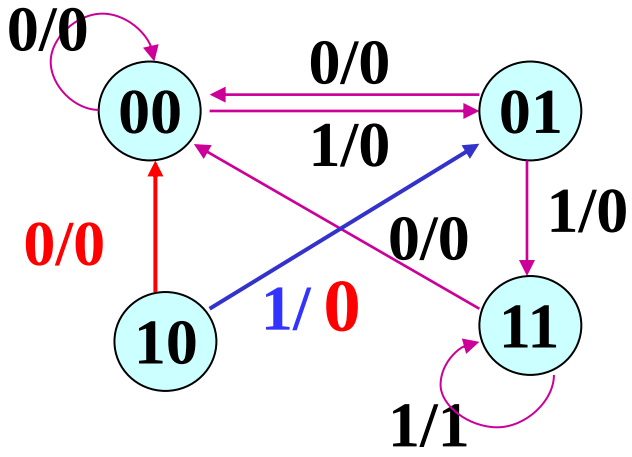
输出应设置为0，才符合题意

检测 连续输入三个及以上个1时，电路输出$Y = 1$。

自启动

让110对应的输出为0.





状态表

X	Q_1^n	Q_0^n	Q_1^{n+1}	Q_0^{n+1}	Y
0	0	0	0	0	0
0	0	1	0	0	0
0	1	0	0	0	0
0	1	1	0	0	0
1	0	0	0	1	0
1	0	1	1	1	0
1	1	0	0	1	0
1	1	1	1	1	1

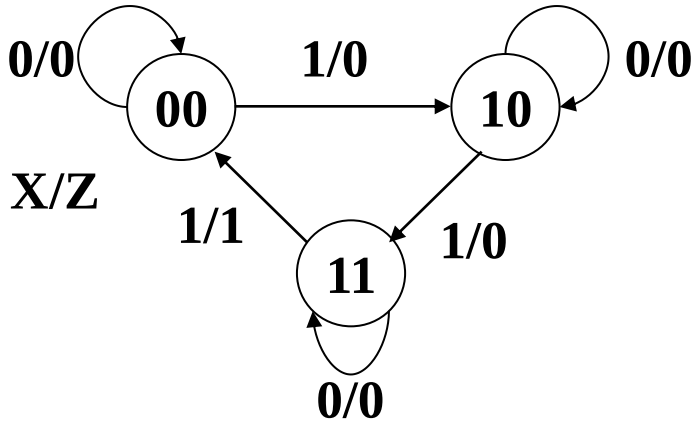
Y / XQ_1^n / Q_0^n	00	01	11	10
0	0	0	0	0
1	0	0	1	0

$Y = XQ_1^nQ_0^n$

既实现自启动，也符合题意.

可以在最初设计时就考虑自启动问题.

例 4. 按照下面状态图设计电路 (例6.6)



状态表 (from state diagram)

X/*Z*

X	Q_2^n	Q_1^n	Q_2^{n+1}	Q_1^{n+1}	Z
0	0	0	0	0	0
0	0	1	ϕ	ϕ	ϕ
0	1	0	1	0	0
0	1	1	1	1	0
1	0	0	1	0	0
1	0	1	ϕ	ϕ	ϕ
1	1	0	1	1	0
1	1	1	0	0	1

确定状态及状态表

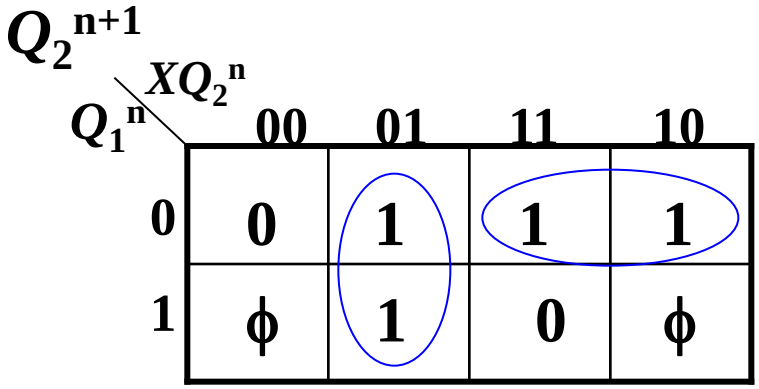
状态数 $\Rightarrow$ (确定) FF 个数

n FFs $\rightarrow 2^n$ 状态

$2^{n-1} \le$ 状态数 $\le 2^n \rightarrow$ *n* FFs

$3 < 2^2$ 需要 2 个 FF

选择 FF (K-map, 圈 1)



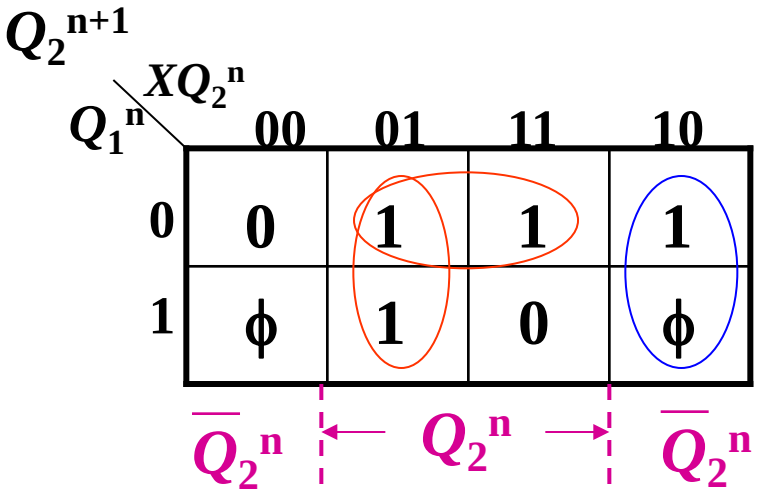
2# FF

$Q_2^{n+1} = X\bar{Q}_1^n + \bar{X}Q_2^n$

2 个圈，选择 JK-FF

$Q_2^{n+1} = J_2\bar{Q}_2^n + \bar{K}_2Q_2^n$　　找到 $J_2 = ?$　$K_2 = ?$

不能按上面方法圈，必须圈成 $Q_2^{n+1} =$ ___ $\bar{Q}_2^n +$ ___ Q_2^n



$$Q_2^{n+1} = X\bar{Q}_2^n + \bar{X}Q_2^n + \bar{Q}_1^nQ_2^n$$

$$= X\bar{Q}_2^n + (\bar{X} + \bar{Q}_1^n)Q_2^n$$

$$= X\bar{Q}_2^n + \overline{XQ_1^n}\,Q_2^n$$

$$\therefore \begin{cases} J_2 = X \\ K_2 = XQ_1^n \end{cases}$$

能找到系数（控制变量）时尽量化简；找不到系数时，牺牲化简也要找到系数。

X	Q_2^n	Q_1^n	Q_2^{n+1}	Q_1^{n+1}	Z
0	0	0	0	0	0
0	0	1	ϕ	ϕ	ϕ
0	1	0	1	0	0
0	1	1	1	1	0
1	0	0	1	0	0
1	0	1	ϕ	ϕ	ϕ
1	1	0	1	1	0
1	1	1	0	0	1

1# FF

Q_1^{n+1}

Q_1^n \ XQ_2^n	00	01	11	10
0	0	0	1	0
1	ϕ	1	0	ϕ

$\rightarrow \overline{Q}_1^n$

$\rightarrow Q_1^n$

JK-FF

$$Q_1^{n+1} = J_1\overline{Q}_1^n + \overline{K}_1 Q_1^n$$

$$= XQ_2^n\overline{Q}_1^n + \overline{X}Q_1^n$$

$$\therefore \begin{cases} J_1 = XQ_2^n \\ K_1 = X \end{cases}$$

输出 Z

Z

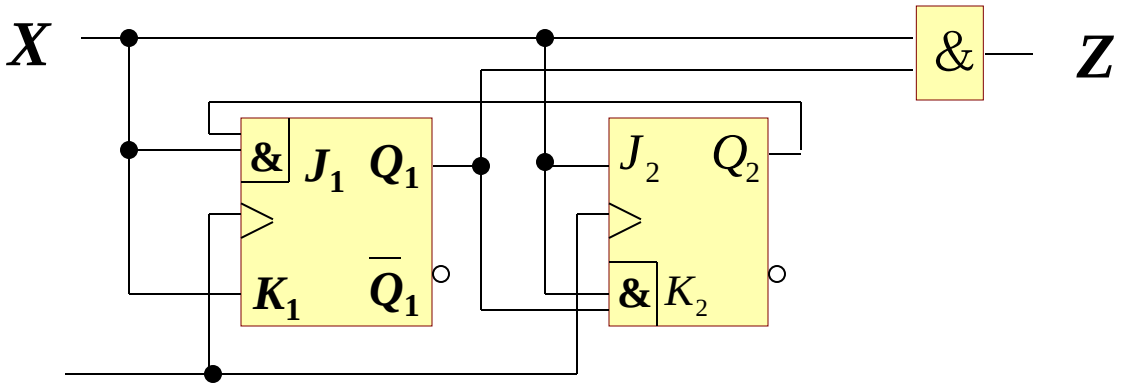
Q_1^n \ XQ_2^n	00	01	11	10
0	0	0	0	0
1	Φ	0	1	Φ

$$Z = XQ_1^n$$

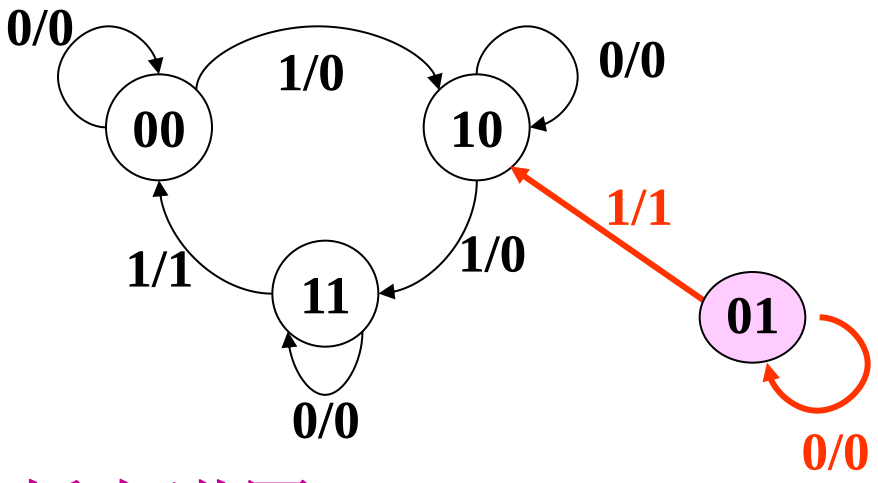
电路

$$\begin{cases} J_2 = X \\ K_2 = XQ_1^n \end{cases}$$

$$\begin{cases} J_1 = XQ_2^n \\ K_1 = X \end{cases}$$



讨论：01 状态



Q_1^n \ XQ_2^n	00	01	11	10
0	0	1	1	1
1	0	1	0	1

Q_2^{n+1}

Q_1^n \ XQ_2^n	00	01	11	10
0	0	0	1	0
1	1	1	0	0

Q_1^{n+1}

Q_1^n \ XQ_2^n	00	01	11	10
0	0	0	0	0
1	0	0	1	1

Z

分析卡诺图 K-map

$XQ_2^nQ_1^n$ = 001, (Z=0)

Next state $Q_2^{n+1}Q_1^{n+1}$ =01,

$XQ_2^nQ_1^n$ = 101时, (Z=1)

Next state $Q_2^{n+1}Q_1^{n+1}$ =10,

实现自启动

例5. 设计一个自动售饮料机的逻辑电路。它的投币口每次只能投入一枚五角或一元的硬币。投入一元五角钱硬币后，机器会自动给出一杯饮料；投入两元（两个一元）硬币后，在给出饮料的同时找回一枚五角硬币。

解：

输入：{ 投入 ¥1.0, A=1；投入 ¥0.5, B=1

输出：{ 出饮料, Y=1；找钱, Z=1

状态：

3 个状态：

S_0 : 初始 (未投币)

S_1 : 投入 ¥0.5,

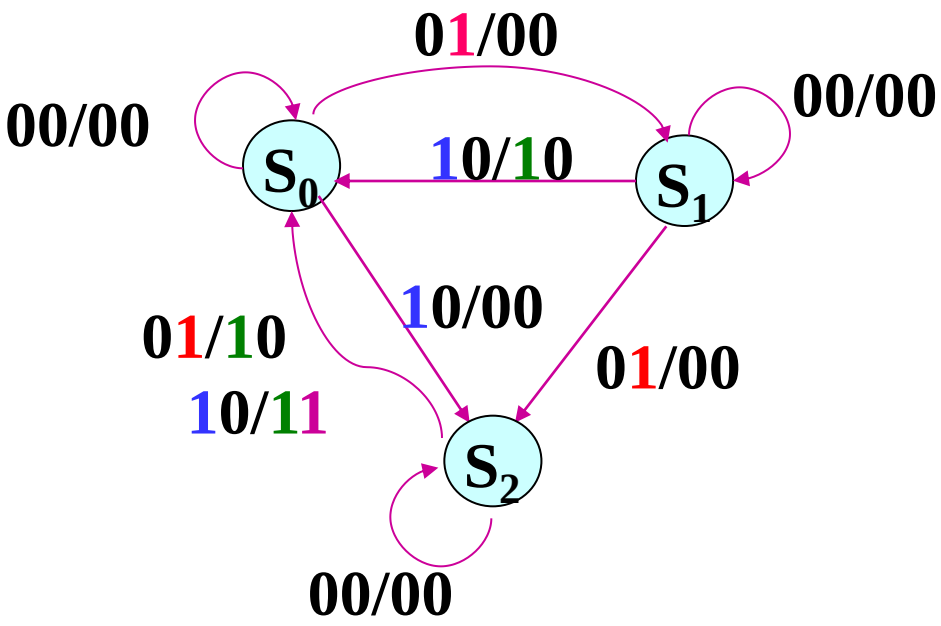
S_2 :投入 ¥1.0 (一个 ¥1.0 或两个 ¥0.5)

再投入 ¥0.5, 返回 S_0, 输出 Y=1, Z=0;

再投入¥1.0, 返回 S_0, 输出 Y=1, Z=1 (找回 ¥0.5).

多输入，多输出

AB /YZ



S_0: 投入 ¥0.5, S_1

S_0:投入¥1.0, S_2

S_1:投入¥0.5, S_2

S_1:投入¥1.0, S_0
饮料

S_2:投入¥0.5, S_0
饮料

S_2:投入¥1.0, S_0
饮料 和 找钱

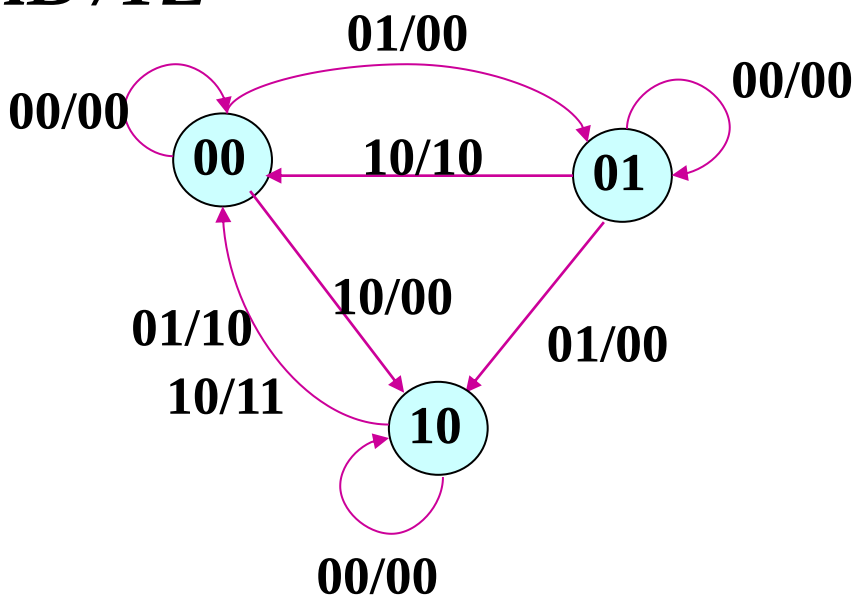
A：¥1.0
B：¥0.5
Y：饮料
Z：找钱

S_0：初始
S_1：¥0.5
S_2：¥1.0

$Q_1^nQ_0^n$

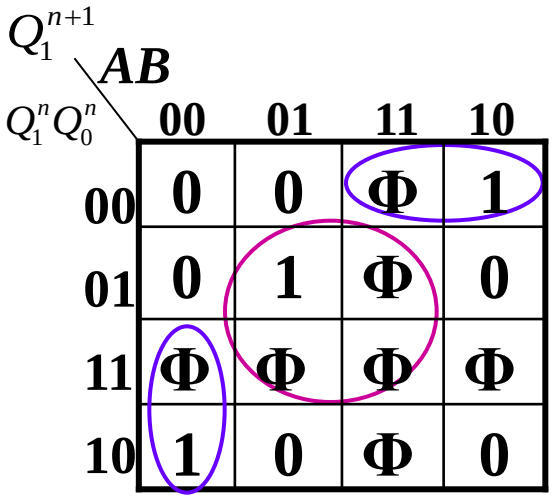
编码 $\begin{cases} S_0 \rightarrow 00 \\ S_1 \rightarrow 01 \\ S_2 \rightarrow 10 \end{cases}$

AB /YZ

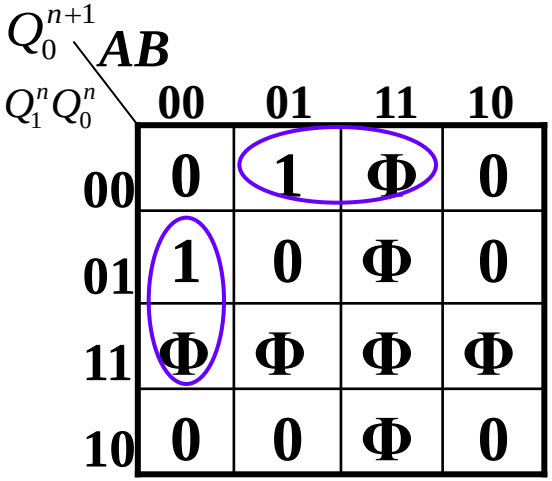


状态表:

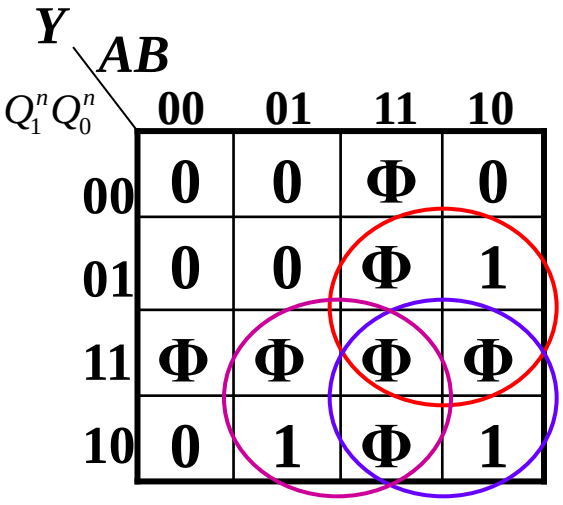
$A\ B\ Q_1^nQ_0^n$	$Q_1^{n+1}Q_0^{n+1}\ Y\ Z$
0 0 0 0	0 0 0 0
0 0 0 1	0 1 0 0
0 0 1 0	1 0 0 0
0 0 1 1	Φ Φ Φ Φ
0 1 0 0	0 1 0 0
0 1 0 1	1 0 0 0
0 1 1 0	0 0 1 0
0 1 1 1	Φ Φ Φ Φ
1 0 0 0	1 0 0 0
1 0 0 1	0 0 1 0
1 0 1 0	0 0 1 1
1 0 1 1	Φ Φ Φ Φ
1 1 0 0	Φ Φ Φ Φ
1 1 0 1	Φ Φ Φ Φ
1 1 1 0	Φ Φ Φ Φ
1 1 1 1	Φ Φ Φ Φ



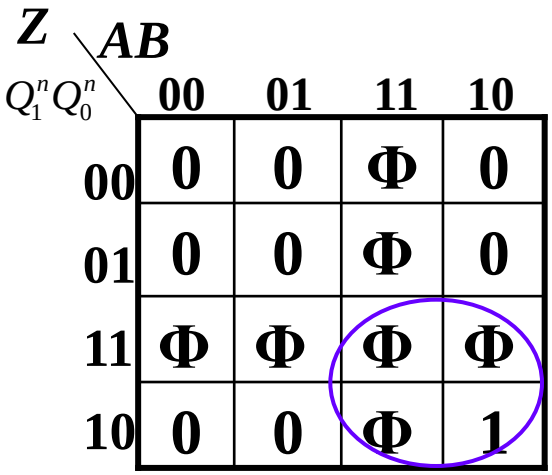
$$Q_1^{n+1} = BQ_0 + A\overline{Q}_1\overline{Q}_0 + \overline{A}\cdot\overline{B}Q_1$$



$$Q_0^{n+1} = B\overline{Q}_1\cdot\overline{Q}_0 + \overline{A}\cdot\overline{B}Q_0$$



$$Y = AQ_0 + AQ_1 + BQ_1$$



$$Z = AQ_1$$

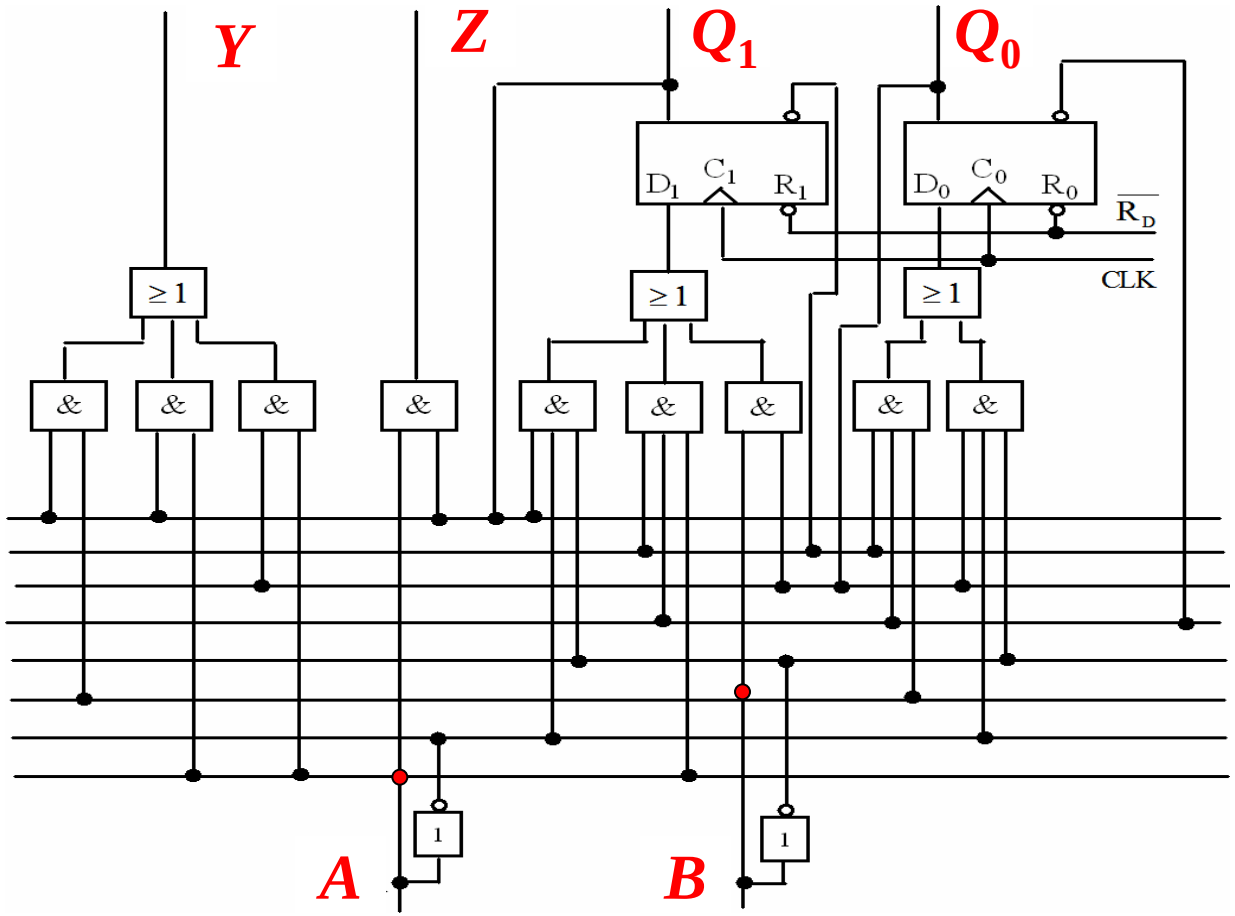
用 D-FF

$$D_1 = BQ_0 + A\overline{Q}_1\overline{Q}_0 + \overline{A}\cdot\overline{B}Q_1$$

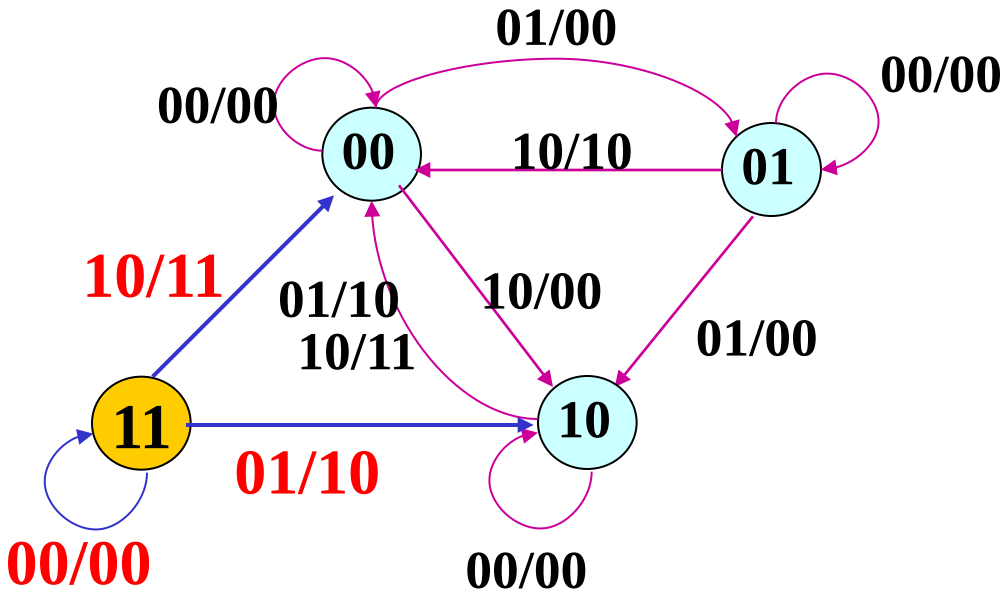
$$D_0 = B\overline{Q}_1\cdot\overline{Q}_0 + \overline{A}\cdot\overline{B}Q_0$$

$$Y = AQ_0 + AQ_1 + BQ_1$$

$$Z = AQ_1$$



由电路得到状态图



A：¥1.0
B：¥0.5
Y：饮料
Z：找钱

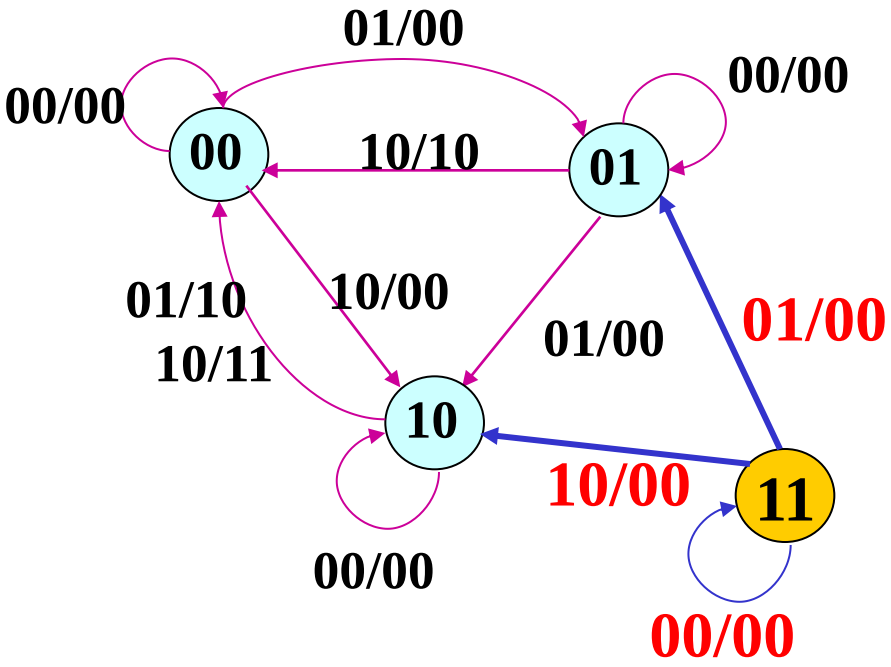
在电路处于“11”状态时，

若 AB = 00 (无输入)，Q_1Q_0 = 11, 电路不能自启动;

若AB = 01 或 10, 电路可以自启动，但是输出错误;

所以，电路初始工作时，应首先将 $\overline{R}_D$ 设置为低电平，电路状态从“00”开始.

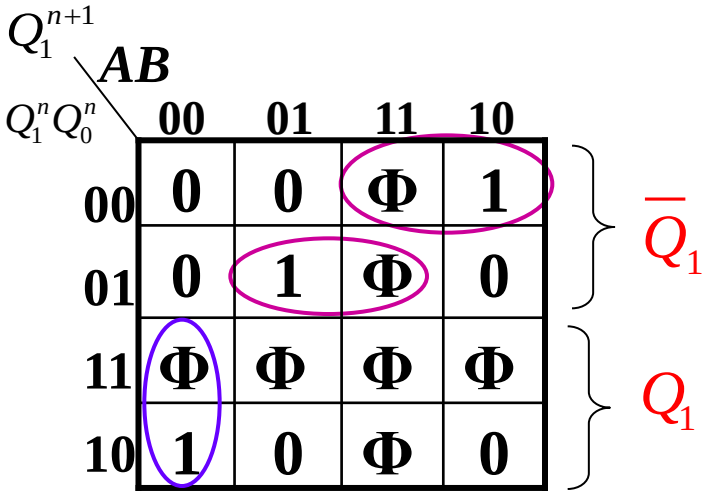
电路状态图应设计成：

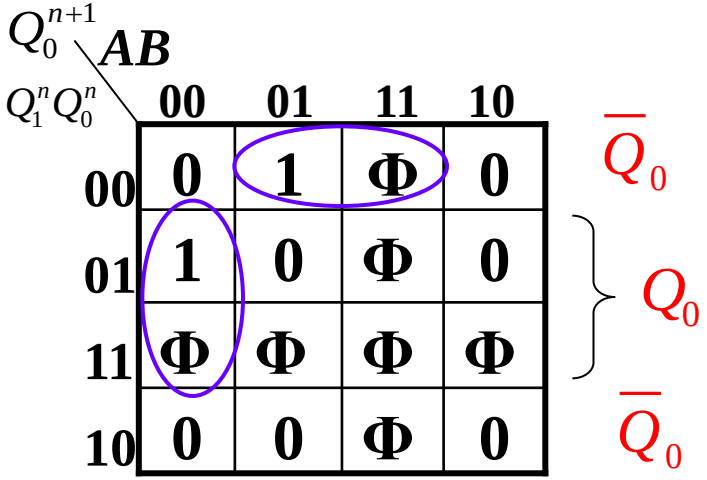


状态表：

A	B	Q_1^n	Q_0^n	Q_1^{n+1}	Q_0^{n+1}	Y	Z
0	0	0	0	0	0	0	0
0	0	0	1	0	1	0	0
0	0	1	0	1	0	0	0
0	0	1	1	1	1	0	0
0	1	0	0	0	1	0	0
0	1	0	1	1	0	0	0
0	1	1	0	0	0	1	0
0	1	1	1	0	1	0	0
1	0	0	0	1	0	0	0
1	0	0	1	0	0	1	0
1	0	1	0	0	0	1	1
1	0	1	1	1	0	0	0
1	1	0	0	Φ	Φ	Φ	Φ
1	1	0	1	Φ	Φ	Φ	Φ
1	1	1	0	Φ	Φ	Φ	Φ
1	1	1	1	Φ	Φ	Φ	Φ

思考：若用JK触发器，应如何圈？





$$Q_1^{n+1} = A\overline{Q}_1\overline{Q}_0 + B\overline{Q}_1Q_0 + \overline{A}\cdot\overline{B}Q_1$$

$$= (A\overline{Q}_0 + BQ_0)\overline{Q}_1 + \overline{A+B}\cdot Q_1$$

$$\begin{cases} J_1 = A\overline{Q}_0 + BQ_0 \\ K_1 = A+B \end{cases}$$

$$Q_0^{n+1} = B\overline{Q}_1\cdot\overline{Q}_0 + \overline{A}\cdot\overline{B}Q_0$$

$$= B\overline{Q}_1\cdot\overline{Q}_0 + \overline{A+B}\cdot Q_0$$

$$\begin{cases} J_0 = B\overline{Q}_1 \\ K_0 = A+B \end{cases}$$

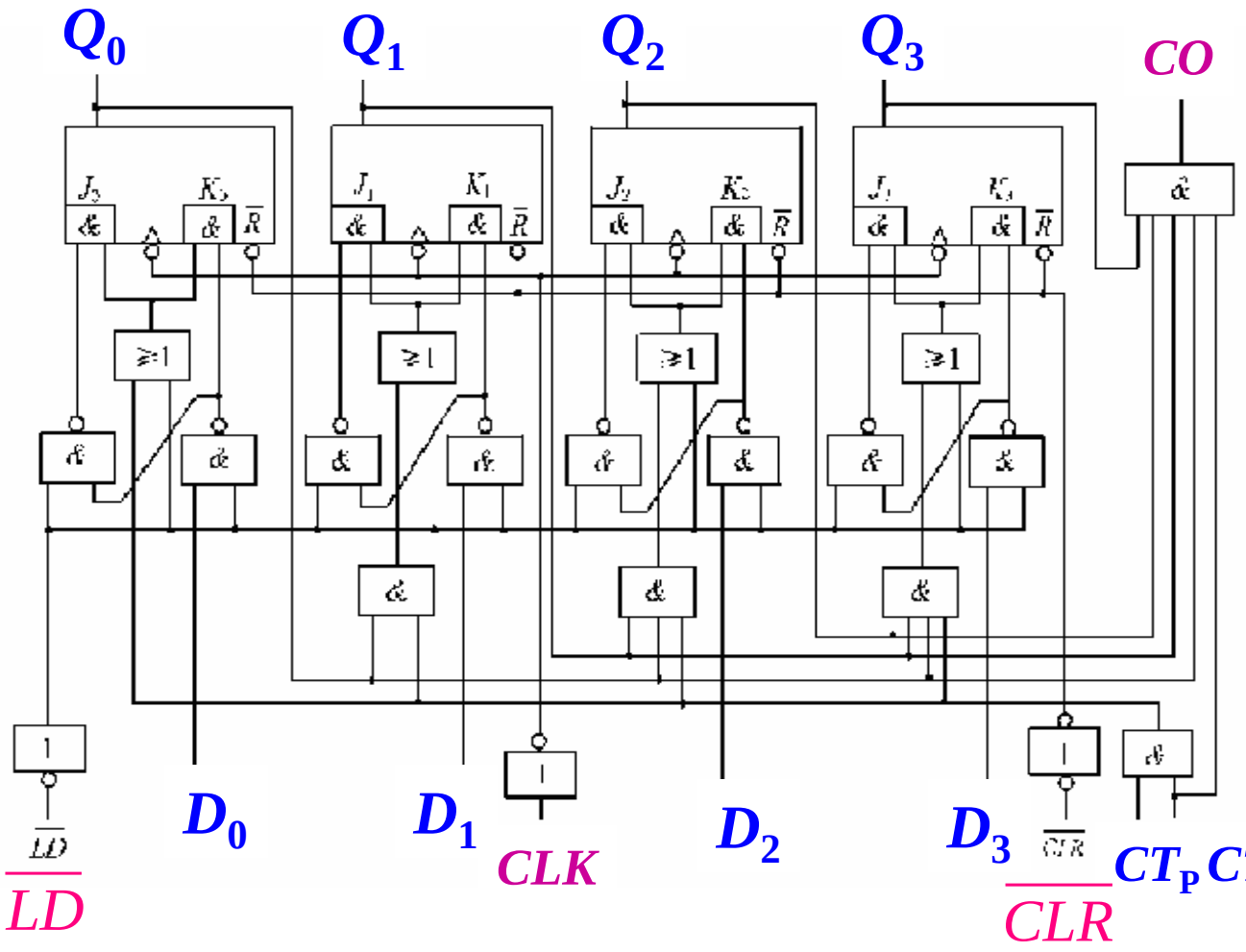
§6.4 集成计数器
Integrated Counter

计数器: 记录*CLK*个数的电路

6.4.1 集成计数器 74161

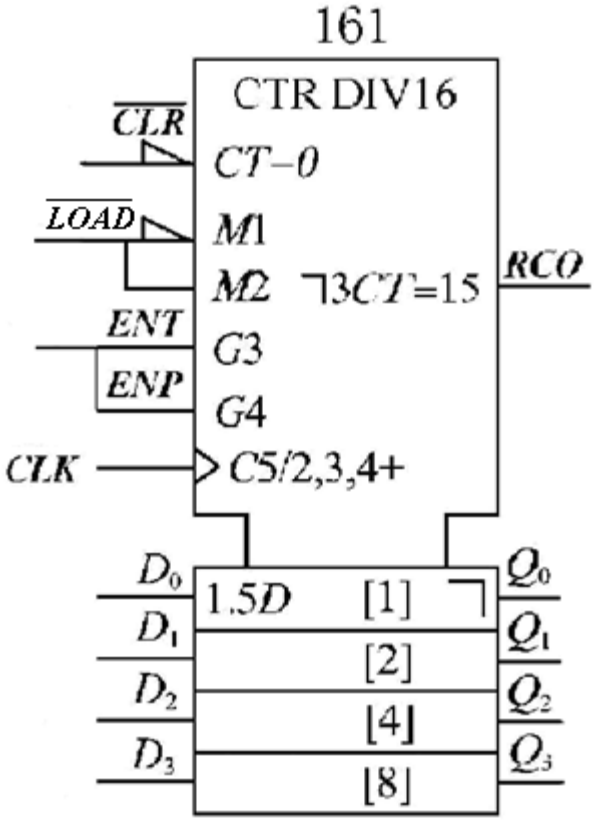
74161: 二进制同步模16加法计数器，
异步清0功能.

计数器74161 电路

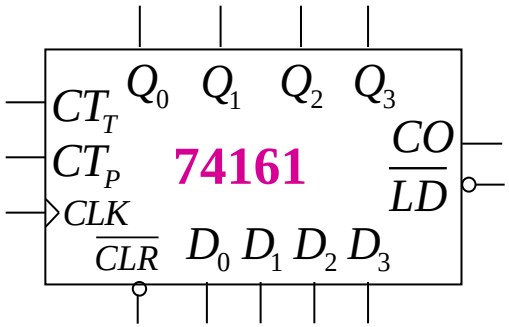

Q_0
Q_1
Q_2
Q_3
CO
$Q_3Q_2Q_1Q_0$
$D_3D_2D_1D_0$
$\overline{LD}$
D_0
D_1
CLK
D_2
D_3
$\overline{CLR}$
CT_P CT_T
$\overline{LD}$
$\overline{CLR}$

符号

IEEE



MSB



输出 $Q_3\,Q_2\,Q_1\,Q_0$

数据输入 $D_3\,D_2\,D_1\,D_0$

异步清零 $\overline{CLR}$

控制端 $ENT\,(CT_T),\ ENP\,(CT_P)$

预置端 $\overline{LOAD}\ \left(\overline{LD}\right)$

进位输出 $RCO\ (CO)$

时钟信号 CLK

74161 功能表

$\overline{CLR}$	$\overline{LD}$	ENT	ENP	CLK	$D_0\,D_1\,D_2\,D_3$	功能
0	X	X	X	X	X X X X	直接置 0
1	0	X	X	↑	$D_0D_1D_2D_3$	Load 预置
1	1	0	X	X	X X X X	保持 RCO=0
1	1	X	0	X	X X X X	保持
1	1	1	1	↑	X X X X	M-16 计数

$Q_3Q_2Q_1Q_0 = 0000$

$Q_3Q_2Q_1Q_0 = D_3D_2D_1D_0$

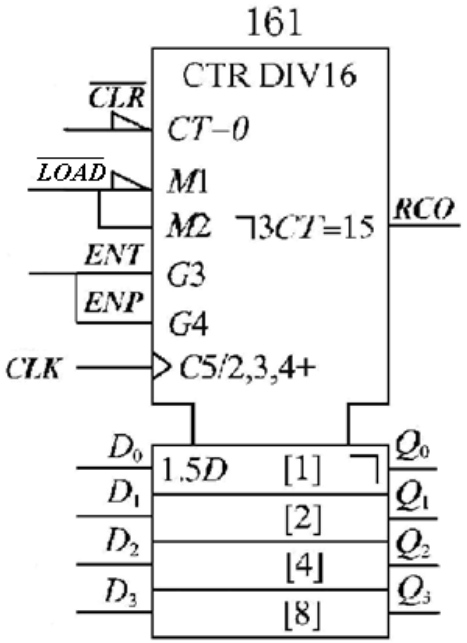
$RCO = ENT \cdot Q_3 \cdot Q_2 \cdot Q_1 \cdot Q_0$

计数时，$ENT = 1$,

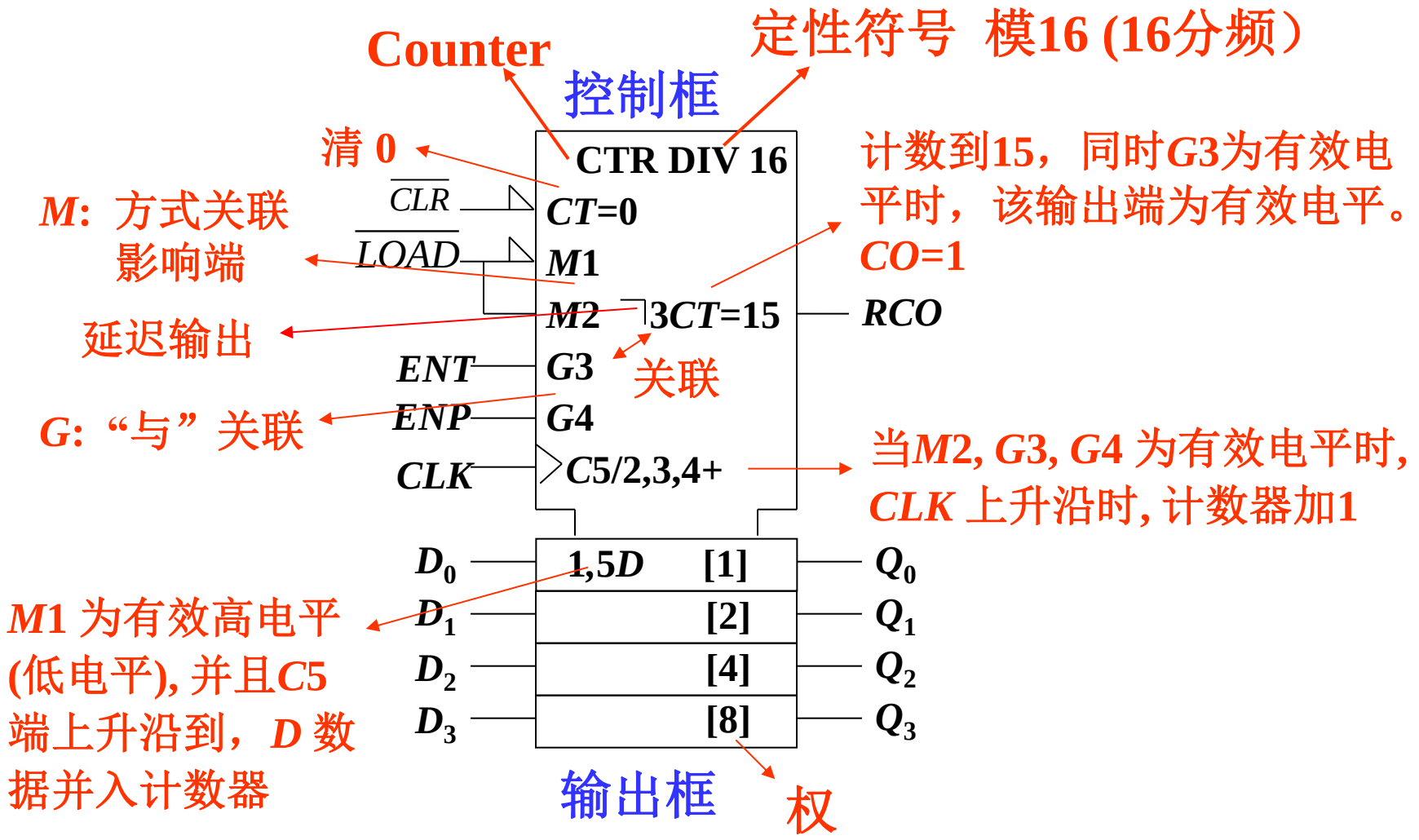
当$Q_3Q_2Q_1Q_0 = 1111$时，（M-16）

$RCO = 1$.

其他时刻，$RCO = 0$



74161 IEEE 符号



例1: 用 74161 实现模11加法计数器　(例6.7)

方法1: 预置归 0 法 （$\overline{LD}$）

$ENT = ENP = 1$, $\overline{CLR} = 1$, $D_3D_2D_1D_0 = 0000$

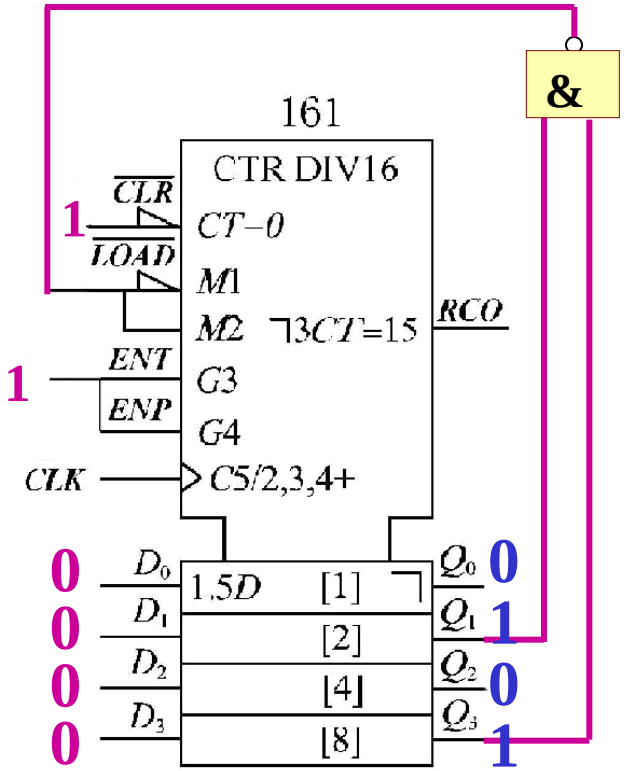
最大状态 1010　最大状态中1端连入一个与非门

输出 $\rightarrow \overline{LD}$

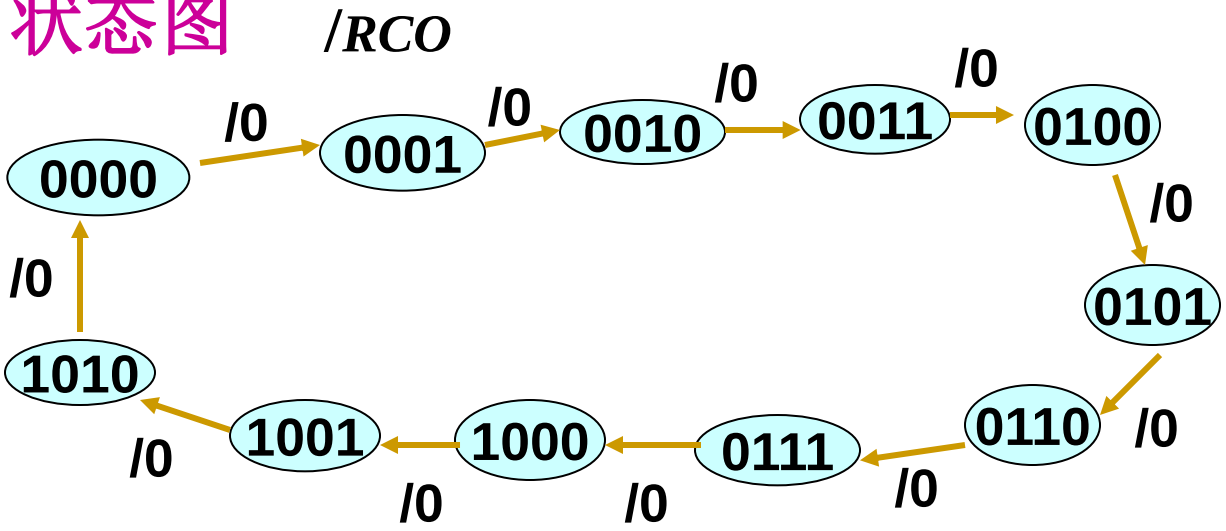
0 →9，与非门 = 1
($\overline{LD}$=1), 计数

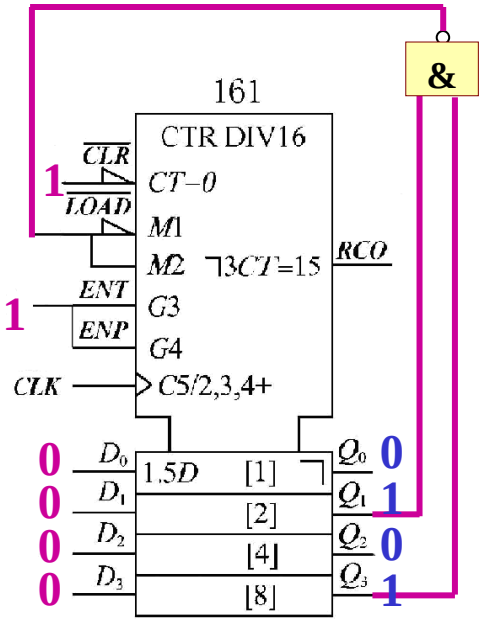
10th CLK 到来,
$Q_3Q_2Q_1Q_0$=1010, $\overline{LD}$=0

下一个 CLK (11th) 到来,
$Q_3Q_2Q_1Q_0 = D_3D_2D_1D_0 = 0000$

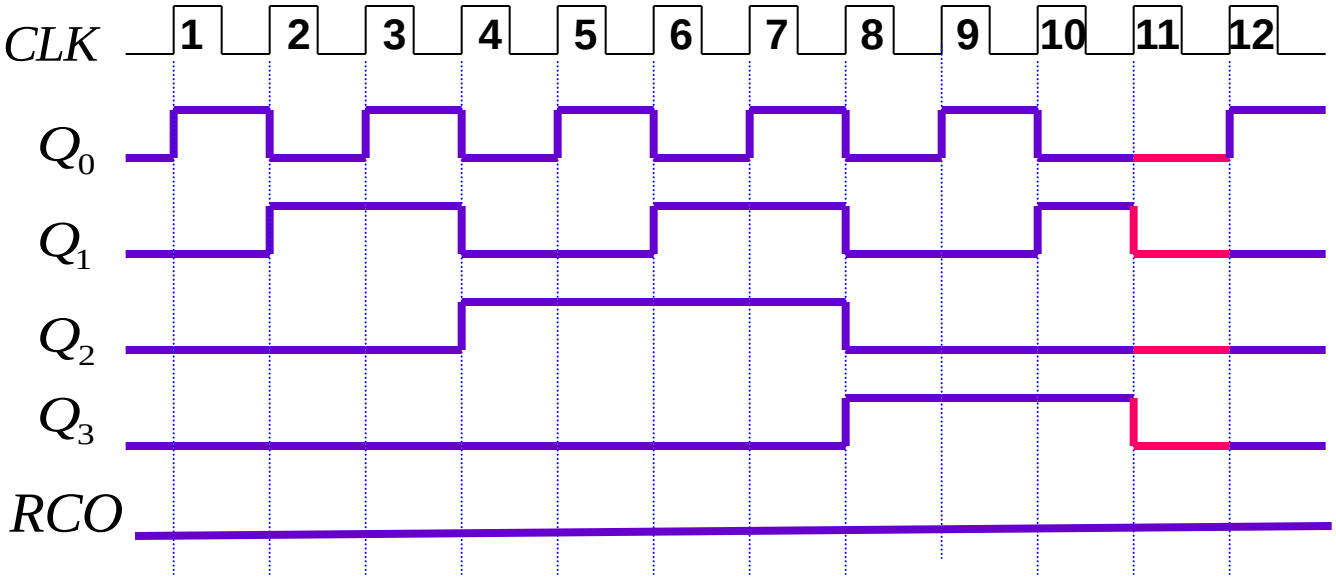


状态图





波形图



Q_0 二分频电路

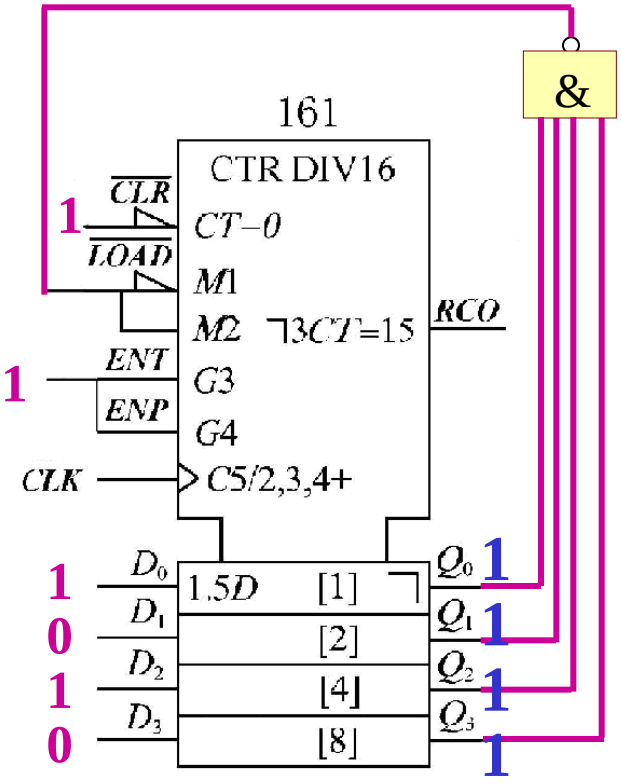
四分频电路

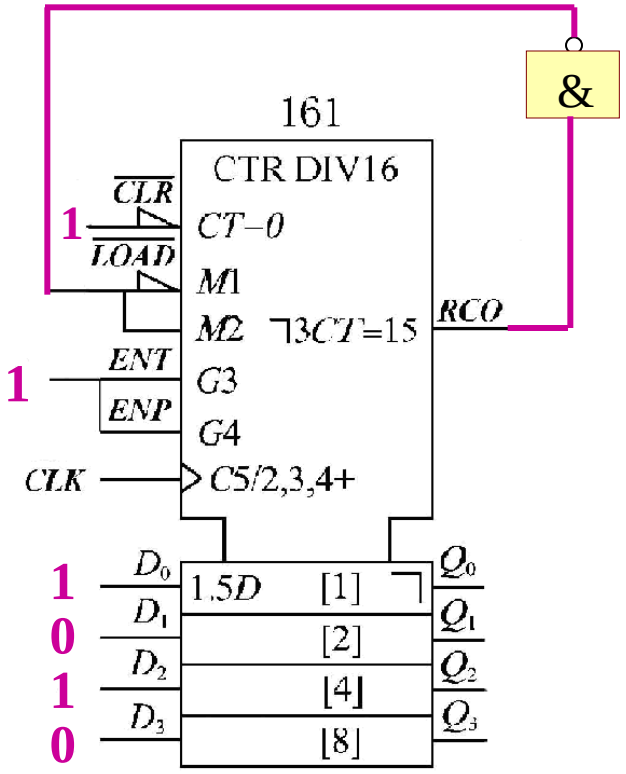
方法 2: 预置补数法 M-11

0000 ~ 1111
16 个状态

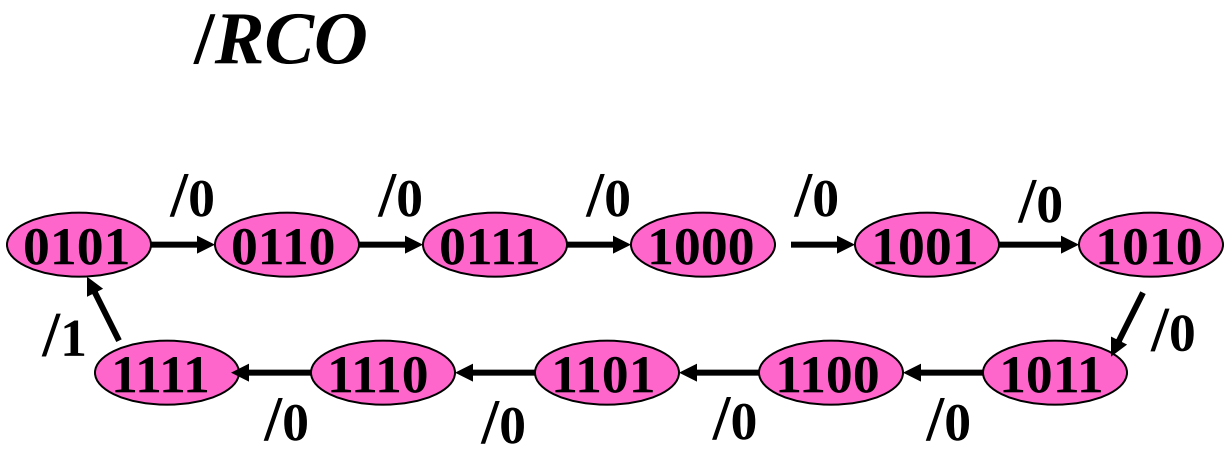
0000 ~ 1010
11 个状态

5 (0101) ~ 15 (1111)
11 个状态





状态图


/RCO
/0
/0
/0
/0
/0
0101
0110
0111
1000
1001
1010
/1
/0
1111
1110
1101
1100
1011
/0
/0
/0
/0

练习接 3 ~ 13

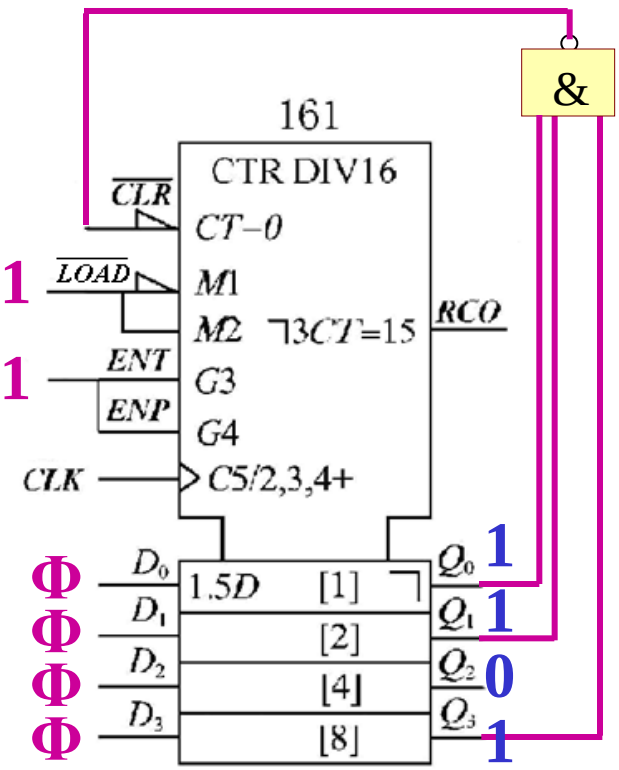
方法 3: 反馈归 0 法M-11 （$\overline{CLR}$ ）

$ENT = ENP = \overline{LD} = 1$

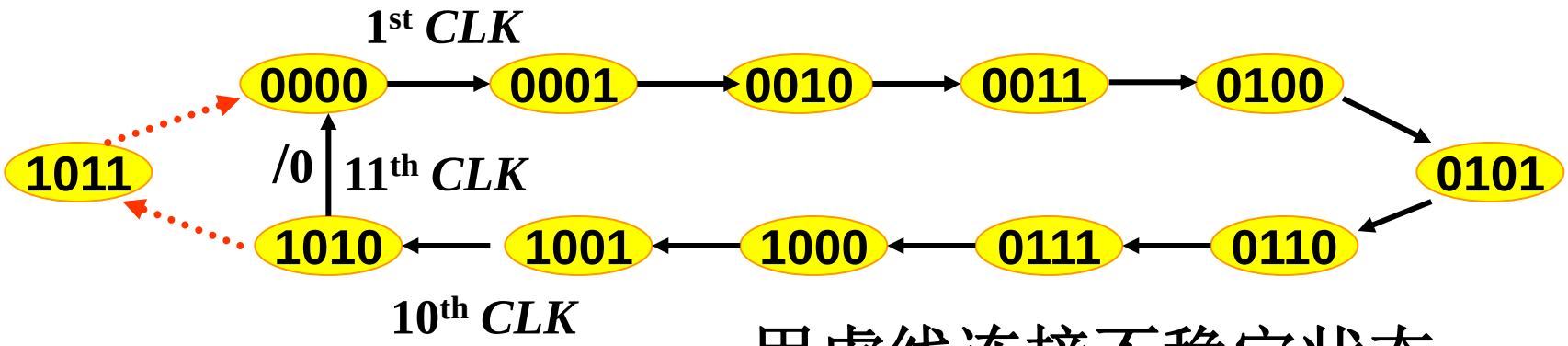
$D_3D_2D_1D_0 = \Phi\ \Phi\ \Phi\ \Phi$

$Q_3Q_2Q_1Q_0 = 1011$

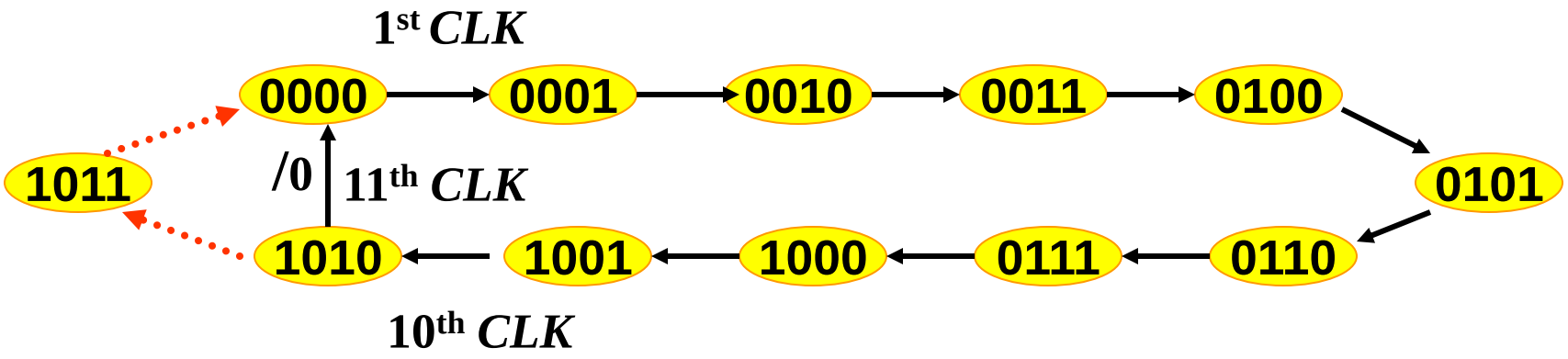
与非门 → $\overline{CLR}$



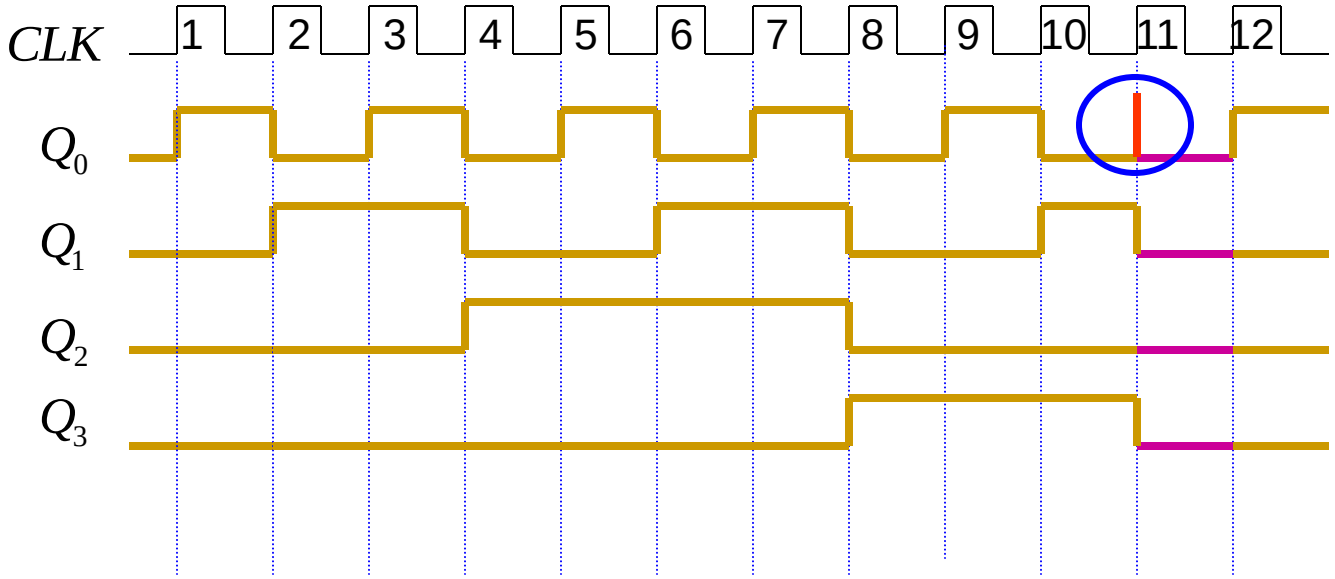
状态图 /RCO



用虚线连接不稳定状态



波形图



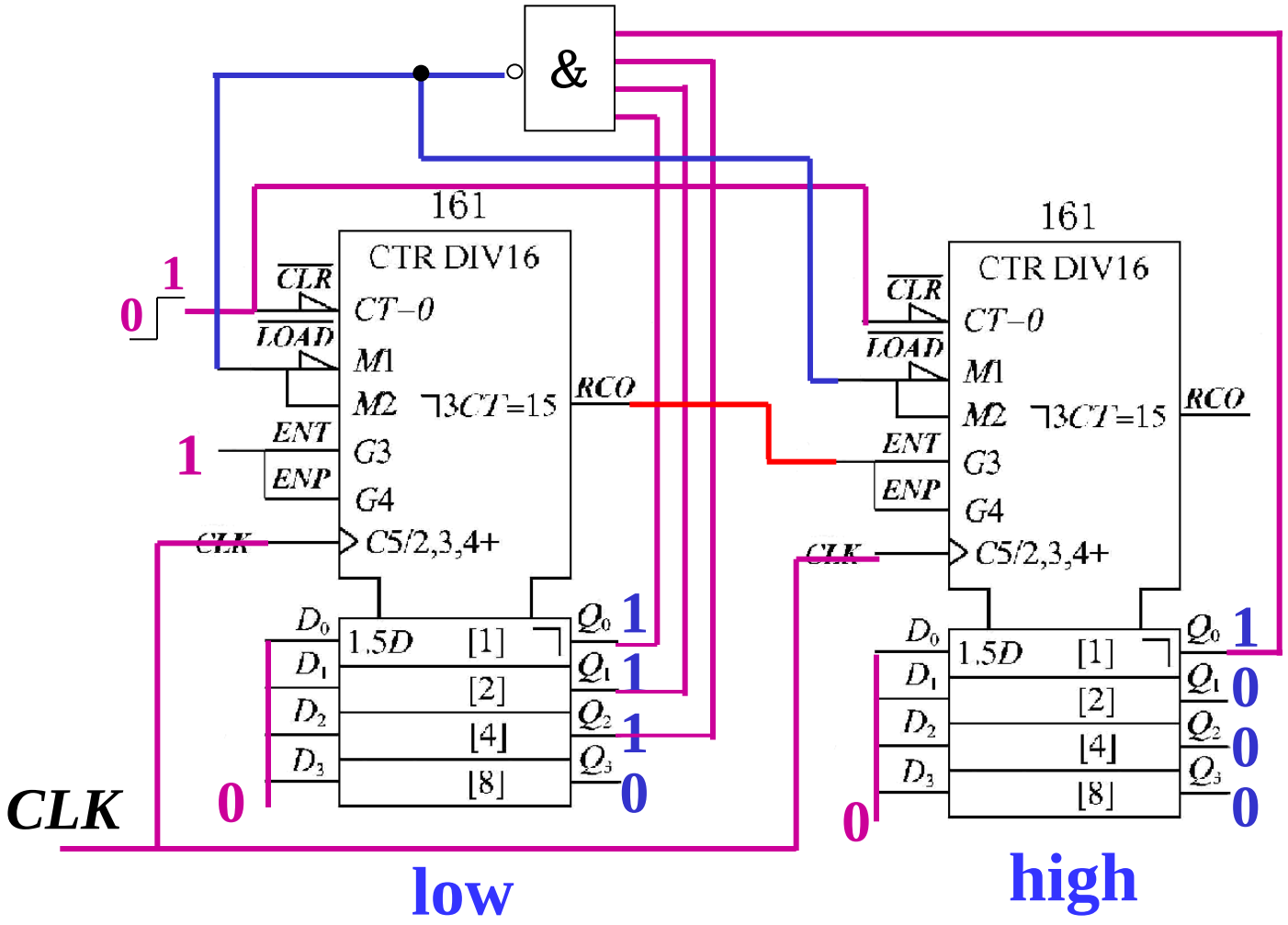
毛刺
（假信号）

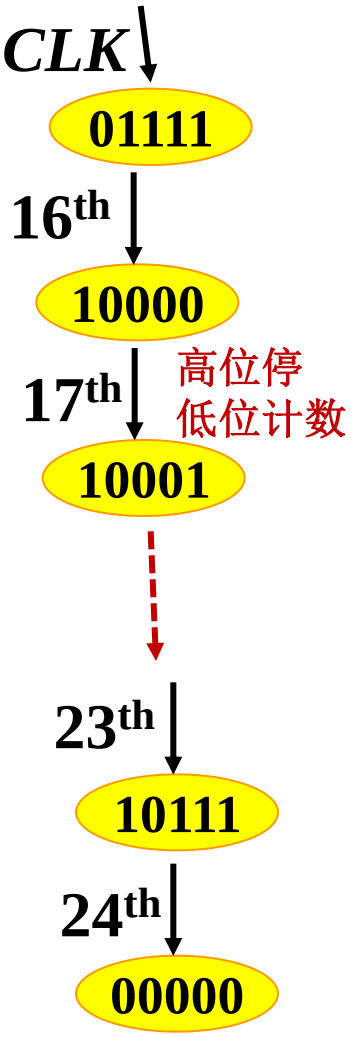
方法1最好, 用$\overline{LD}$端归0。

例 2：用74161 设计模 24 计数器

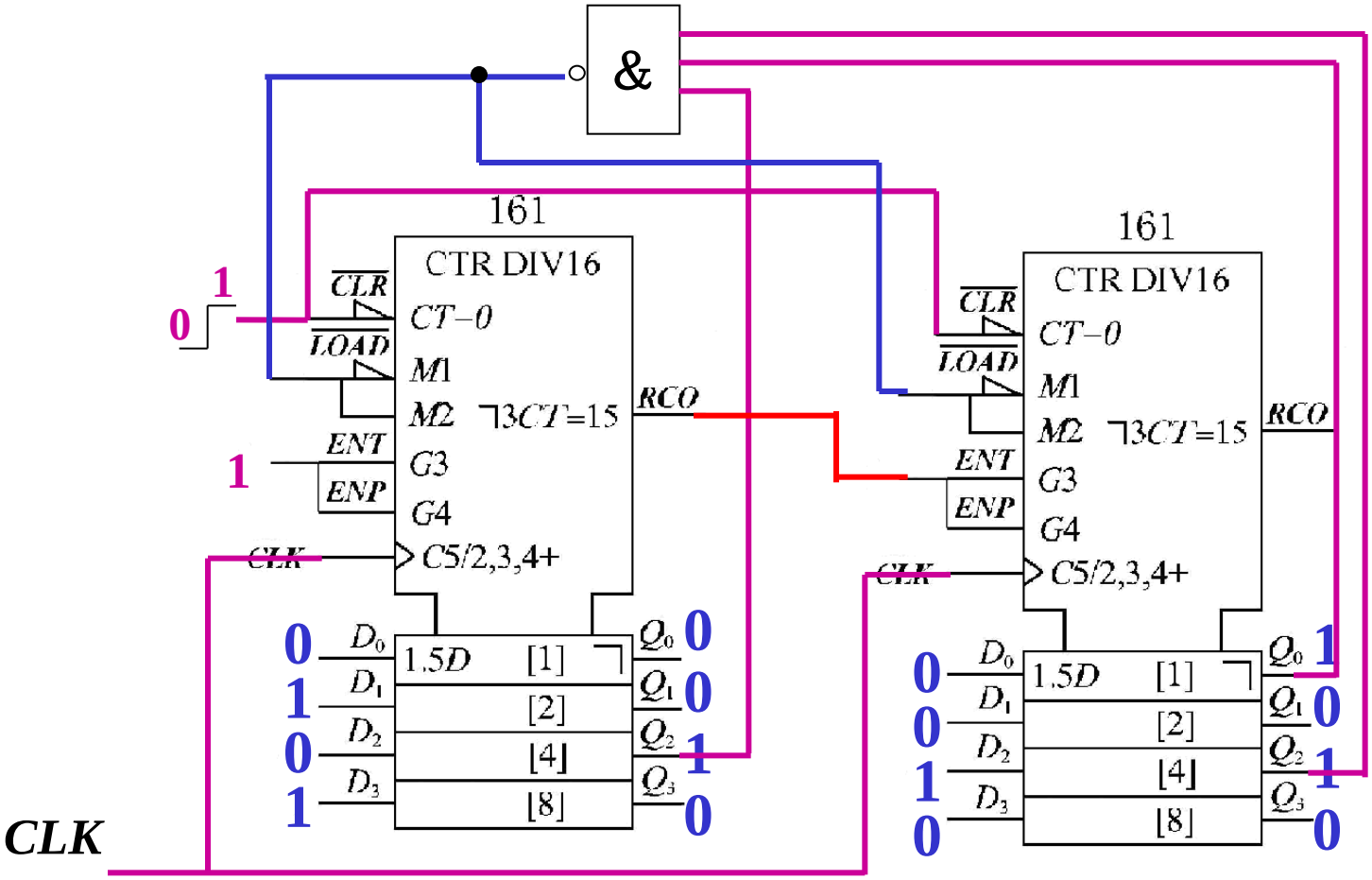
状态图

最大状态: 23 (10111)　　两个 74161





例 3：求下图计数器电路的模值.　M = ?



终点：　01010100 ＝ 84

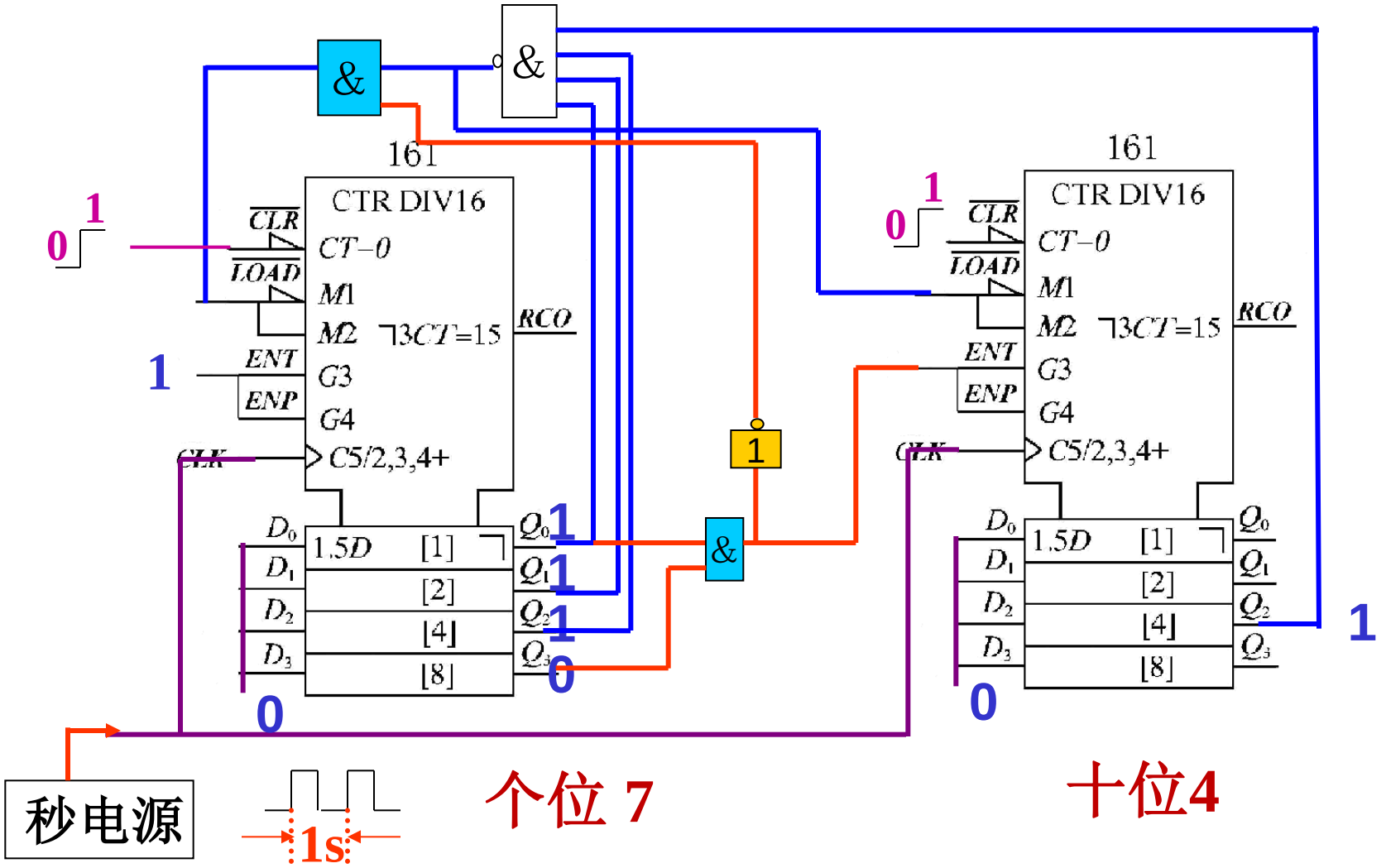
补：　01001010 ＝ 74

M = 84 – 74　+ 1 = 11

例 4：用74161设计十进制 48 秒计数器

计数：二进制
进位：十进制

最大状态 47



6.4.2 IC 计数器 74160 (M-10)

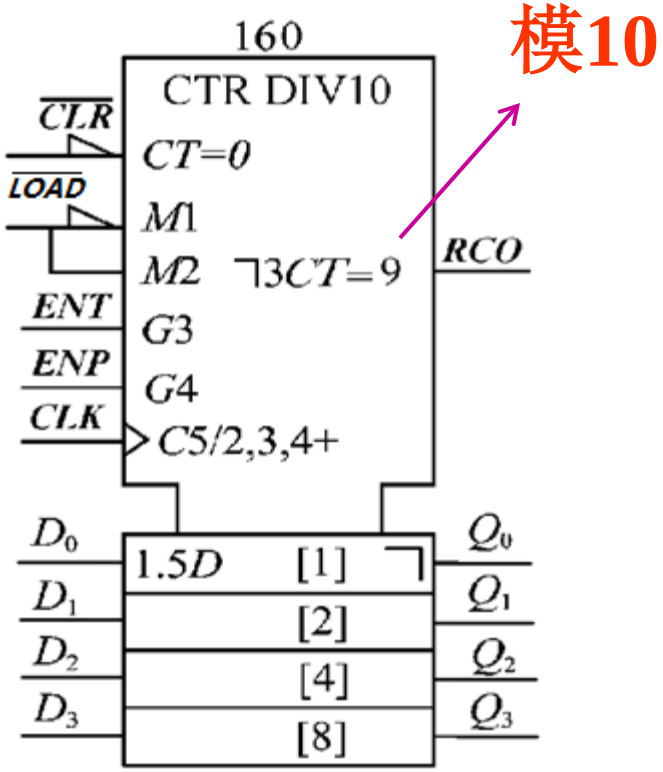
(8421BCD码同步加法计数器)

模10, 其它与74161相同. 异步清0

$$RCO = ENT \cdot Q_3 \cdot Q_0$$

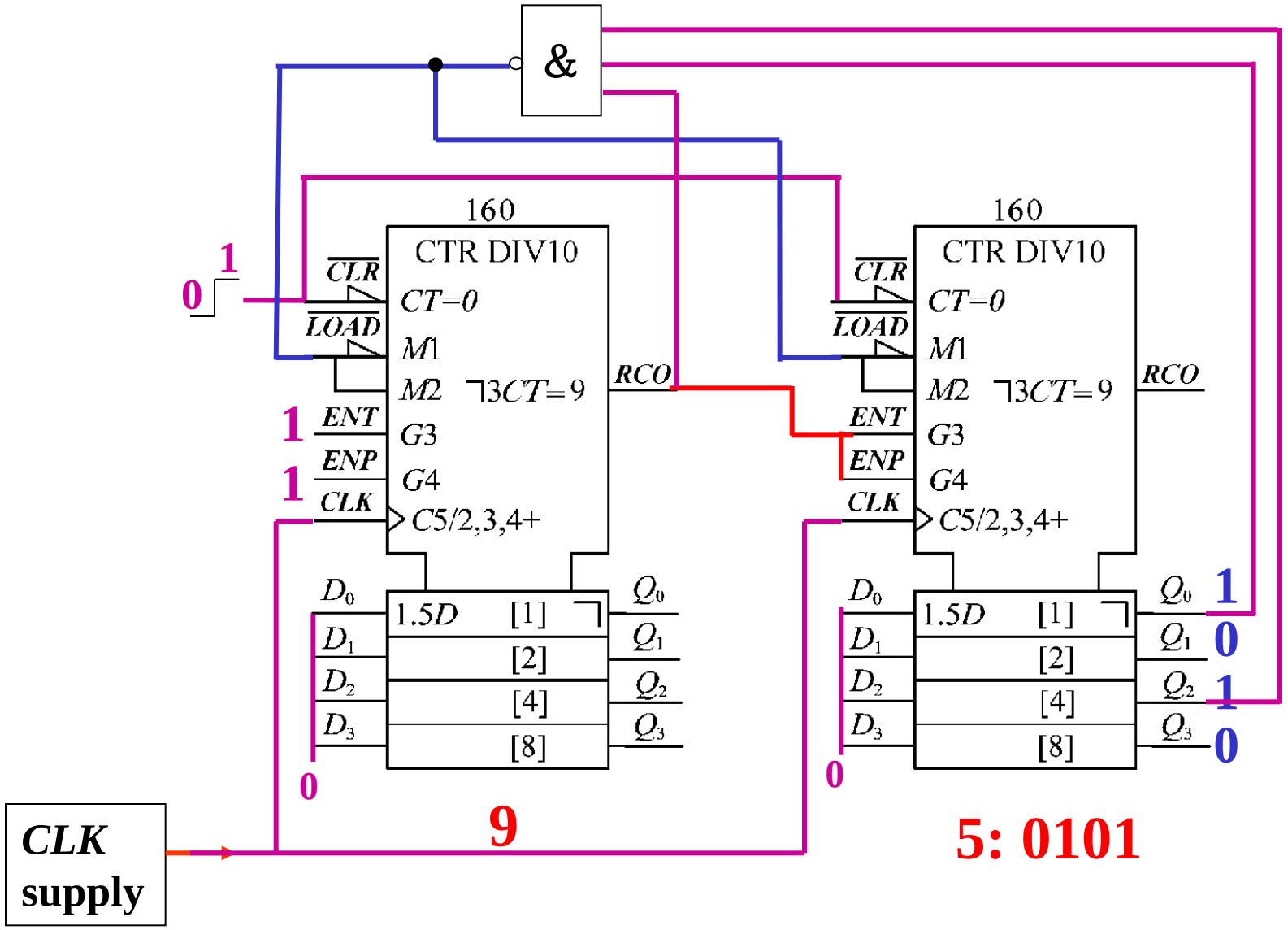
当 $Q_3Q_2Q_1Q_0 = 1001$,

$RCO = 1$

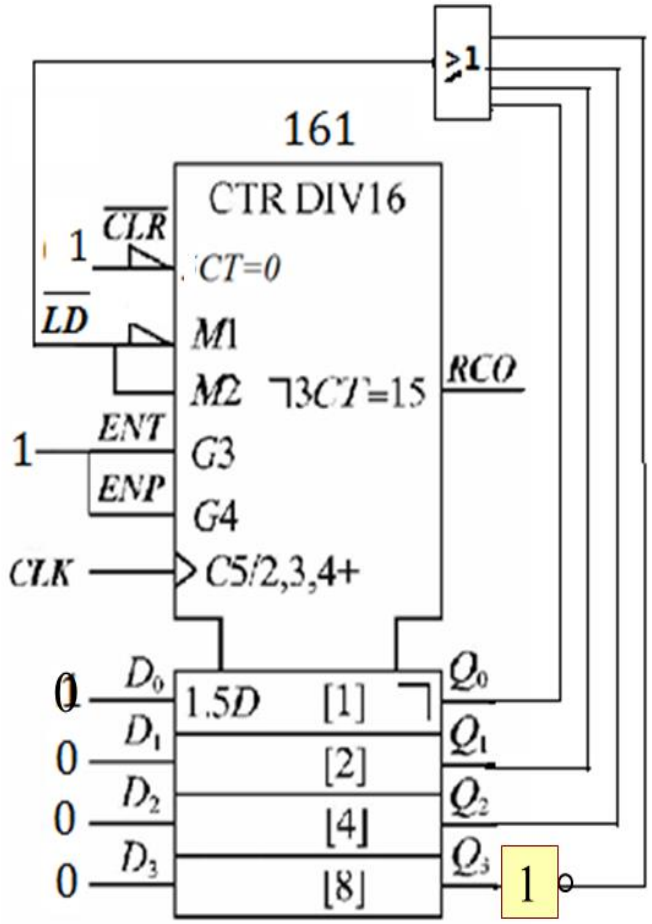


例: 用74160 设计一个 60 s 计数器.

59

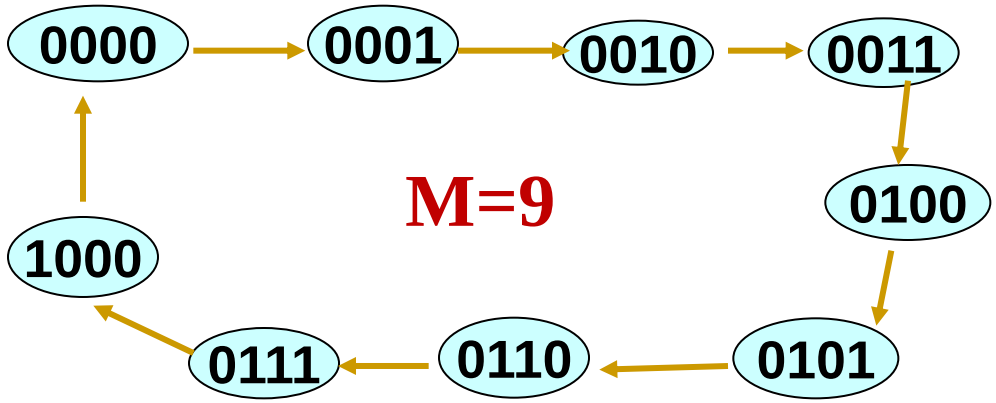


习题：分析下图电路的计数模值 M=？画出状态图.
初始状态$Q_3Q_2Q_1Q_0$=0000



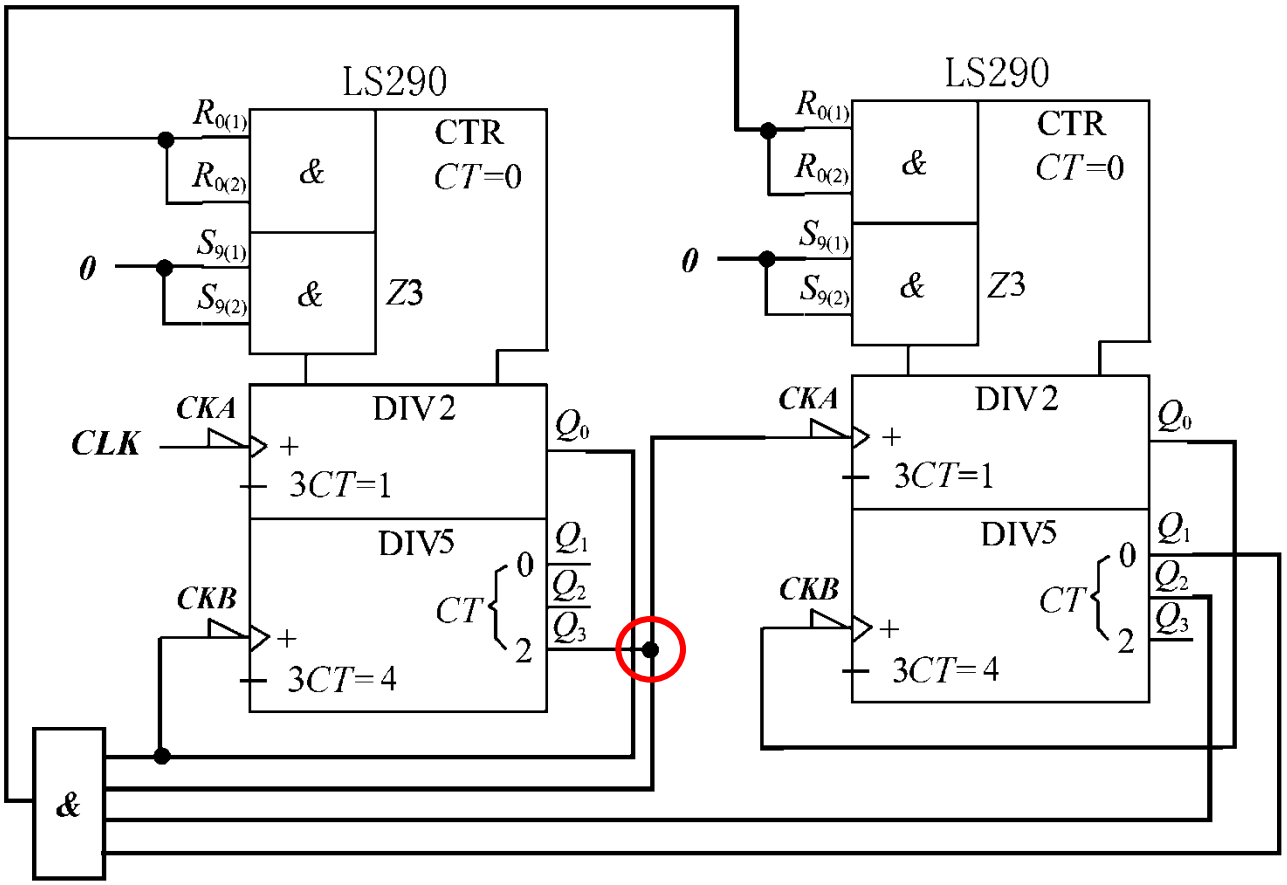
$Q_3Q_2Q_1Q_0$

$$\overline{LD} = \overline{Q_3} + Q_2 + Q_1 + Q_0$$

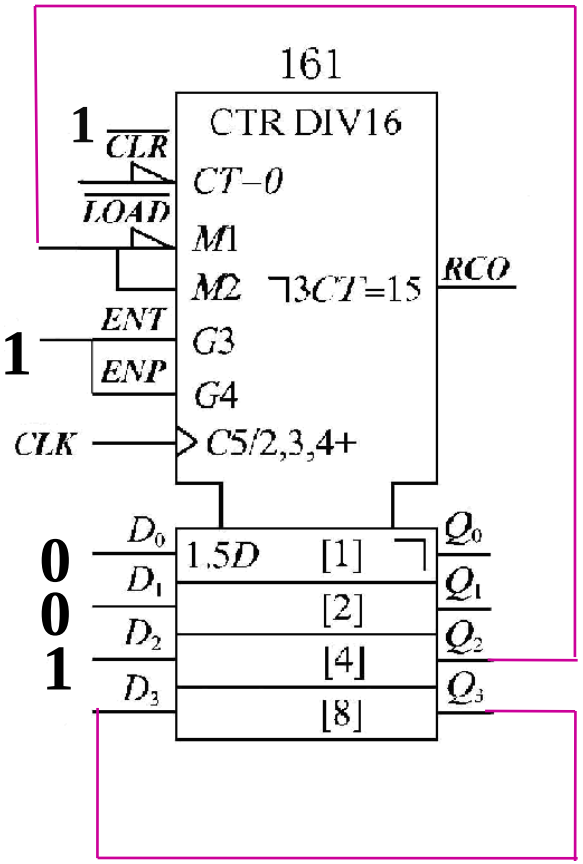


6.19 请指出74290如题图6.19所示电路图的模值为多少？

（改错：P.153）



例：74161电路连接如图，分析电路计数长度M=？画出状态图。初态$Q_3Q_2Q_1Q_0$=0000.

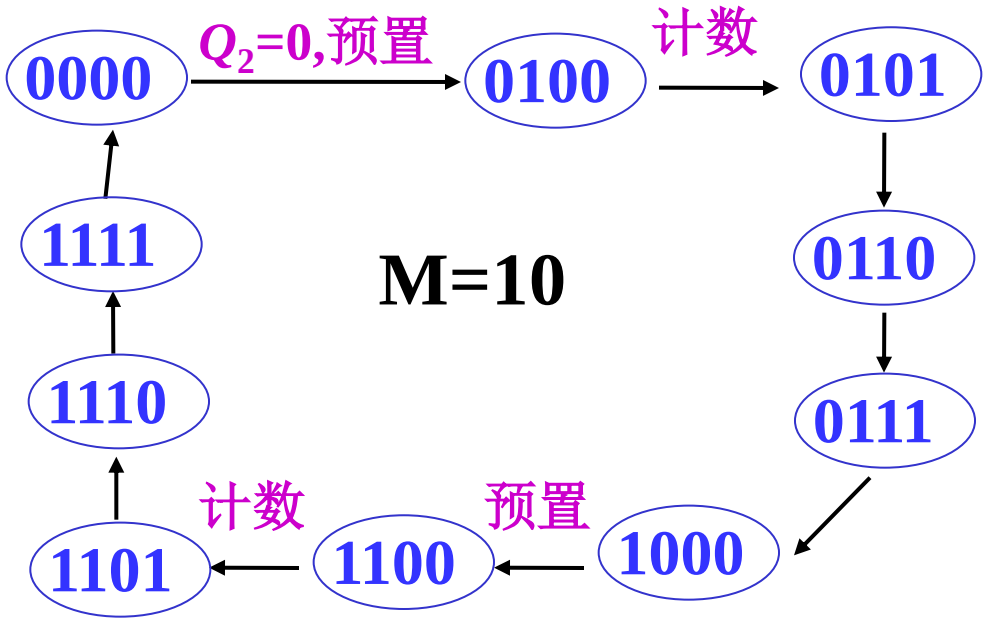


多次置数

解：

Q_2 连 $\overline{LD}$ 端 $\begin{cases} Q_2=0，\text{预置} \\ Q_2=1，\text{计数} \end{cases}$

状态图 $Q_3Q_2Q_1Q_0$

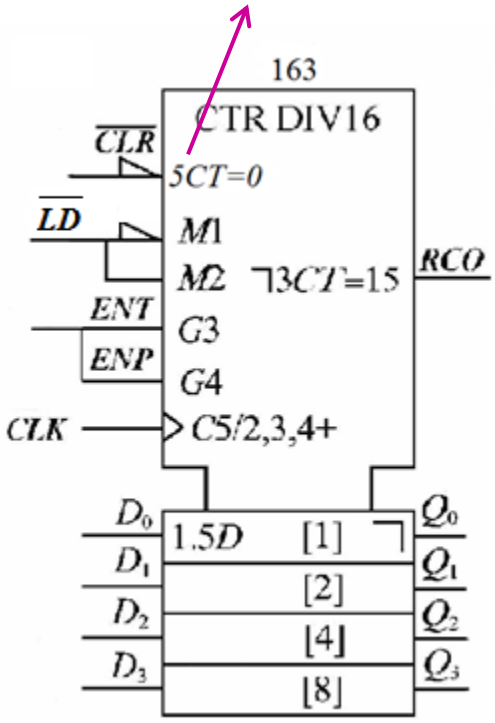


6.4.3 IC 计数器 74163 (M-16)

同步清0，其它与 74161相同（模16）

图中：5*CT*=0 在 5 端有效时清0

*CLK*有效边沿



74163 功能表

$\overline{CLR}$	$\overline{LD}$	ENT	ENP	CLK	$D_0\,D_1\,D_2\,D_3$	功能
0	X	X	X	↑	X X X X	直接置 0
1	0	X	X	↑	$D_0D_1D_2D_3$	Load 预置
1	1	0	X	X	X X X X	保持 *RCO*=0
1	1	X	0	X	X X X X	保持
1	1	1	1	↑	X X X X	M-16 计数

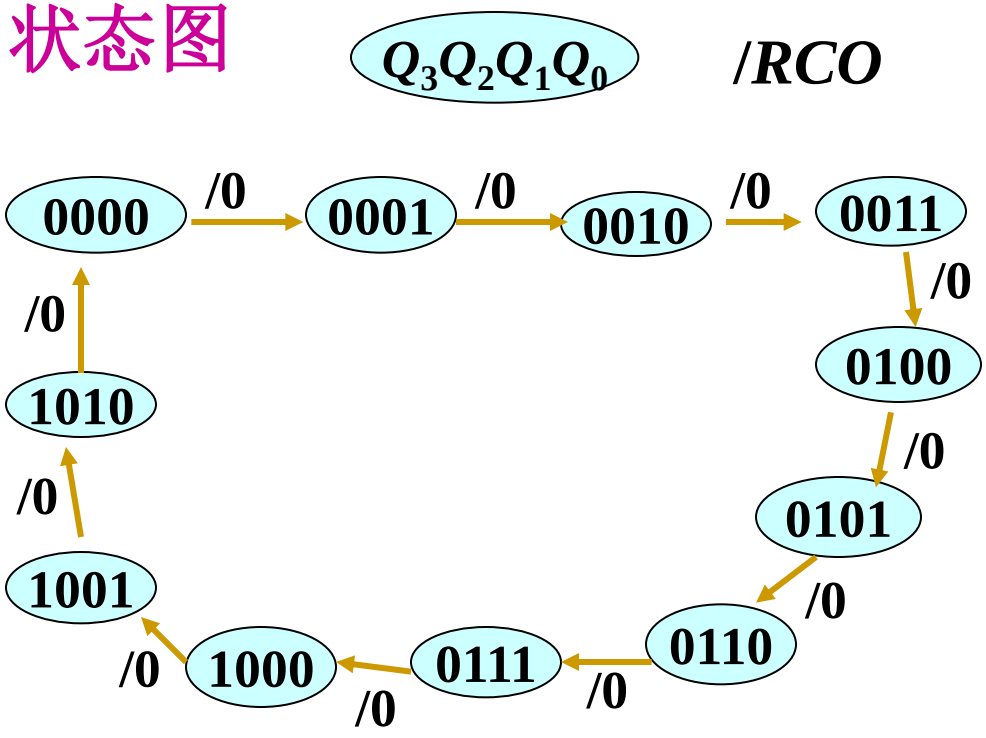
同步清零0: 当$\overline{CLR}$ =0 时，下一个 *CLK* 到达，

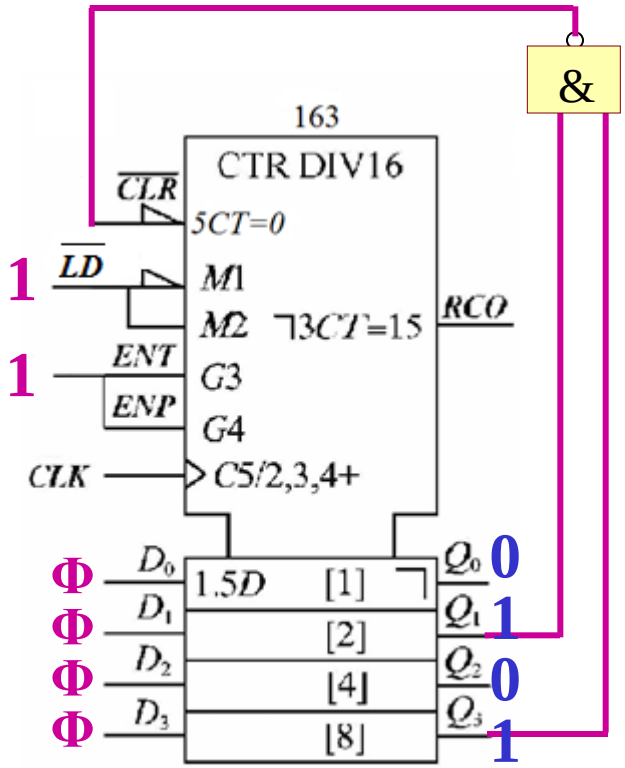
$Q_3Q_2Q_1Q_0$ = 0000

例：用74163的同步清零功能设计一个模11计数器.
($\overline{CLR}$)

最大状态 1010

状态图





没有毛刺

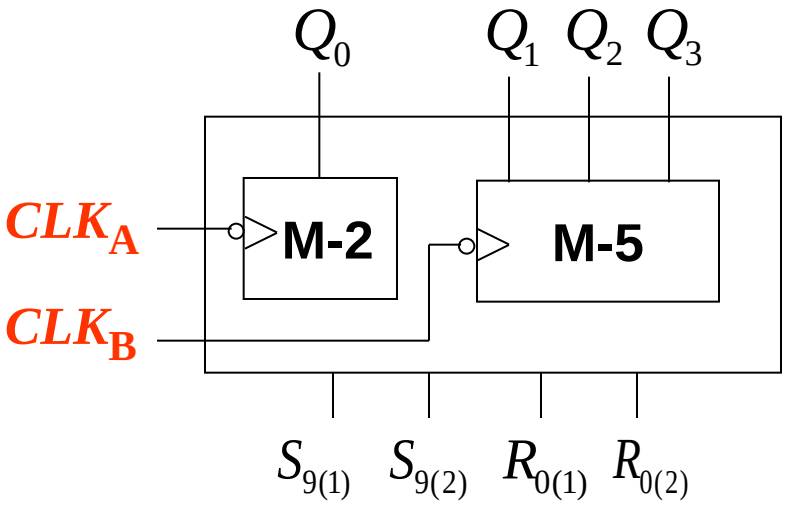
用 $\overline{LD}$ 端，与74161相同，初始为0000.

6.4.4 IC 计数器 74290

1. 74290 功能

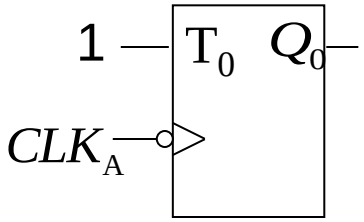
模 2-5-10 异步计数器

框图



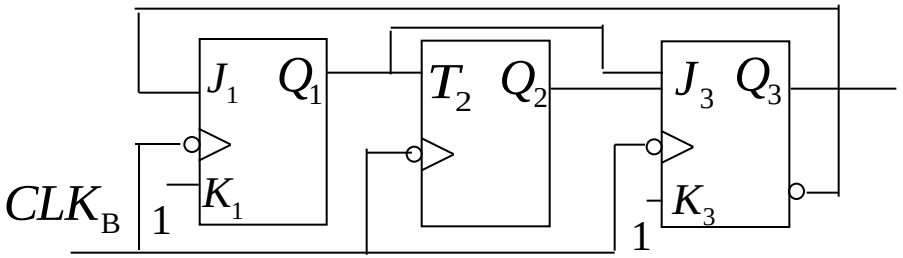
两个独立的下降沿FF

M-2 计数器，输出 Q_0



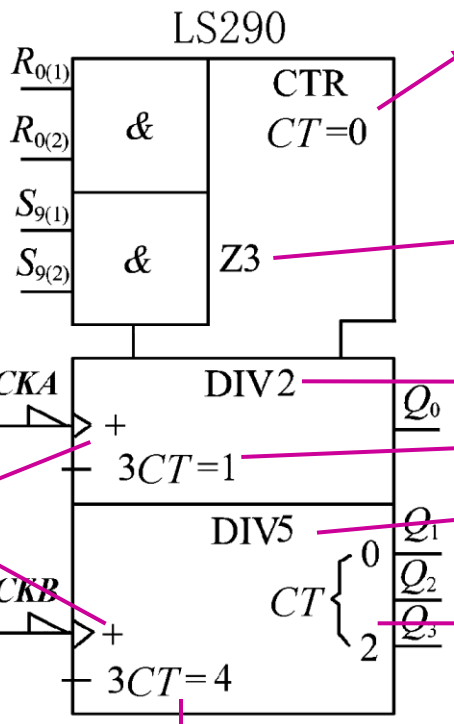
T'触发器，
二分频电路

M-5 计数器，输出 $Q_3Q_2Q_1$



符号

异步输入



$R_{(0)}$“与”为1时，计数器置0

Z：互连置位. 3：3端.

2分频

当$S_{(9)}$“与”为1时，即3端有效时Q_0置1

加计数

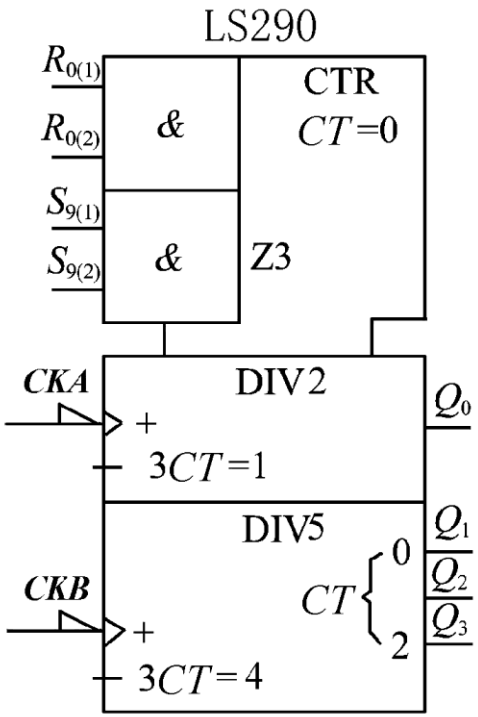
5分频

位权（高地位顺序）

3端有效时，$Q_3Q_2Q_1$＝100 (置4)；

即$Q_3Q_2Q_1Q_0$ ＝1001. (计数器置9).

功能



(1) 异步清0

当 $\begin{cases} S_{9(1)} \cdot S_{9(2)} = 0 \ \text{(low)} \\ R_{0(1)} = R_{0(2)} = 1 \ \text{(high)} \end{cases}$

$Q_3Q_2Q_1Q_0 = 0000$　　高电平有效

(2) 异步置9

当 $S_{9(1)}=S_{9(2)}=1$

$Q_3Q_2Q_1Q_0 = 1001$

(3) 计数

当 $\begin{cases} S_{9(1)} \cdot S_{9(2)} = 0 \\ R_{0(1)} \cdot R_{0(2)} = 0 \end{cases}$

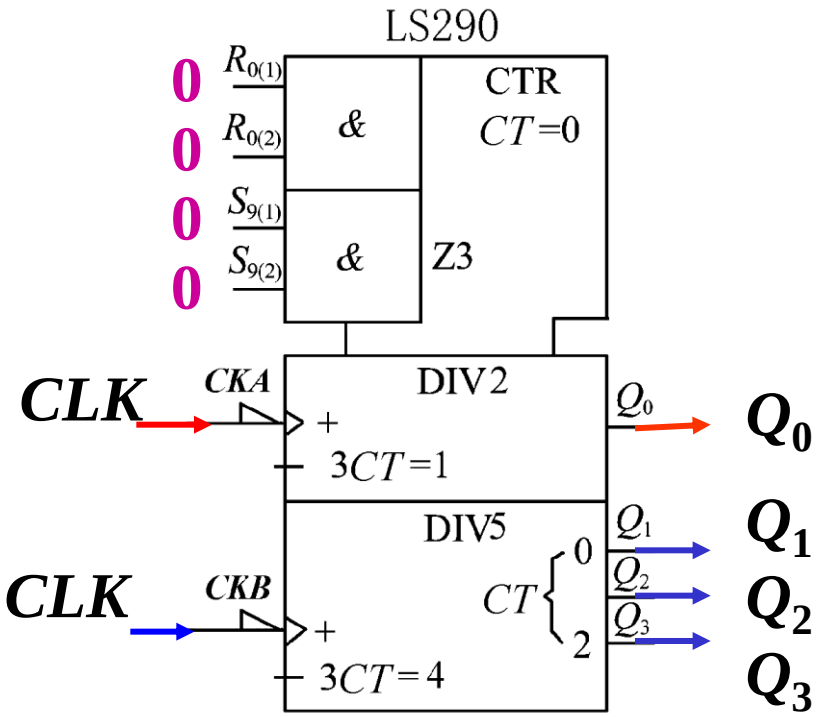
同时满足，*CLK*下降沿实现计数

2. 74290应用

(1) 模 2 计数器

$$\begin{cases} S_{9(1)} \cdot S_{9(2)} = 0 \\ R_{0(1)} \cdot R_{0(2)} = 0 \end{cases}$$

CLK 从 CLK_A 接入, Q_0 输出, 实现 模 2加计数



(2) 模 5 计数器

$$\begin{cases} S_{9(1)} \cdot S_{9(2)} = 0 \\ R_{0(1)} \cdot R_{0(2)} = 0 \end{cases}$$

CLK 从 CLK_B 接入, $Q_3Q_2Q_1$ 输出, 实现 模 5加计数

两种用法完全独立。构成更大模数时需外接线连接

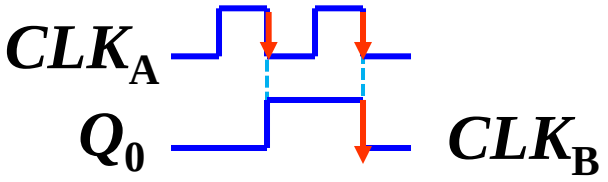
(3) 8421BCD 码模10 计数器

$S_{9(1)} \cdot S_{9(2)} = 0$，　$R_{0(1)} \cdot R_{0(2)} = 0$

CLK 接 CLK_A，

$Q_0 \longrightarrow CLK_B$

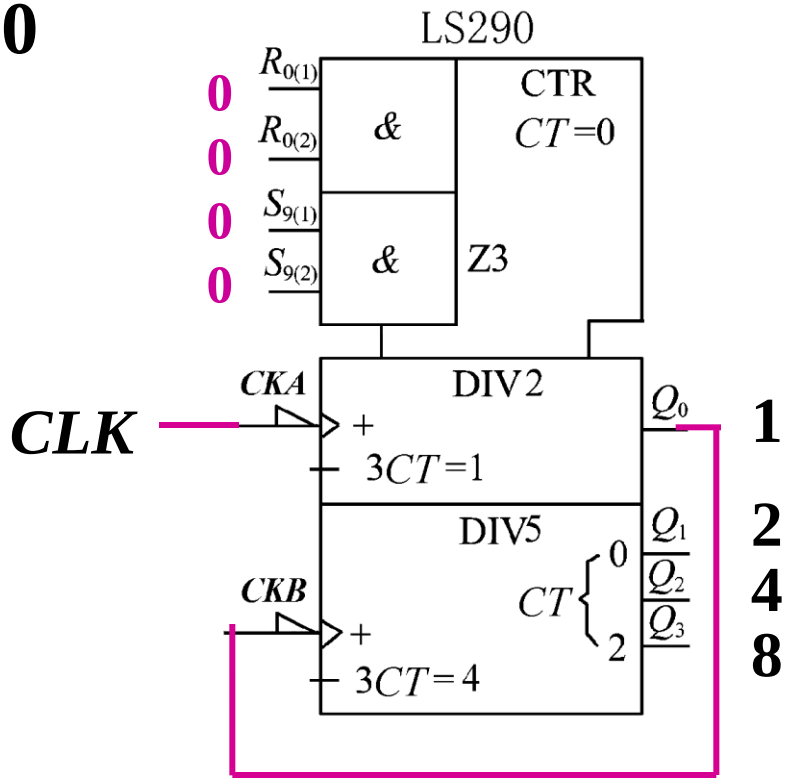
在 Q_0 下降沿(CLK_B $1 \rightarrow 0$), M-5 计数.

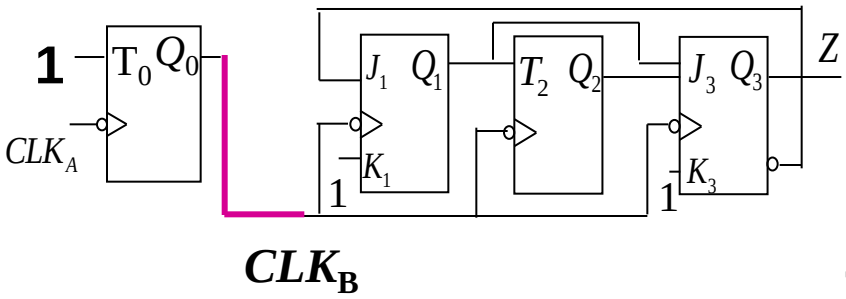


触发 M-5 计数

输出位权

$Q_3Q_2Q_1Q_0$: 8 4 2 1





(4) 8421 BCD码任意进制计数器

“直接置 0 R_0 ”高电平清 0

例 : M-7 计数器

① $S_{9(1)} = S_{9(2)} = 0$

② $CLK \rightarrow CLK_A$

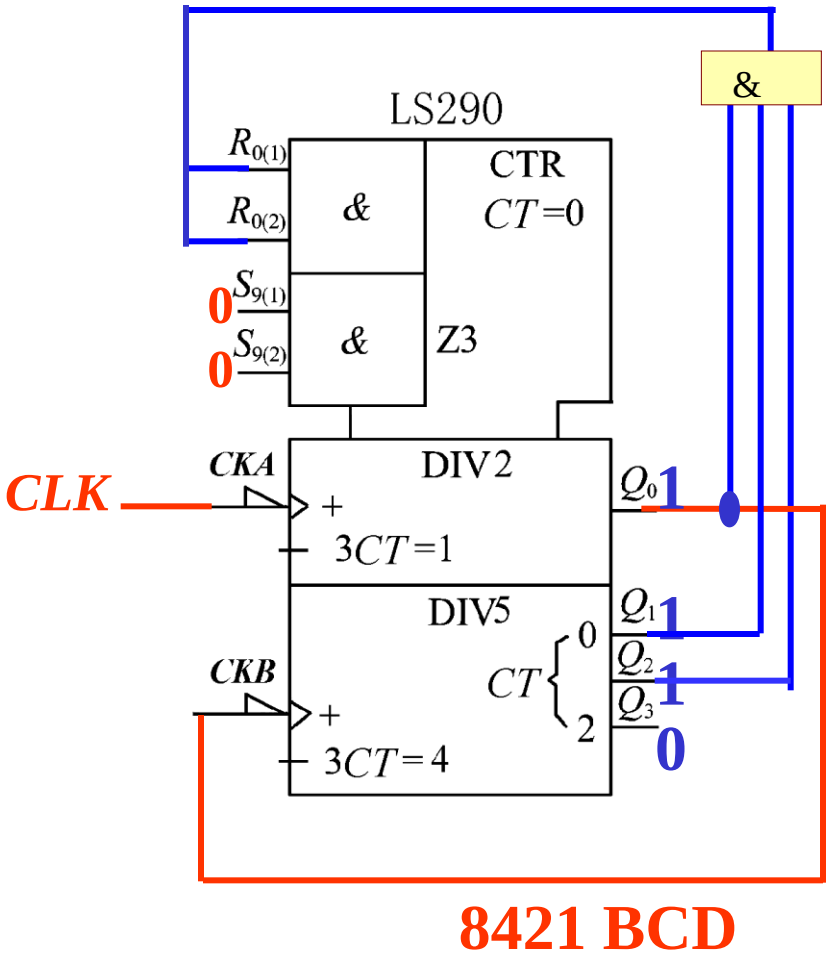
③接: 8421 BCD 模10

$Q_0 \rightarrow CLK_B$

④ 输出 $Q_3Q_2Q_1Q_0 = 0111$

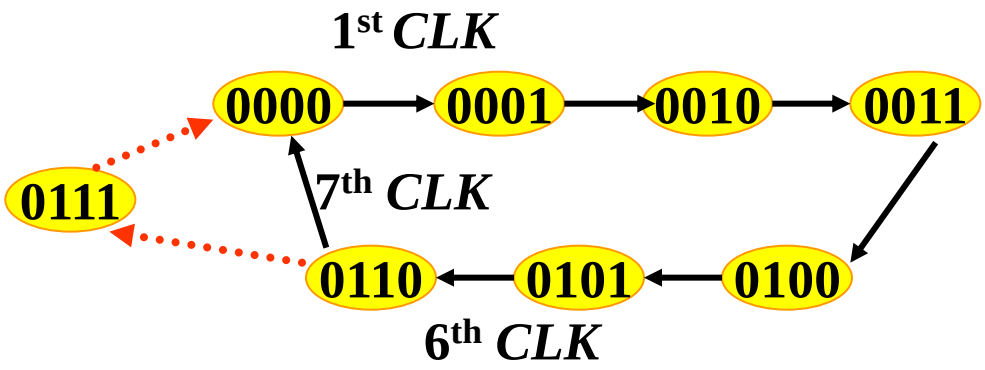
→ 与门

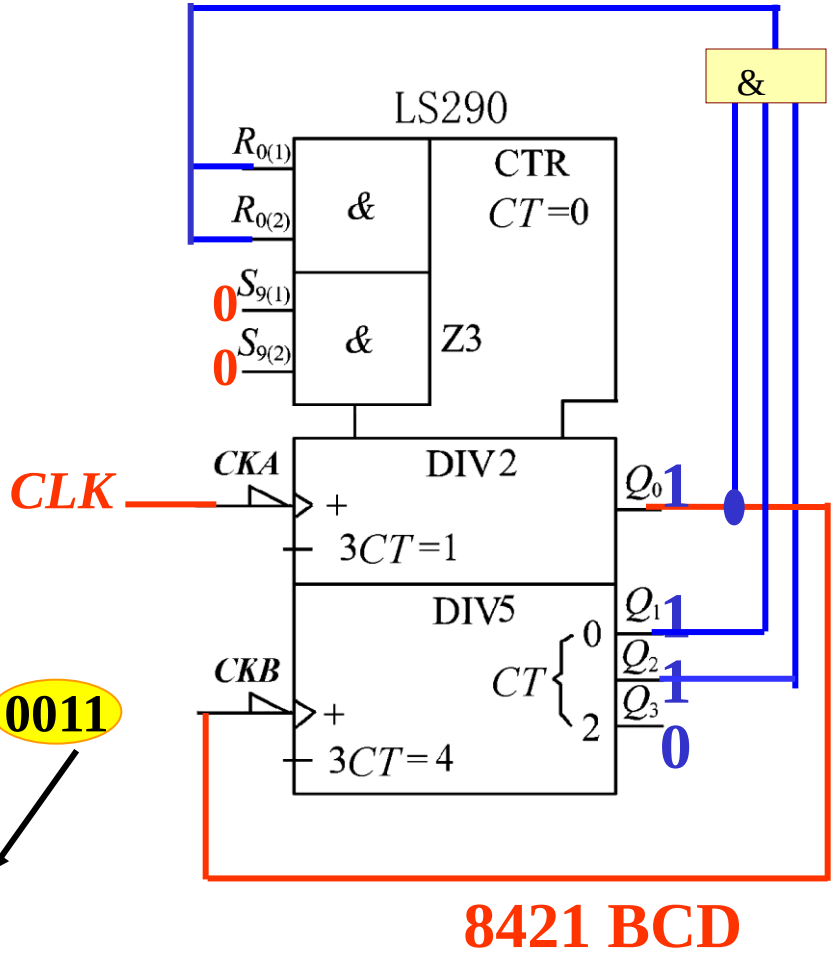
⑤ 与门 → R_0 (直接清0)



当$Q_3Q_2Q_1Q_0$ = 0111时，立即清0，0111只是一闪，出现 毛刺

主循环7个状态：
0000 ~ 0110.





不稳定状态用虚线连接

74290没有进位输出端

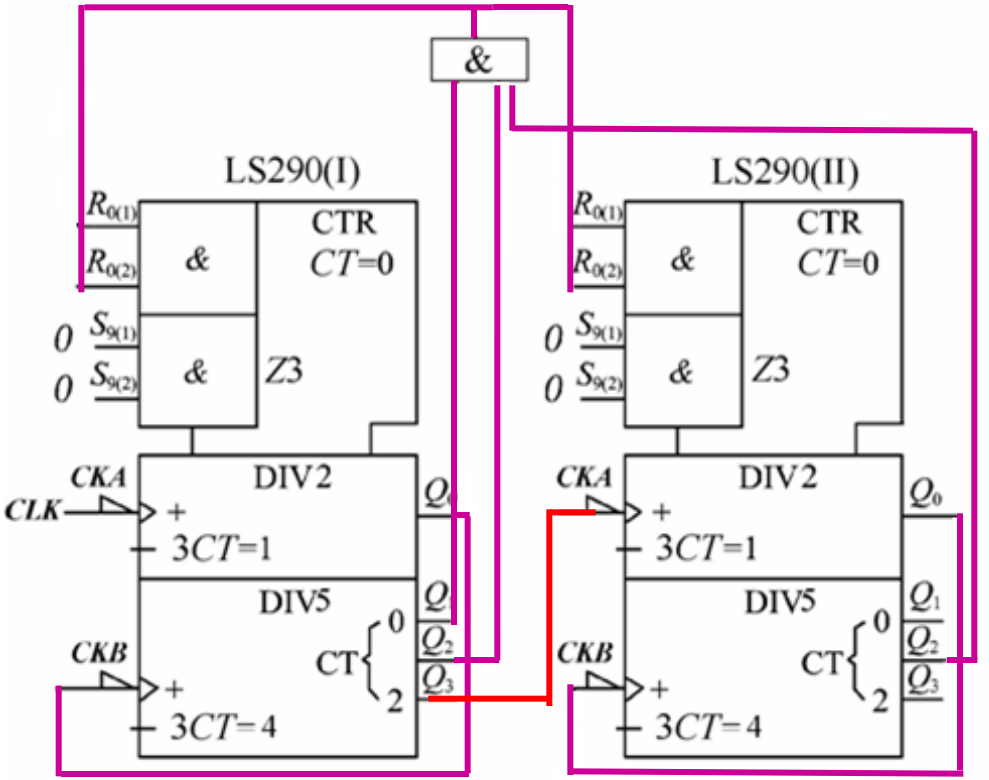
(5) 8421 BCD 级联计数器

当计数模值>10　　74290 级联

例: 用74290 设计一个 8421BCD 码模 46 计数器.

8421 十进制

进位

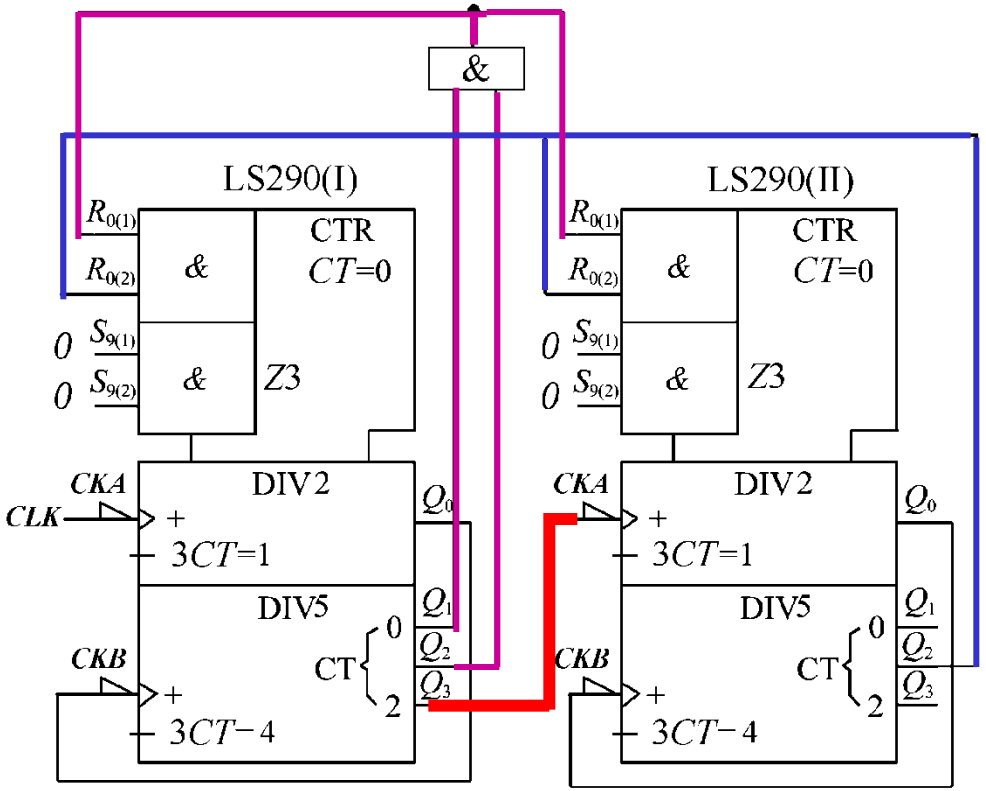


个位：6 (0110)　　十位：4 (0100)

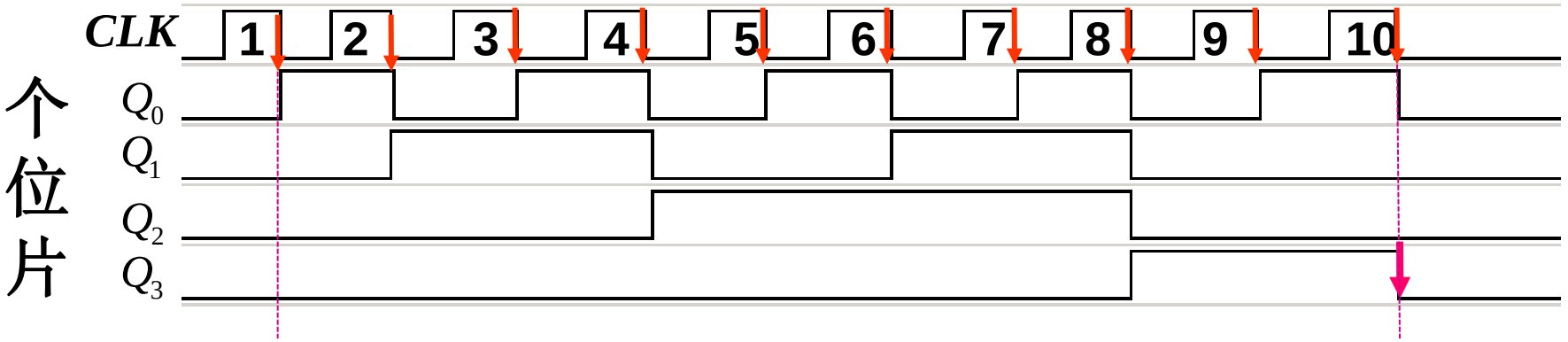
电路连接方式也可以

注意：进位

利用Q_3第10个CLK下降沿触发十位片的CLK_A (不用连Q_0Q_3)



波形：



§6.5　寄存器　Registers

寄存器用于寄存一组二进制代码。主要由触发器构成。

一个触发器能够存储 1 位二进制代码，所以用 ***n*** 个触发器组成的寄存器可以存储一组 ***n*** 位二进制信息。

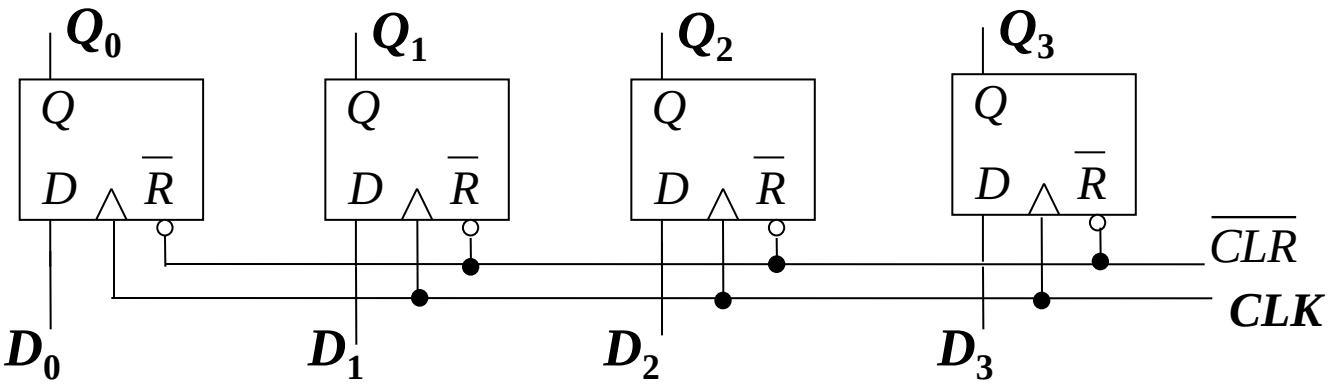
寄存器广泛地应用于各类数字系统和数字计算机中。移位型寄存器在数字通信中的应用极其广泛。如在计算机串行数据通信中，需要发送的信息总是先放在发送端的移位寄存器中，然后由移位寄存器将其逐位移出。接收端的寄存器逐位从线路上接收信息，收完一个完整的数据后才从移位寄存器中取走数据。

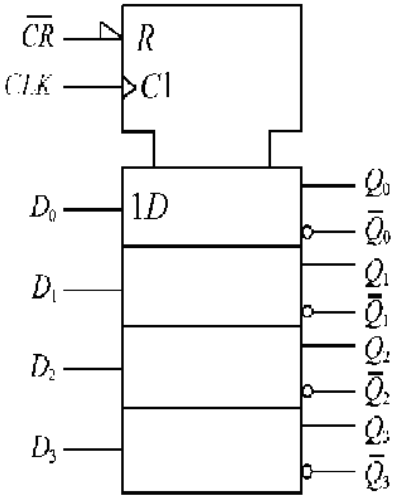
6.5.1 寄存器分类

Classifications of Registers

1. 并入/并出型寄存器　Parallel In/Out

例如，4个 D-FFs 构成寄存器　　74LS175



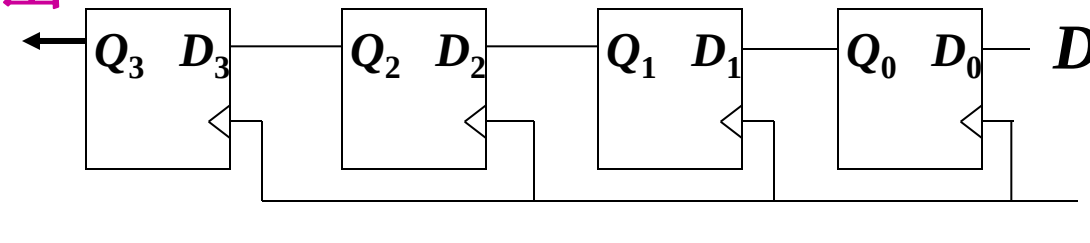


在*CLK* 正边沿, 4 个数据并行输入，状态 $Q_3Q_2Q_1Q_0 = D_3D_2D_1D_0$ 并行输出

2. 左移串入/串出型寄存器

Serial In /Serial Out Shift Left Registers

数据串出



数据串入

shift left

一个*CLK* 到来, 左移一位.

例：

初始 $Q_3Q_2Q_1Q_0$ =1001

串入: 1011 (D_{SL}),

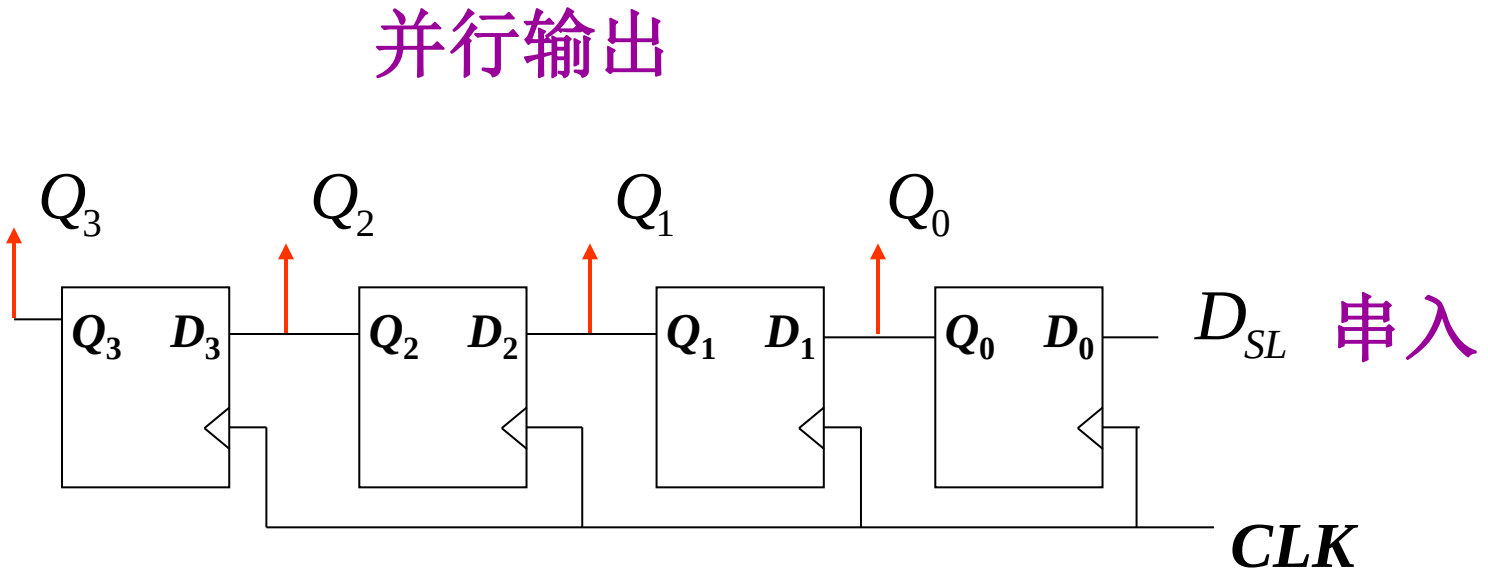
4 个*CLK* 后,

$Q_3Q_2Q_1Q_0$ =1011

CLK	串出	$Q_3Q_2Q_1Q_0$ 串入
	1	1 0 0 1 1 0 1 1
1	0	0 0 1 1
2	0	0 1 1 0
3	1	1 1 0 1
4	1	1 0 1 1

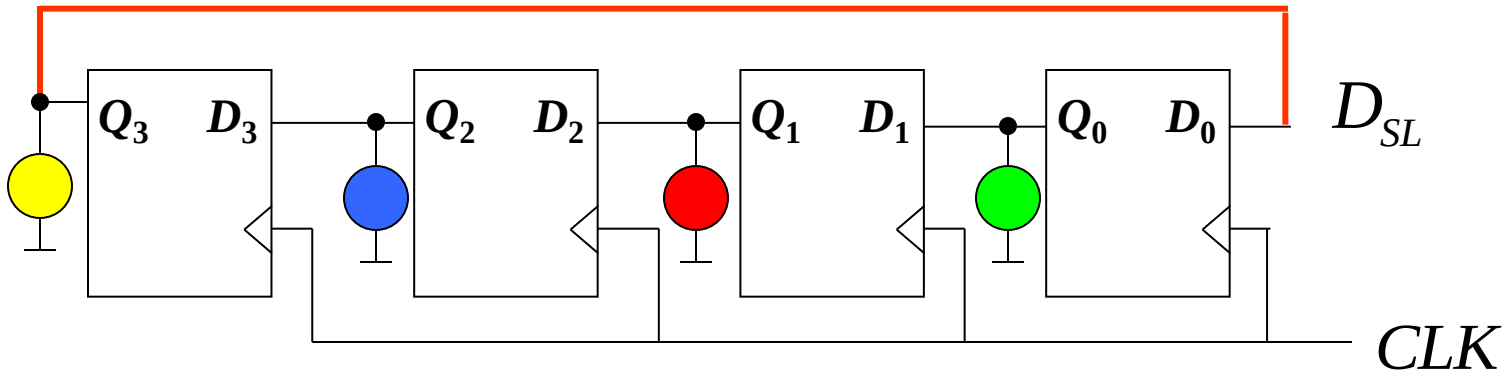
3. 左移串入/并出型寄存器

Serial In / Parallel Out Shift Left Registers



4. 左移环型寄存器 Shift Left Ring Registers

串出端与串入端相连



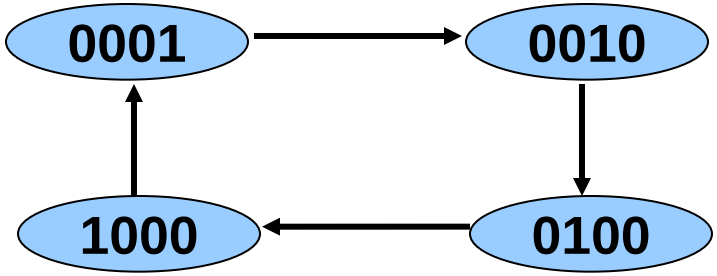
各FF 输出接彩灯

当输出为 0001时，接高电平的灯亮

灯亮时间：

取四位中只有一个1 的状态为主循环

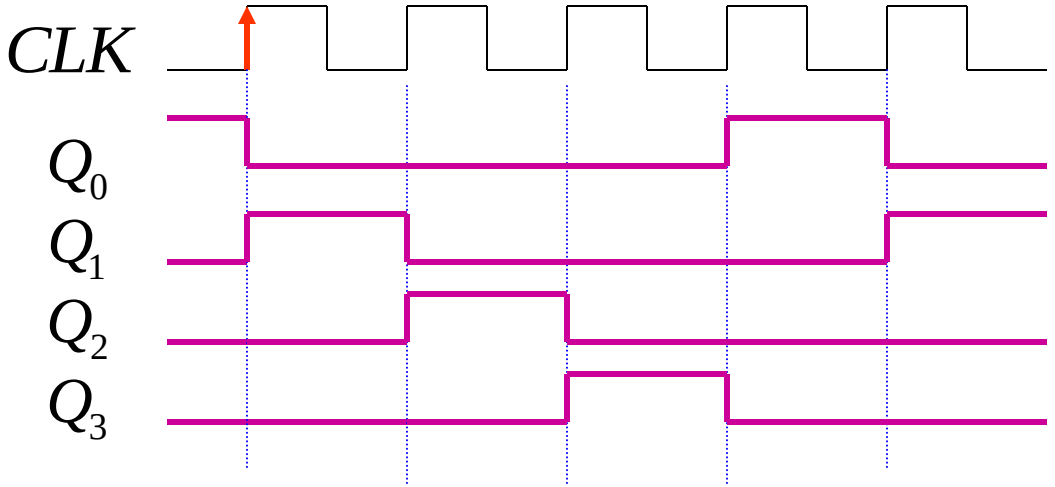
状态图



环形计数器

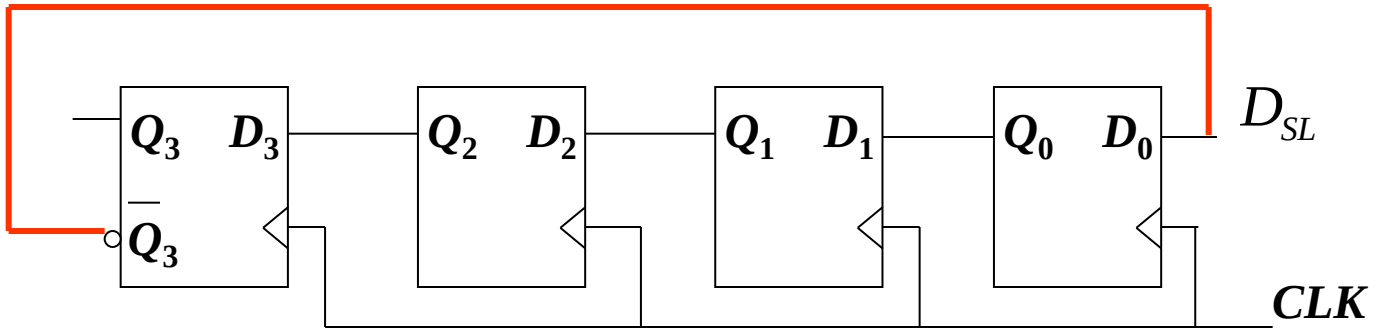
注意：*n* FFs → *n* 个状态 → 模 *n*

波形图



节拍发生器

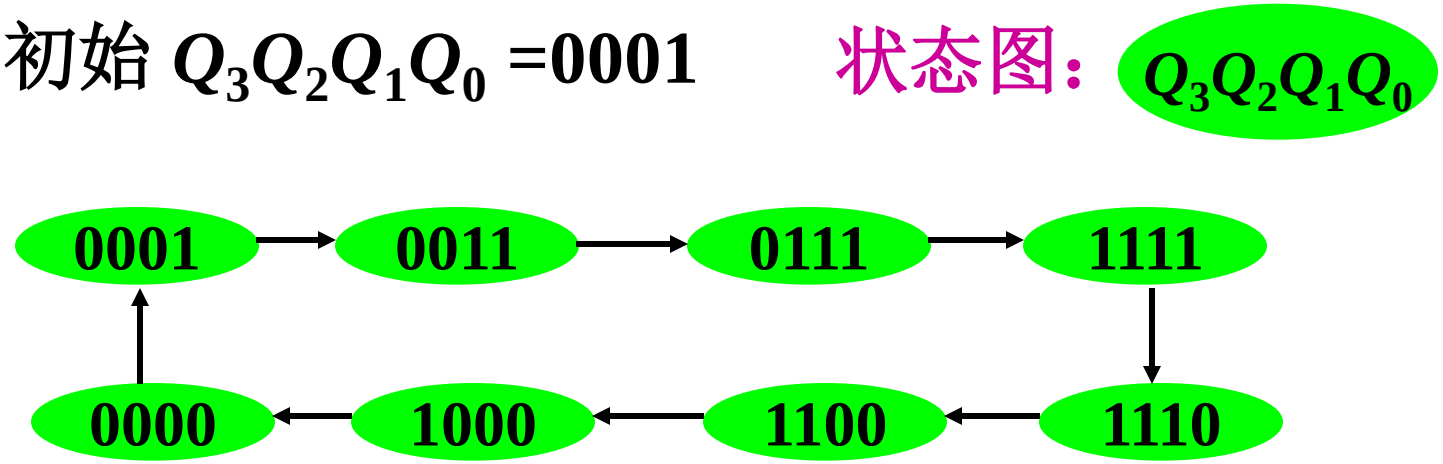
5. 左移扭环寄存器 Shift Left Twisted-Ring Registers



$\overline{Q_3} \rightarrow D_{SL}$

初始 $Q_3Q_2Q_1Q_0$ =0001

状态图：



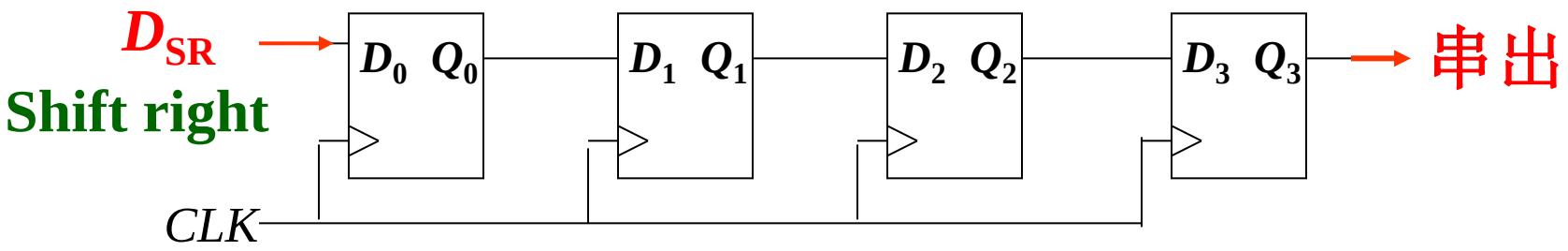
扭环计数器

n FF → 模 2*n*

Johnson counter

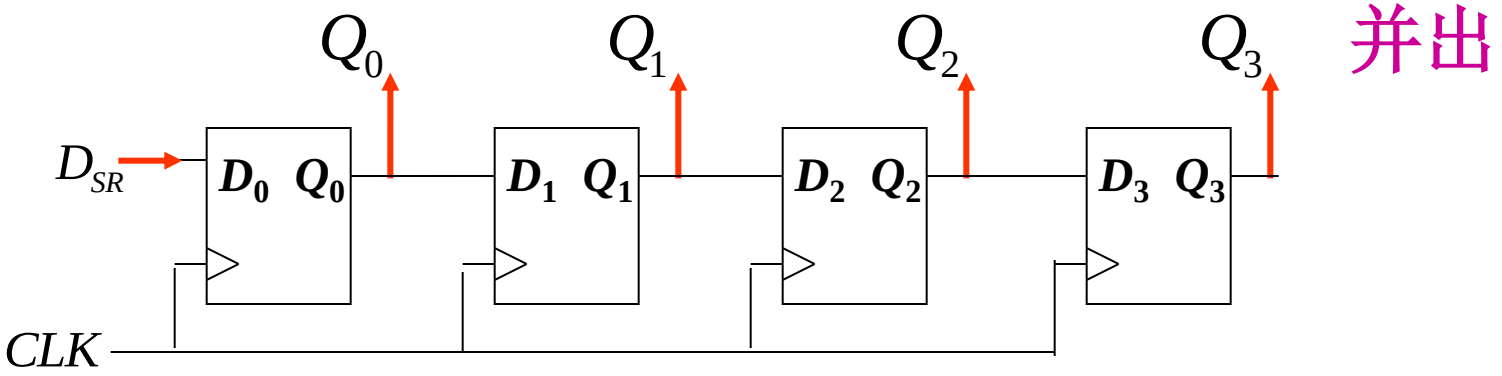
6. 右移串入/串出寄存器

Serial In /Serial Out Shift Right Registers

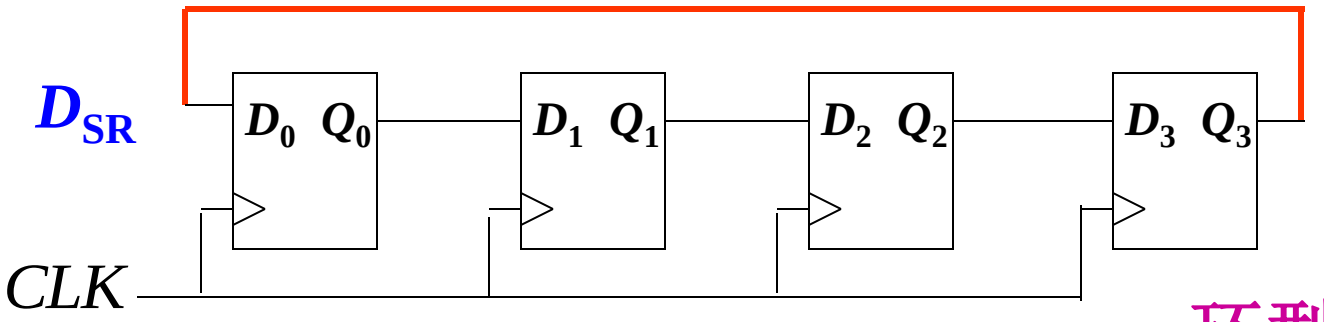


7. 右移串入 /并出寄存器

Serial In /Parallel Out Shift Right Registers

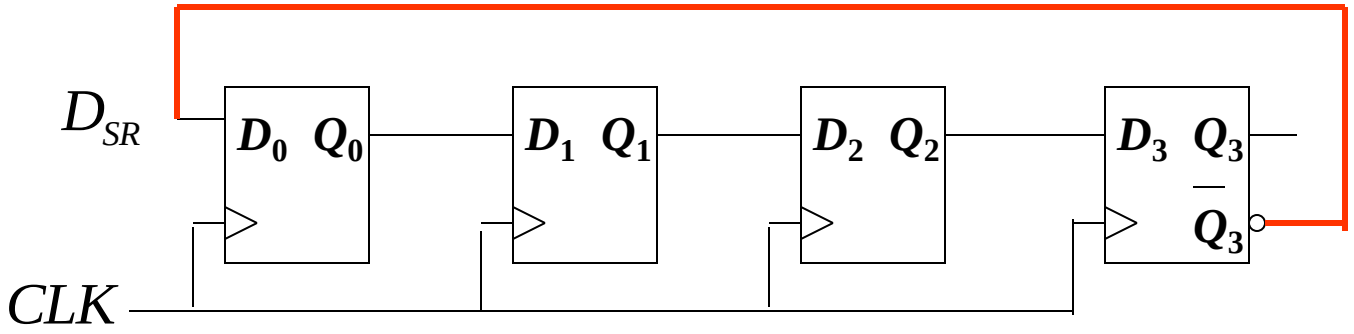


8. 右移环型寄存器 Shift Right Ring Registers



环型计数器

9. 右移扭环寄存器 Shift Right Twisted-Ring Registers

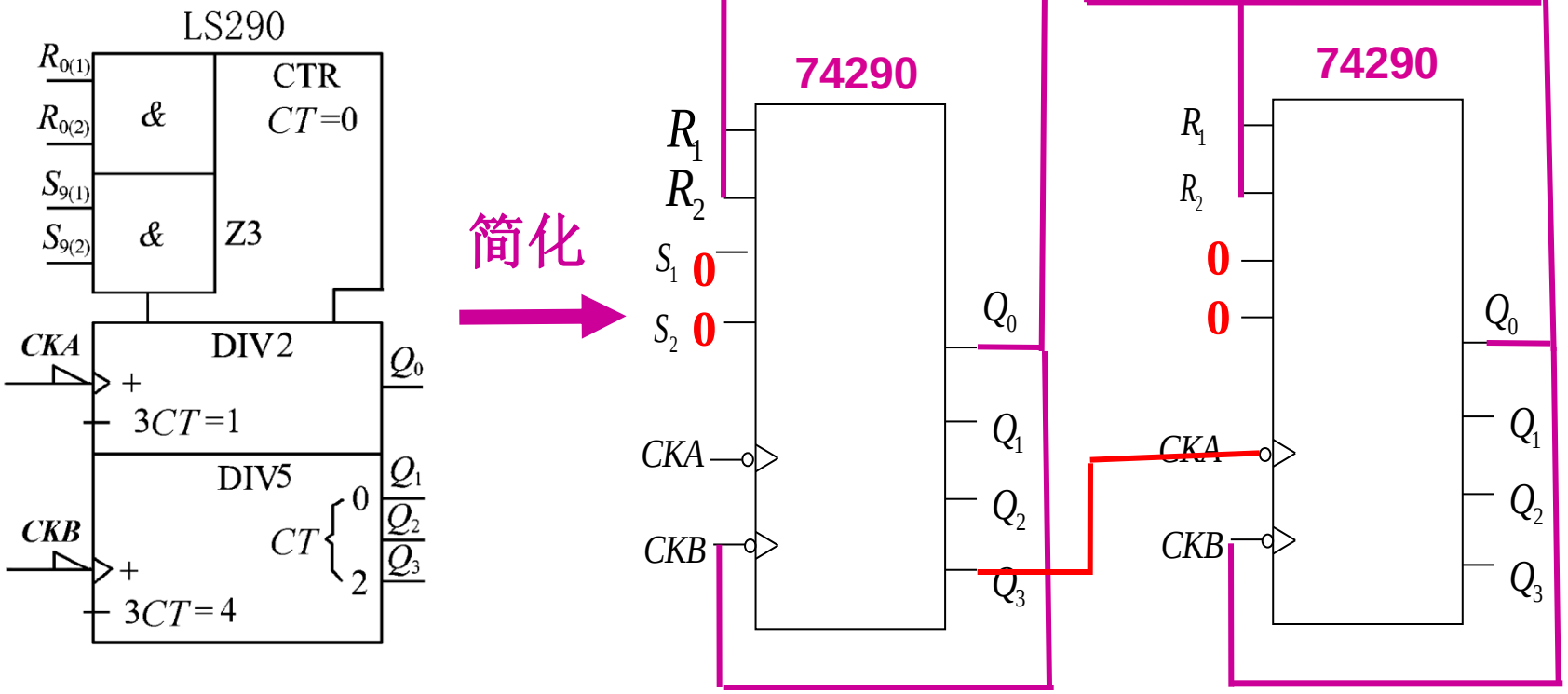


扭环计数器

课堂练习

74290是十进制计数器

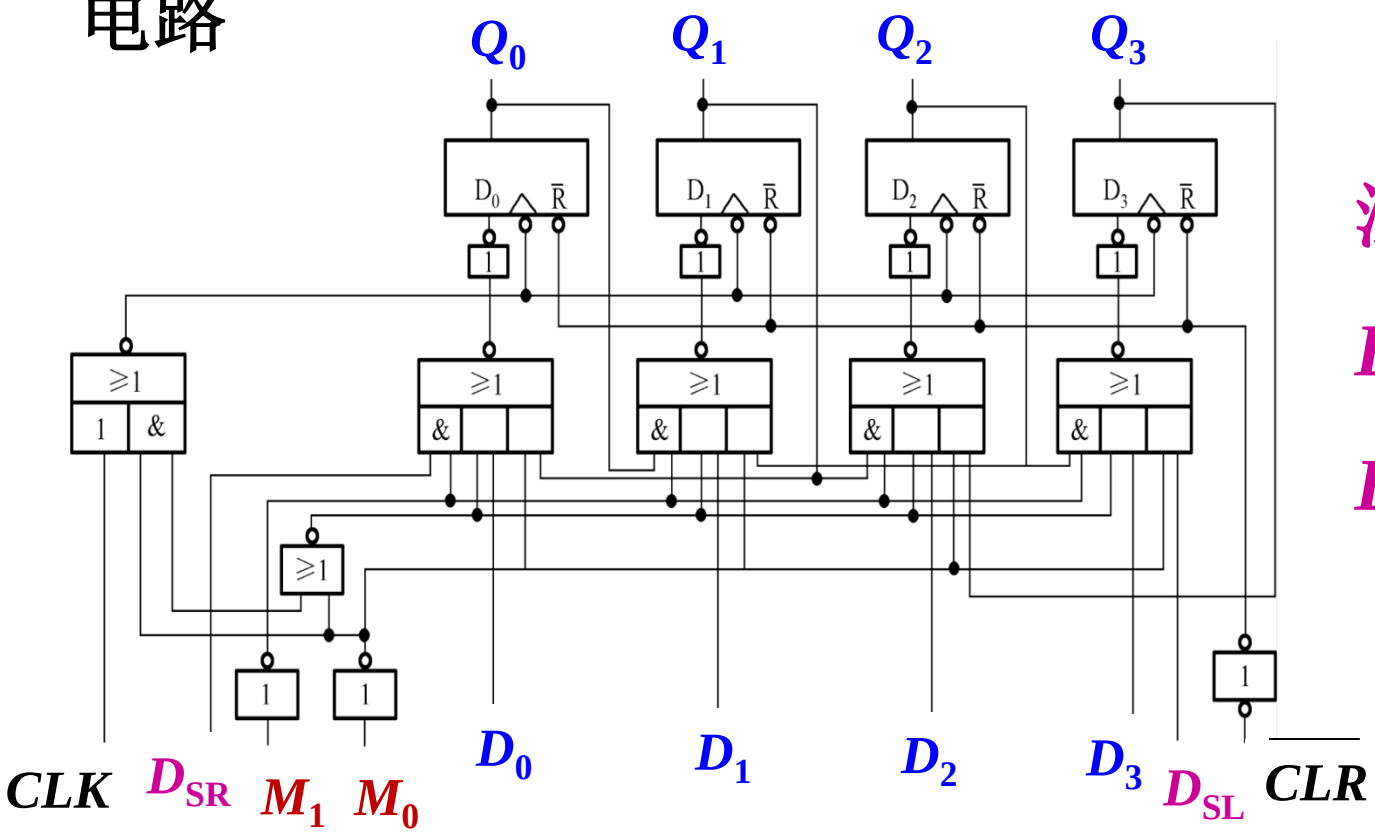
用74290连接8421BCD模11计数器电路，画出状态图。



6.5.3 集成寄存器 74194　IC Register 74194

多功能寄存器：四位并行存取双向移位寄存器

电路

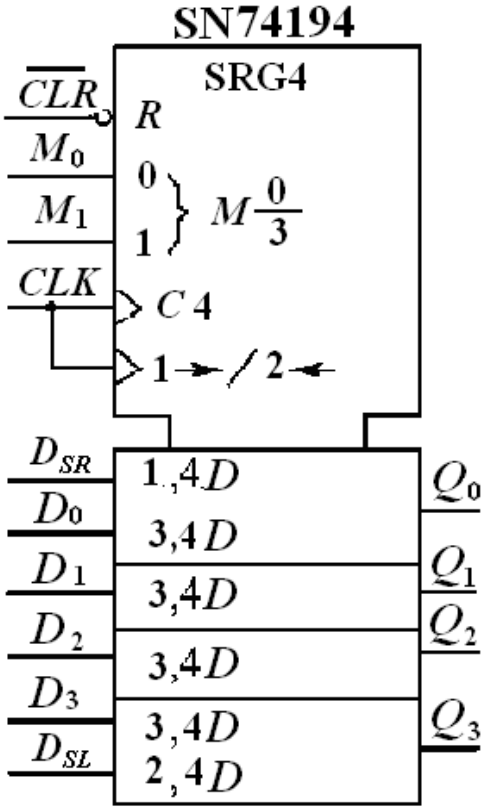


注意：

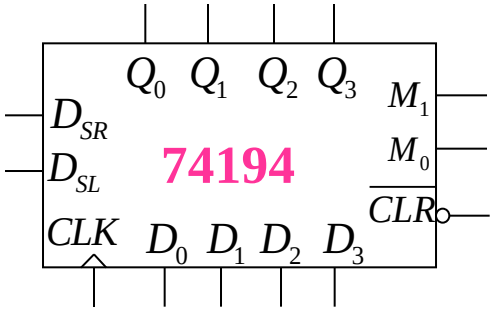
D_{SR}在Q_0一侧

D_{SL}在Q_3一侧

符号



IEEE



$Q_3Q_2Q_1Q_0$ 数据输出

$D_3D_2D_1D_0$ 数据输入

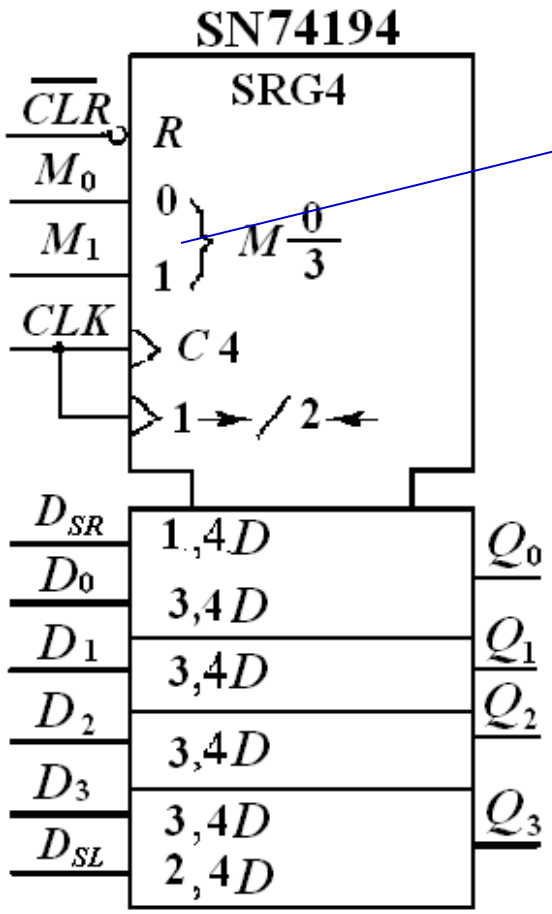
D_{SR} D_{SL} 串入

1→ 右移 shift right

2← 左移 shift left

$\overline{CLR}$ = 0, 异步清0

CLK 正边沿触发



$M_1\,M_0$ 控制 (模式)，M_1 高位

$M_1\,M_0$ 组成4种模式

74194 功能

M_1M_0	功 能
0 0	保持
0 1	右移
1 0	左移
1 1	并入

$Q_0\ Q_1\ Q_2\ Q_3$

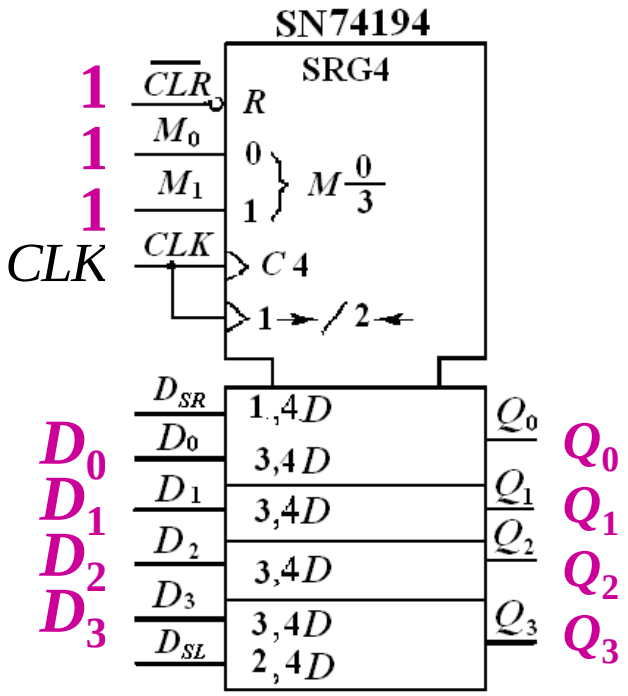
↑ ↑ ↑ ↑

$D_0\ D_1\ D_2\ D_3$

实现前面 9 种功能

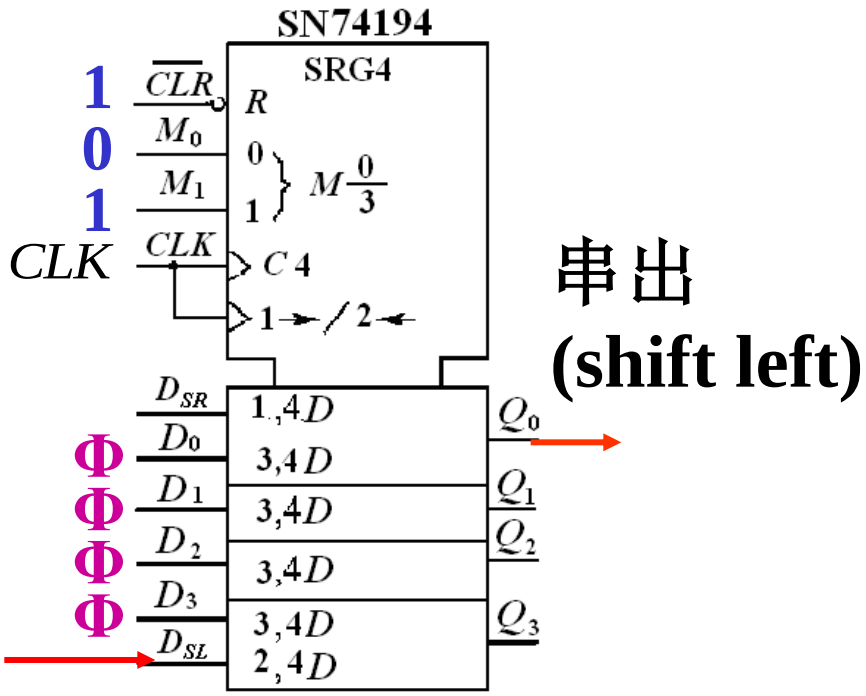
注：$Q_0Q_1Q_2Q_3$ 只有排列顺序，没有高、低位。

(1) 并入/并出



并入　　并出

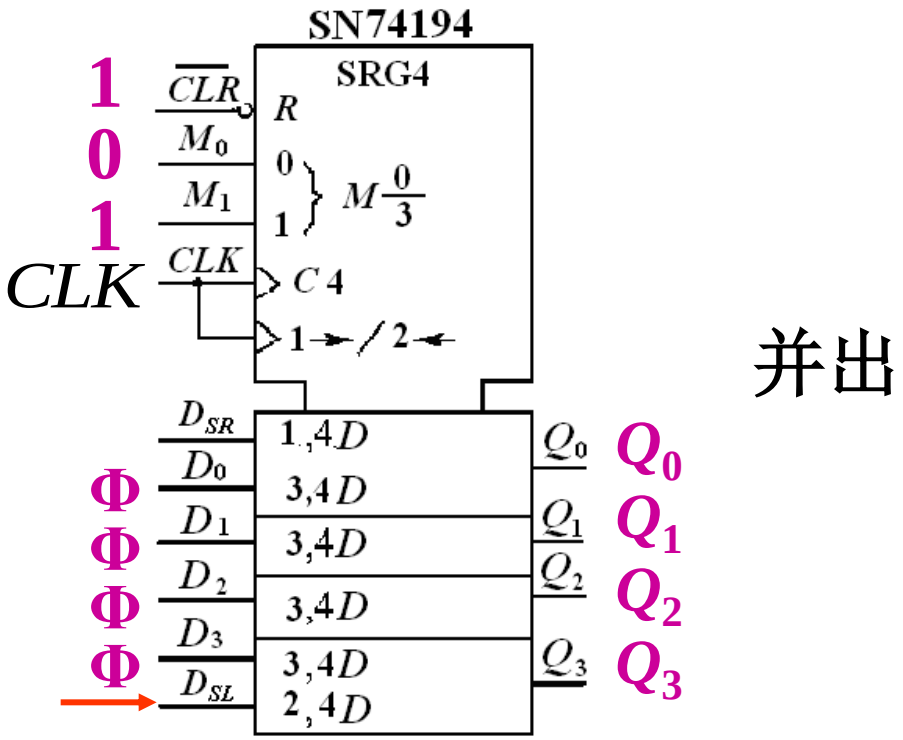
(2) 左移串入/串出



串入
shift left

经过4个触发器

(3) 左移串入 / 并出



并出

串入

shift left

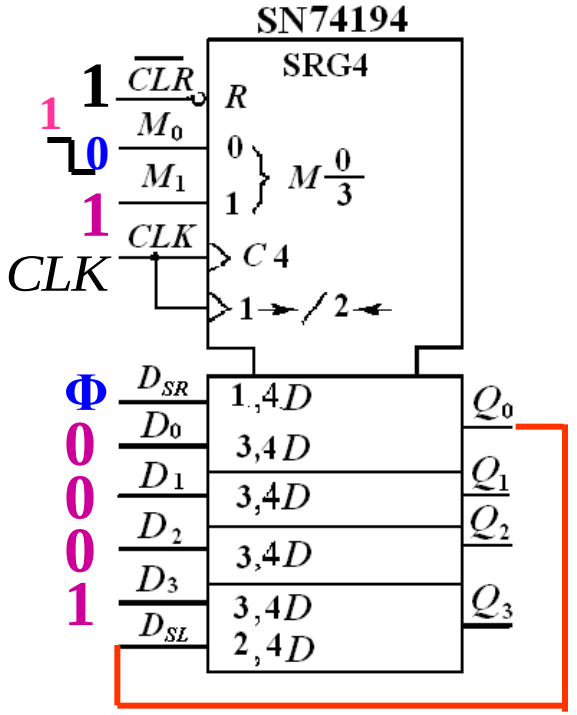
(4) 左移环形

$Q_0 \rightarrow D_{SL}$

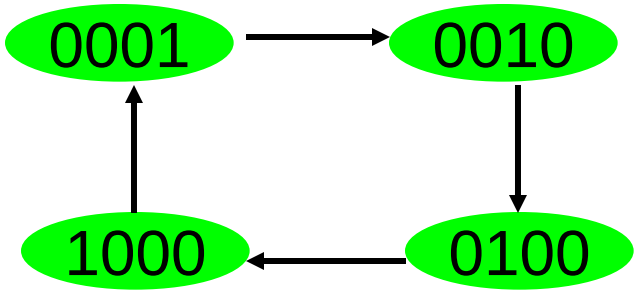
先置 M_1=1, M_0=1,
在 CLK 上升沿并入,
$Q_0Q_1Q_2Q_3$ =
$D_0D_1D_2D_3$ =0001

再置 M_0 =0, CLK
边沿到来 → 左移
→ M-4 计数

接彩灯



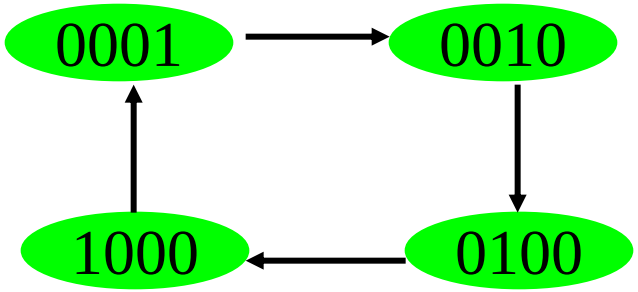
一个1为主循环 $Q_0Q_1Q_2Q_3$

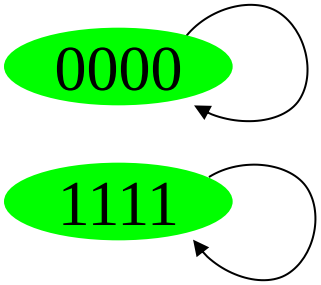


一个1为主循环

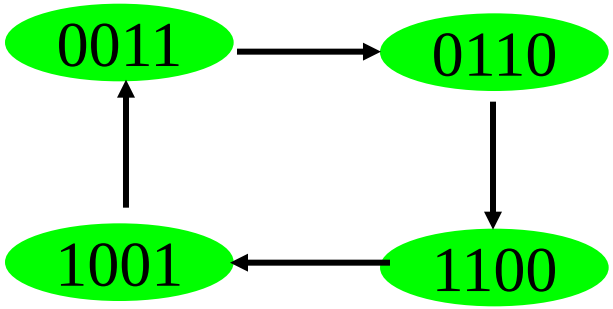
$Q_0Q_1Q_2Q_3$

左移环形的其他置数方式：

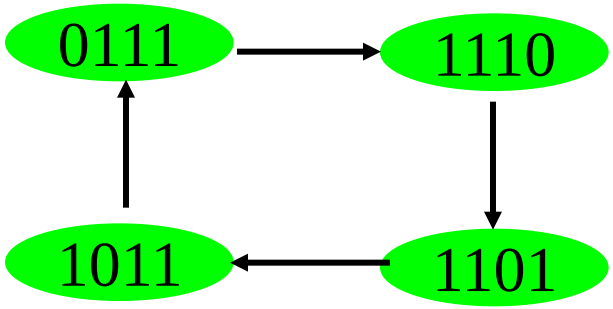




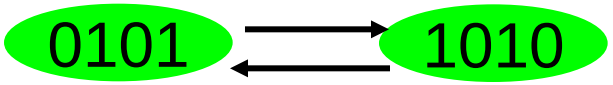
保持



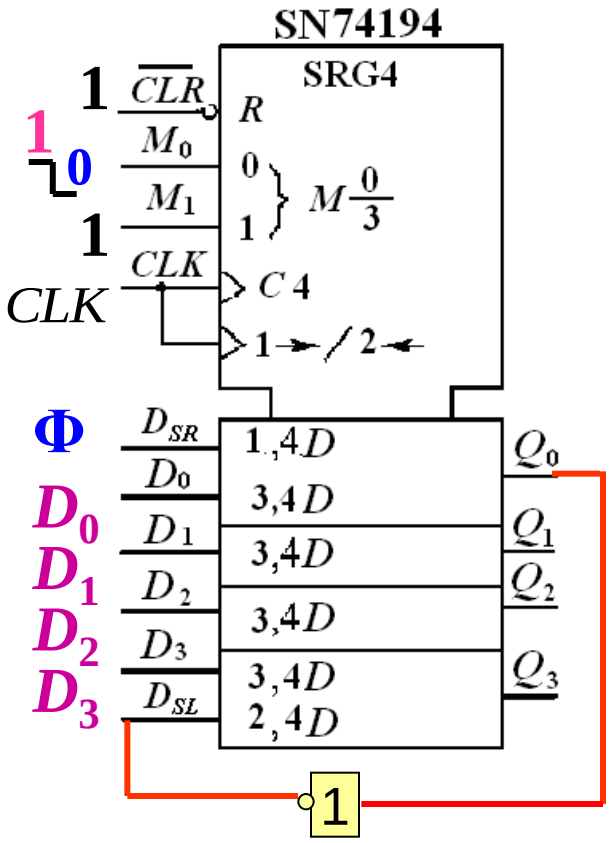
M-4环形计数器



M-4环形计数器



(5) 左移扭环寄存器



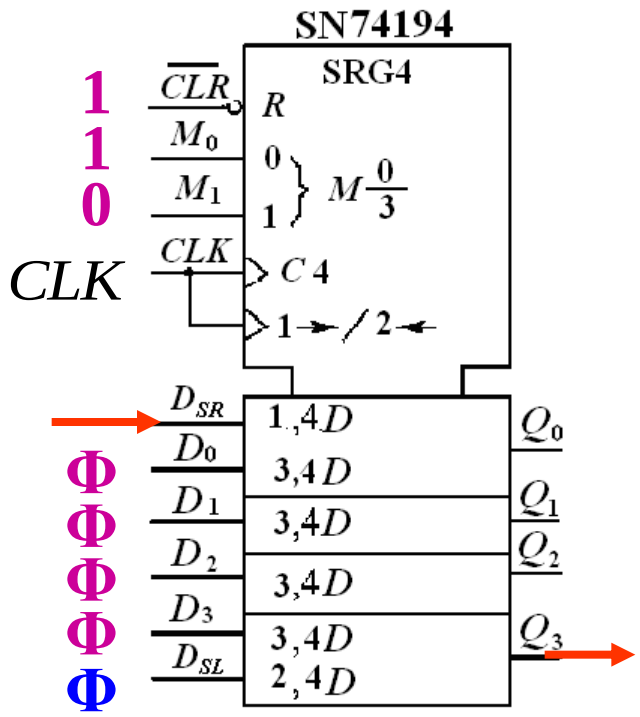
$M_1 = 1$,

$$M_0 = \begin{cases} 1, \text{并入} \\ \quad Q_0Q_1Q_2Q_3 = D_0D_1D_2D_3 \\ 0, \text{左移扭环} \end{cases}$$

$\overline{Q}_0$ 连接 D_{SL}

$D_0D_1D_2D_3$ 接Φ，都可以构成扭环

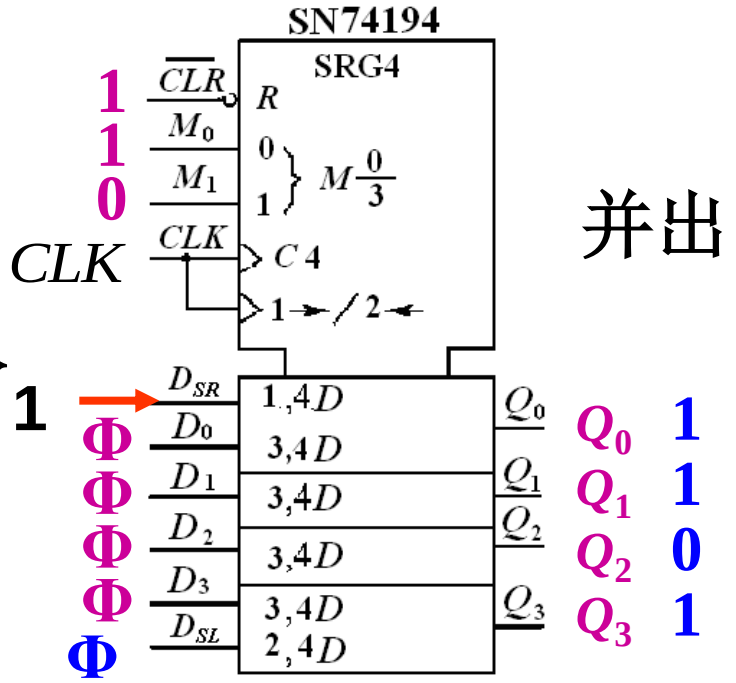
(6) 右移串入/串出寄存器



串入
shift right

串出
shift right

(7) 右移串入/并出寄存器



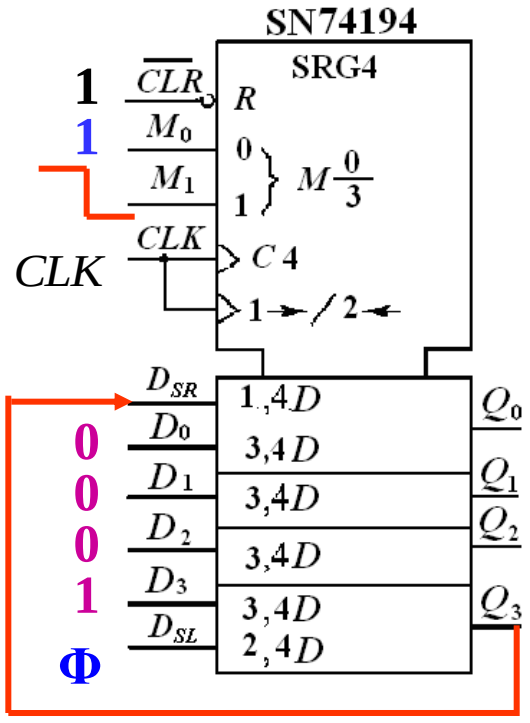
串入
1 1 0 1

并出

Q_0 1
Q_1 1
Q_2 0
Q_3 1

4个 *CLK* 后

(8) 右移环形寄存器



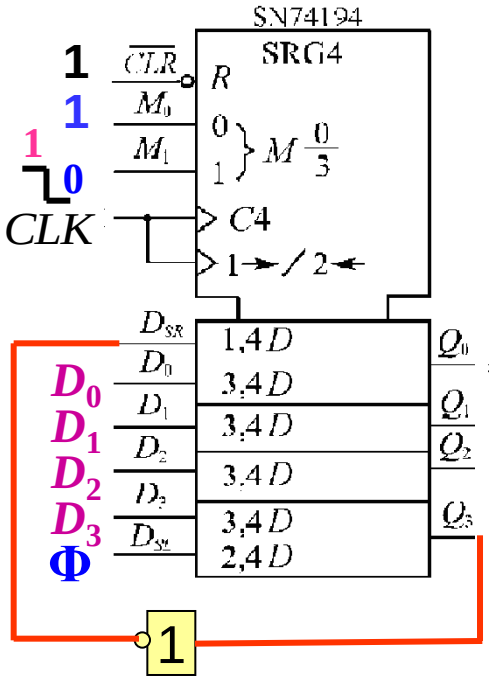
$$Q_3 \rightarrow D_{SR}$$

$$M_0 = 1$$

$$M_1 = \begin{cases} 1,\ CLK\ 到 \\ \quad Q_0Q_1Q_2Q_3 = D_0D_1D_2D_3 \\ 0,\ CLK\ 到 \\ \quad 右移环形 \end{cases}$$

模 4 计数器

(9) 右移扭环寄存器



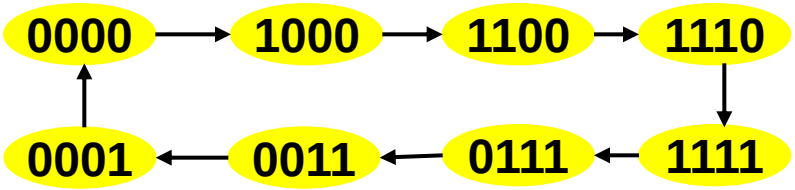
$\overline{Q}_3$ 接 D_{SR}

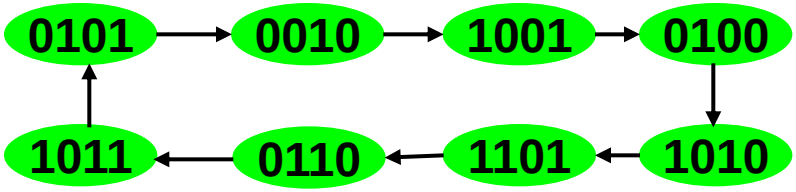
M_0= 1,

$$M_1=\begin{cases}1, \text{并入} \\ \quad Q_0Q_1Q_2Q_3 = D_0D_1D_2D_3 \\ 0, \text{右移扭环}\end{cases}$$

$D_0D_1D_2D_3$ 接Φ，都可以构成扭环

只有两种状态图 $Q_0Q_1Q_2Q_3$



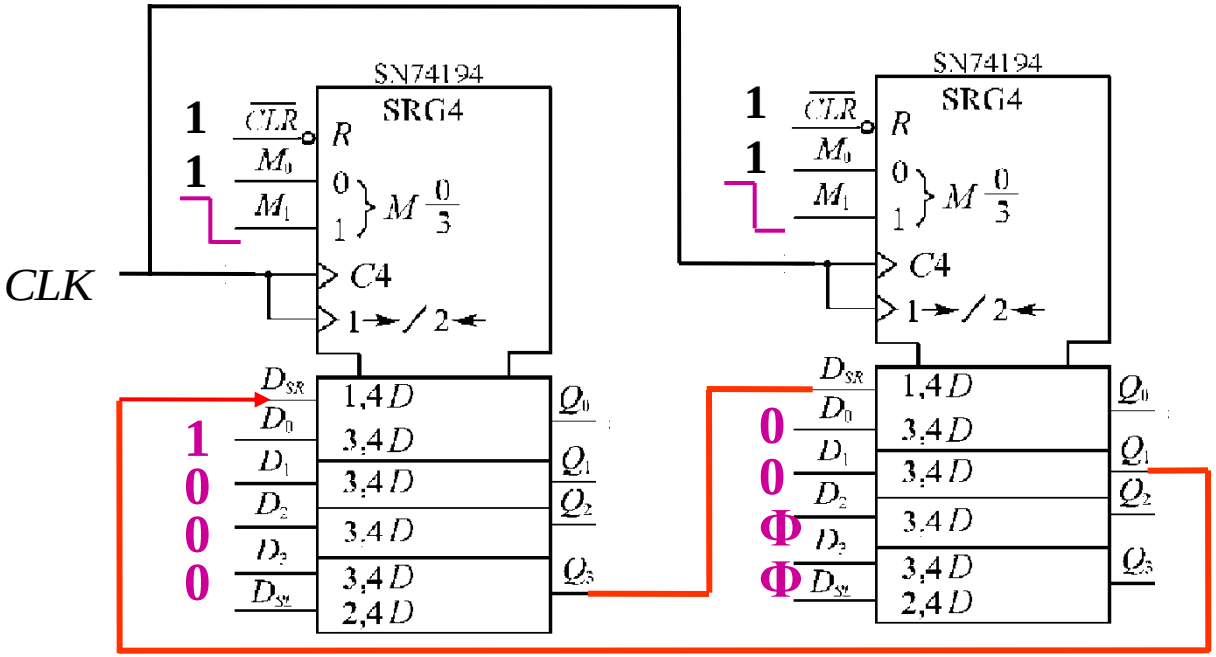


注意：从并入的$D_0D_1D_2D_3$开始循环　　模 8 计数器

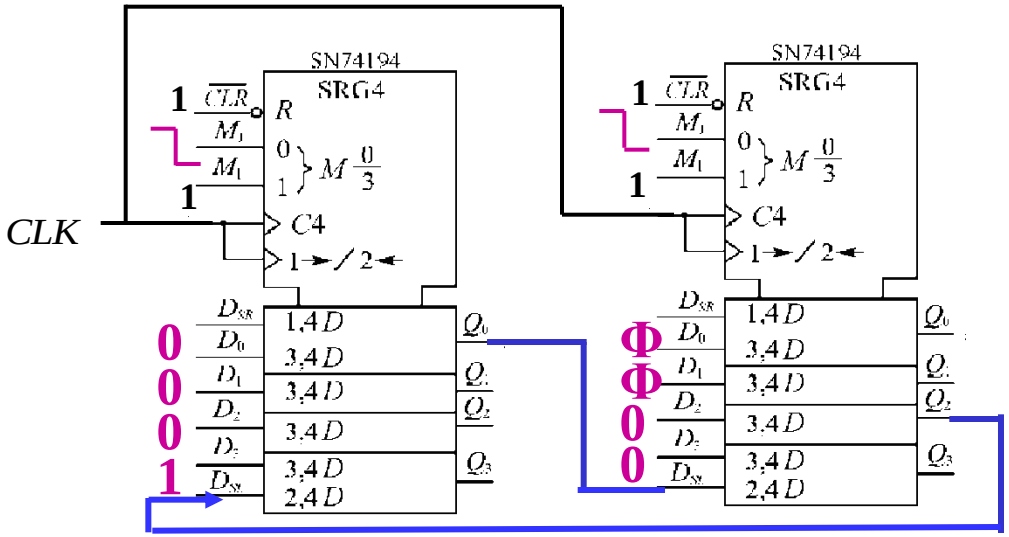
例1. 用74194 设计模 6 环形计数器

6 FFs

右移

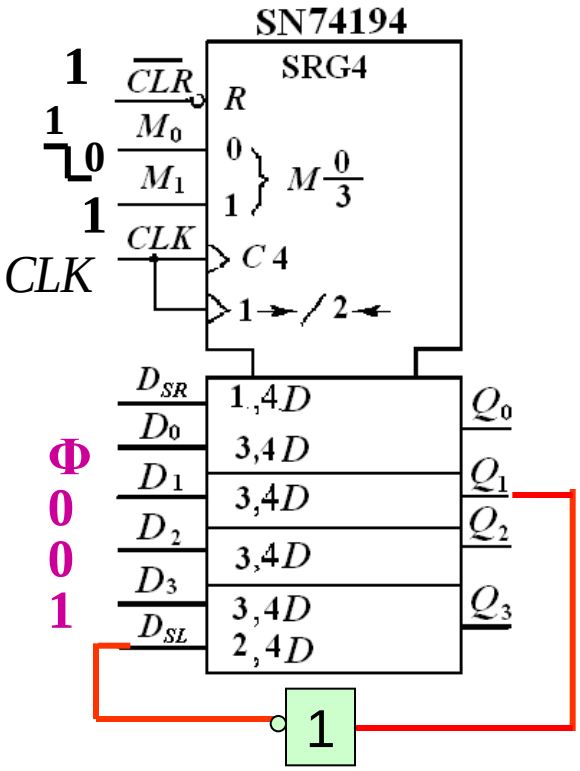


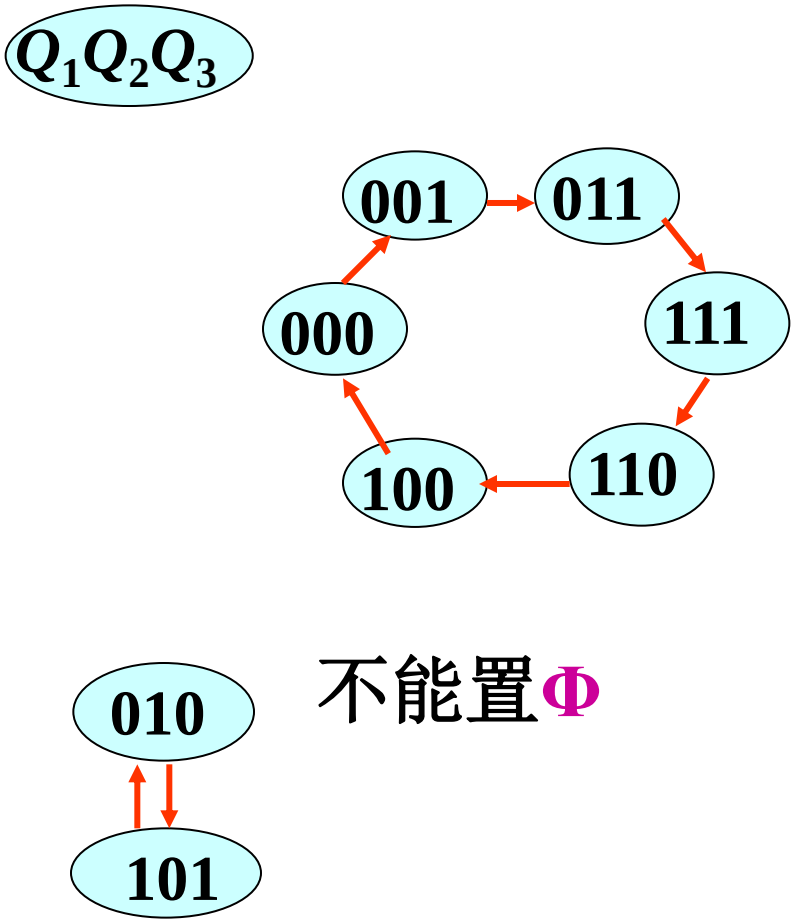
左移



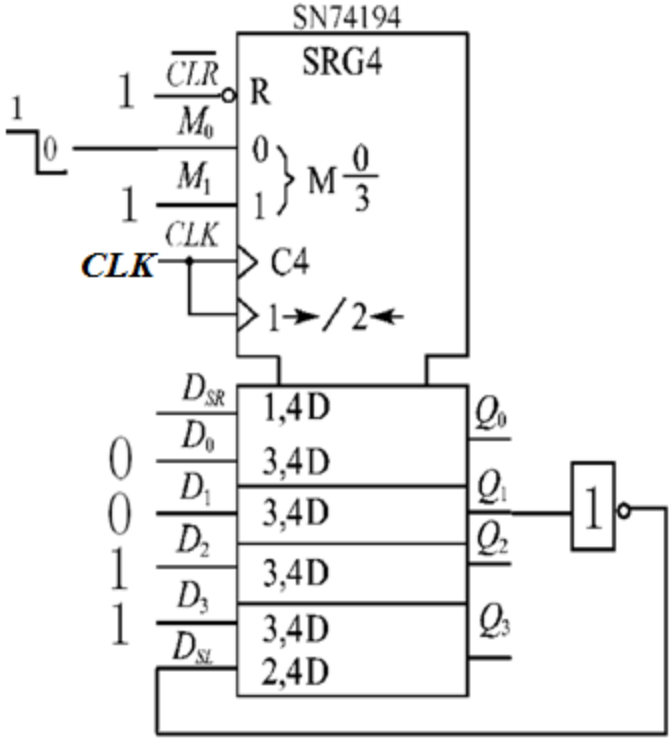
例2. 用74194设计模6 扭环计数器，画出状态图

3 FFs　　左移

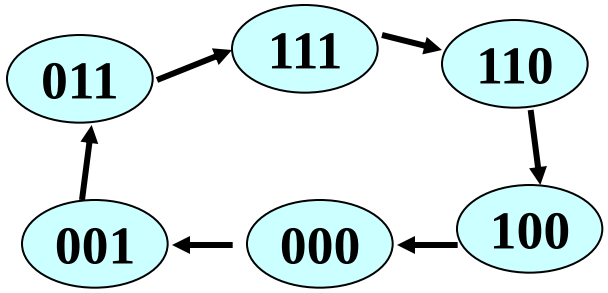




例3：分析如图所示的芯片功能，画出状态图。

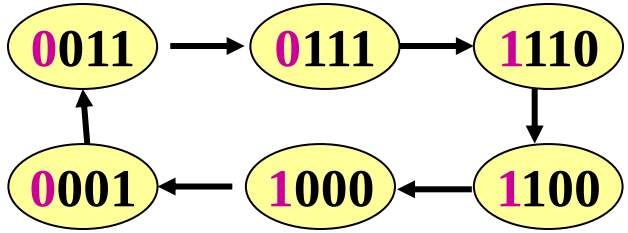


左移扭环 $Q_1Q_2Q_3$

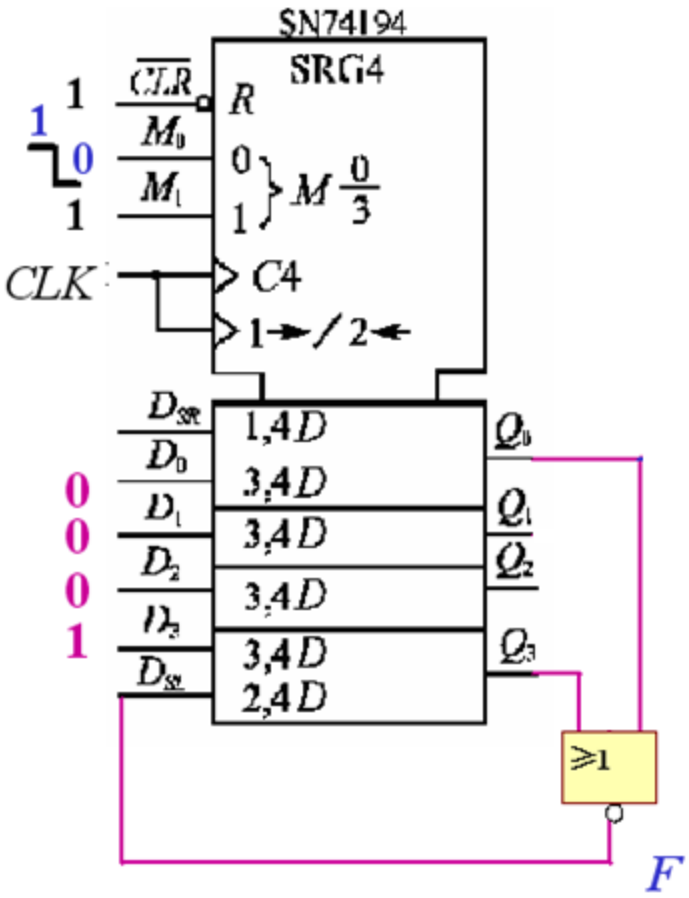


M - 6

$Q_0Q_1Q_2Q_3$

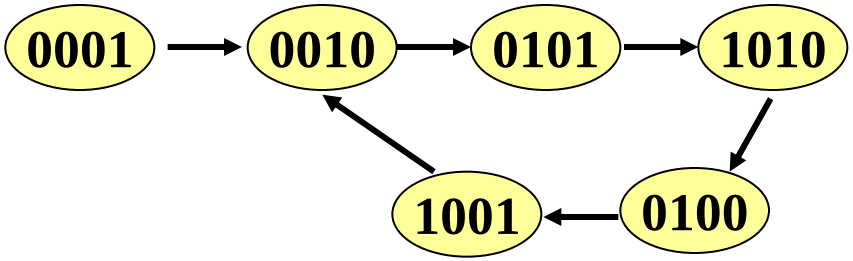


例4：画出下图74194连接电路的状态图



$$D_{SL} = F = \overline{Q_0 + Q_3}$$

$$Q_0Q_1Q_2Q_3 \rightarrow Q_1Q_2Q_3D_{SL}$$



§6.6 序列信号发生器

Series Signal Generator

Applications of Counters and Registers

序列信号：一组特定的循环数字信号

序列信号发生器：产生一组序列信号的时序电路

序列信号发生器分类：

- 计数型 **Counter-type**
- 移位型 **Shift-type**

6.6.1 计数型序列信号发生器

Counter-type Series Signal Generator

Counter + MUX

例：

设计一个产生7位序列信号 **1010110** （时间顺序：从左到右）的序列信号发生器.

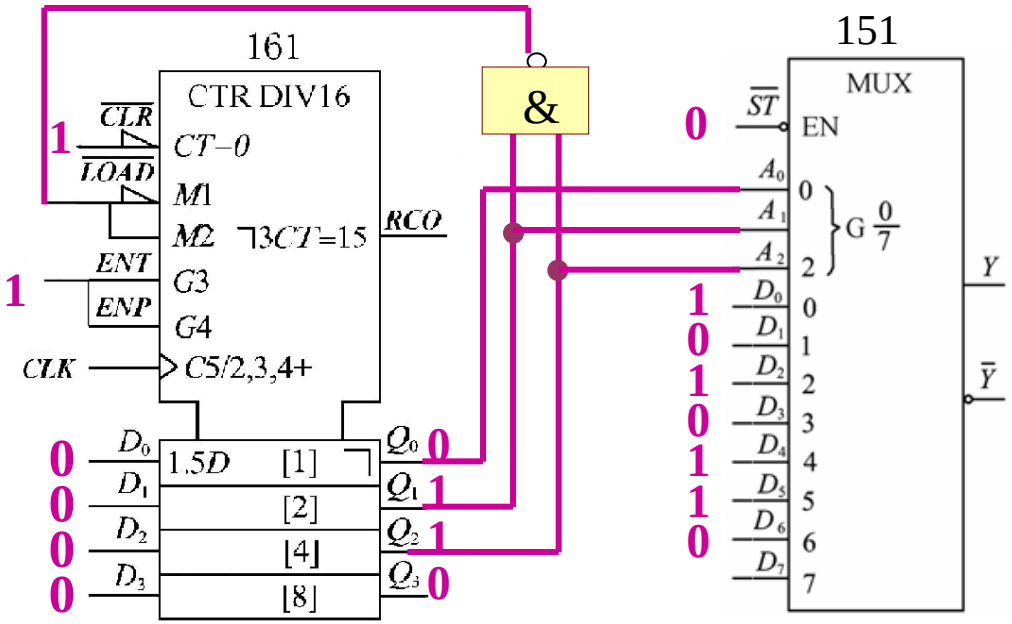
结构：

M-7 Counter → 7 Bit → 74161

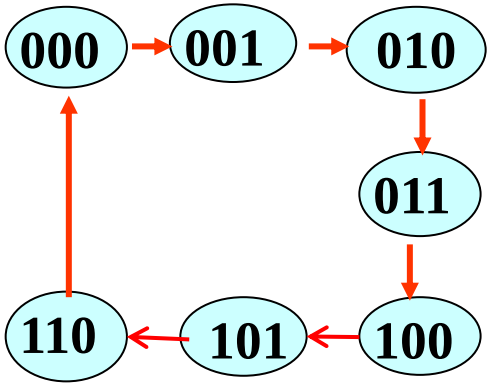
8-1 MUX → Select 1010110 → 74151

1010110

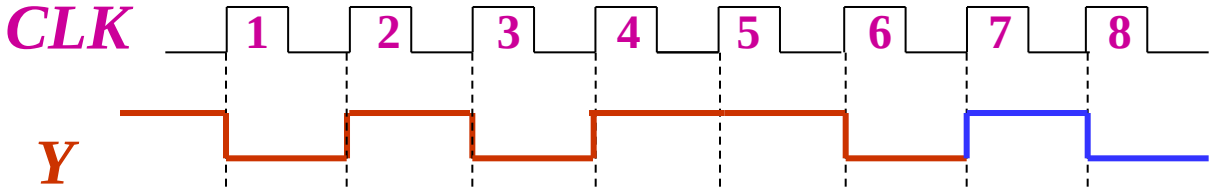
(M-7)



0110101 = *Y*



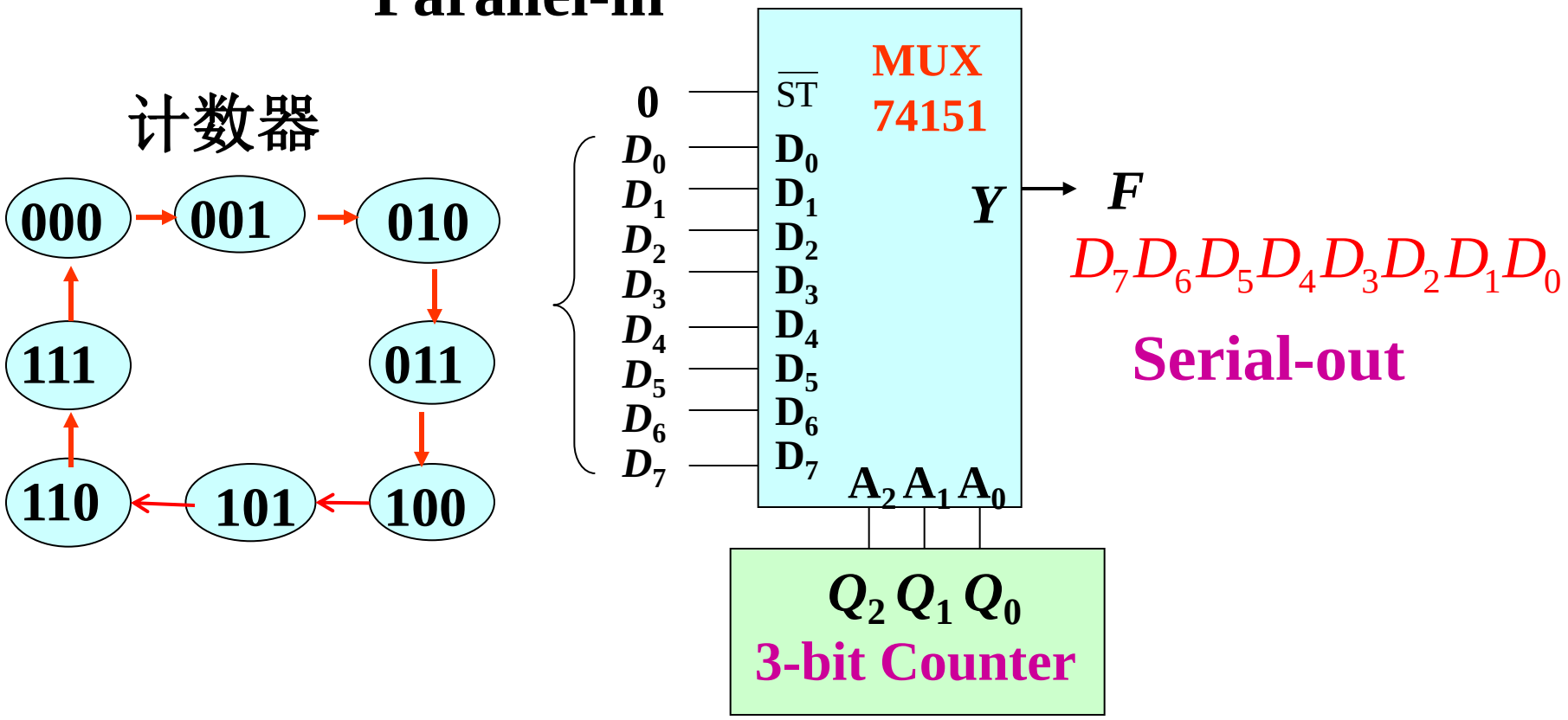
波形图：



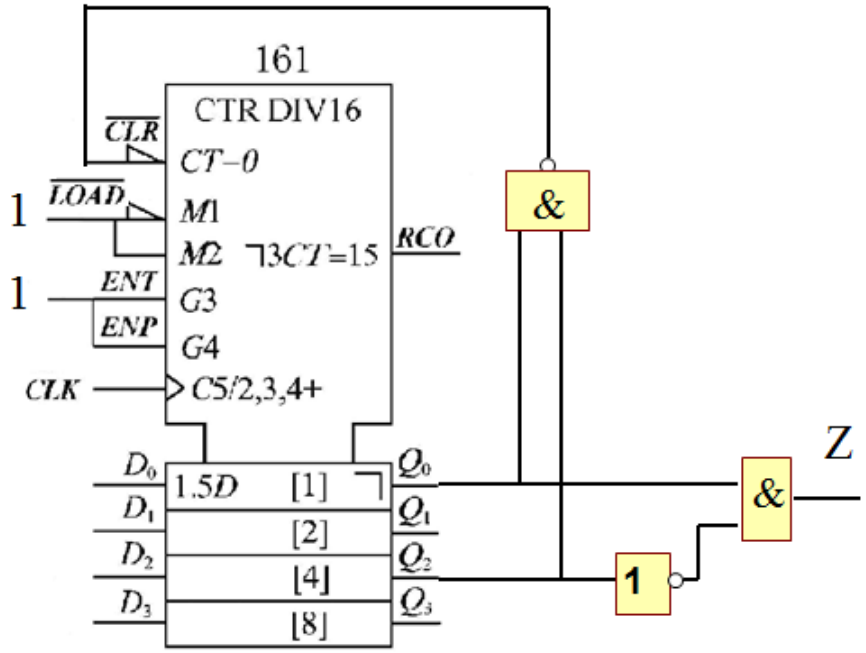
例2. 用序列信号发生器实现数据并/串转换

Counter + MUX

Parallel-in



例3 . 分析下图电路.



The counter from 000 to 100 in cycle, the output is 01010.

74161 : M-5 counter

(000) ~ (100) (101) glitch

$$Z = Q_0 \cdot \overline{Q}_2$$

输出为原状态的输出

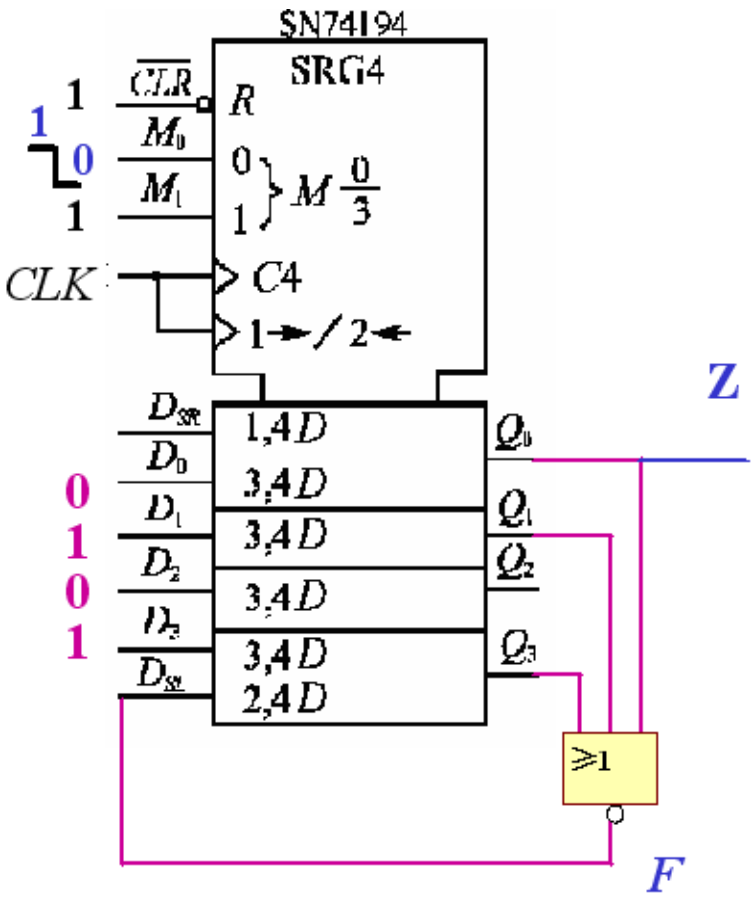
状态表

Q_2^n	Q_1^n	Q_0^n	Q_2^{n+1}	Q_1^{n+1}	Q_0^{n+1}	Z
0	0	0	0	0	1	0
0	0	1	0	1	0	1
0	1	0	0	1	1	0
0	1	1	1	0	0	1
1	0	0	0	0	0	0

The function of the circuit is 01010 series signal generator.

用计数器74161和组合电路构成序列信号发生器

例4：分析如图所示的芯片功能，
（1）画出状态图；（2）求输出Z。



分析：

左移计数器

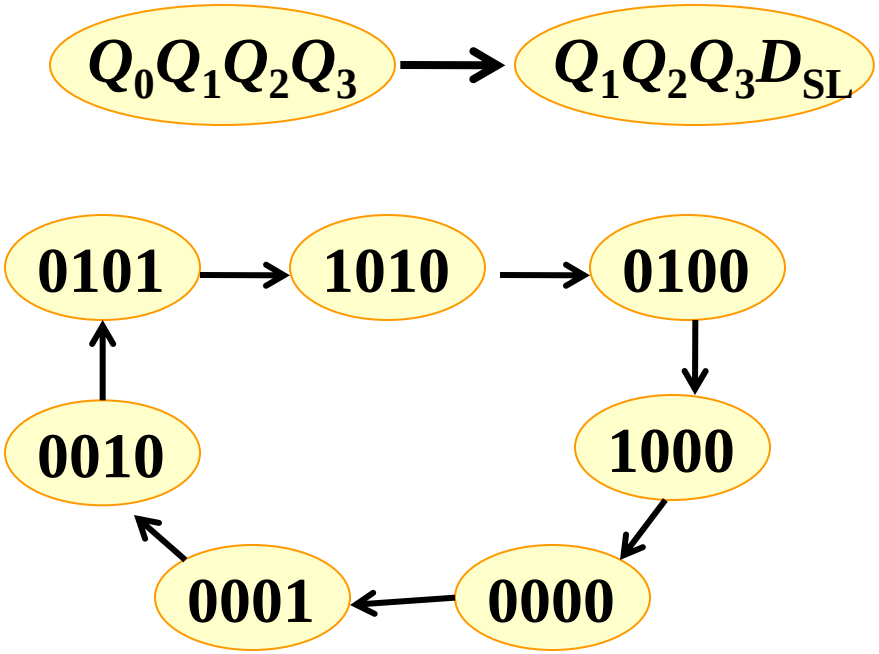
$$D_{SL} = F = \overline{Q_0 + Q_1 + Q_3}$$

$Q_0Q_1Q_2Q_3 \longrightarrow Q_1Q_2Q_3D_{SL}$

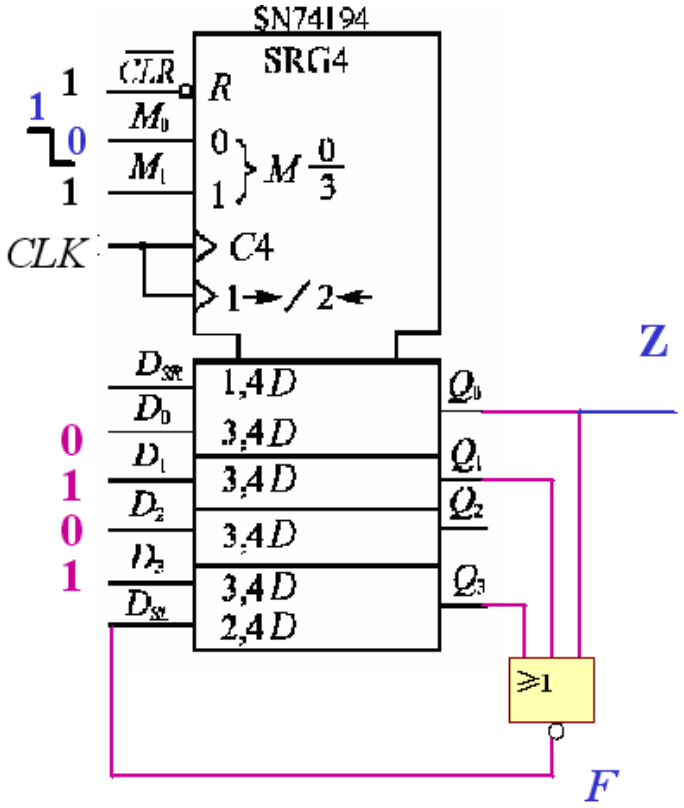
初始状态：0101

（1）画出状态图

$$D_{SL} = F = \overline{Q_0 + Q_1 + Q_3}$$



M-7 计数器



（2）求输出Z

$Z = Q_0 = 0101000$

循环输出

电路功能：Z 端输出7位序列信号0101000.

作业

6.3　　6.8　　6.9

6.12　　6.15（b）为163

6.19（改错）　　6.21

6.25*（选作）